

INTRODUCTION TO PROBABILITY

By
Nephath Jeshurun Mwanza

MAT2901 Lecture Notes Draft.

UNIVERSITY OF ZAMBIA

2026

TABLE OF CONTENTS

TABLE OF CONTENTS	iii
CHAPTER 1: Introduction To Probability	1
1.1 Introduction	1
1.2 Notation and Terminology	1
1.3 Definitions and Axioms of Probability	3
1.3.1 Axioms of probability	4
1.4 Independent Events, Condition Probability and Bayes Theorem	6
1.4.1 Independent Events	6
1.4.2 Conditional Probabilities	8
1.4.3 Bayes theorem	9
1.4.4 Contingency Tables	11
1.5 Counting Techniques	13
1.5.1 Multiplication Principle	13
1.5.2 Permutations	14
1.5.3 Combinations	16
CHAPTER 2: Random Variables and Probability Distributions	19
2.1 Random Variables	19
2.2 Discrete Random Variables	20
2.3 Continuous Random Variables	21
2.4 Mathematical Expectations and Generating Functions of Random Variables .	23
2.4.1 Expectations and Moments	24

2.4.2	Probability Generating Functions (PGFs)	30
2.4.3	Moment Generating Functions (MGFs)	33
2.4.4	Characteristic Functions	36
2.5	Some Important Discrete Probability Mass Function	37
2.5.1	The Bernoulli Distribution	37
2.5.2	Discrete Uniform	40
2.5.3	The Binomial Distribution	42
2.5.4	The Hypergeometric Distribution	46
2.5.5	The Geometric Distribution	49
2.5.6	The Negative Binomial Distribution	52
2.5.7	The Poisson Distribution	54
2.5.8	The Zeta (or Zipf) Distribution	59
2.6	Some Important Continuous Probability Density Functions	61
2.6.1	The Uniform Distribution	61
2.6.2	Exponential Distribution	64
2.6.3	Gamma Distribution	67
2.6.4	Beta Distribution	69
2.6.5	Normal Distribution	72
2.6.6	Cauchy Distribution	74
2.6.7	The Weibull Distribution	75
2.6.8	Rayleigh Distribution	76

2.6.9	The Pareto Distribution	77
2.7	Cumulative Distribution Function (CDF)	79
2.8	Markov's and Chebyshev's Inequality	89
CHAPTER 3: Jointly Distributed Random Variables		93
3.1	Introduction	93
3.2	Joint Distributions	93
3.2.1	Joint Cumulative Distribution Function	99
3.2.2	Expected Values	101
3.3	Conditional Probability Functions and Independence of Random Variables .	104
3.3.1	Special Joint Distribution Function (Bivariate Normal)	114
3.4	Joint Moment Generating Function	118

CHAPTER 1 : INTRODUCTION TO PROBABILITY

1.1. Introduction

In the scientific investigation of physical phenomena, it is essential to develop mathematical models that facilitate the prediction of observed characteristics. Broadly, these models are categorized into two distinct types: *deterministic* and *non-deterministic (stochastic)*.

Example 1.1.1. Deterministic Models A deterministic model is one in which the output is uniquely determined by the initial conditions and parameter values, without any element of randomness.

- (i) **Classical Mechanics:** The displacement s of an object under constant acceleration a after time t is given by $s = ut + \frac{1}{2}at^2$.
- (ii) **Geometry:** The volume V of a sphere is a direct function of its radius r , defined by $V = \frac{4}{3}\pi r^3$.
- (iii) **Ohm's Law:** The voltage V across a conductor is the product of the current I and the resistance R , such that $V = IR$.

Example 1.1.2 (Non-deterministic (Stochastic) Models). These models account for inherent randomness or uncertainty; even with identical initial conditions, the observed outcomes may vary.

- (i) **Telecommunications:** The number of data packets arriving at a network server during a peak hour.
- (ii) **Reliability Engineering:** The operating lifespan of a semiconductor before failure under standard thermal conditions.
- (iii) **Finance and Insurance:** The daily fluctuations of a stock market index or the number of claims filed with an insurer in a fiscal quarter.
- (iv) **Public Health:** The transmission rate and number of new infections of a pathogen within a specific population over a six-month period.

Note 1.1.3. The primary objective of probability is to provide a rigorous mathematical framework for analyzing and modeling these non-deterministic situations

1.2. Notation and Terminology

Definition 1.2.1. The following are probability terms used:

1. Random experiment: (also called act, trial, operation, or process) is an activity that leads to the occurrence of one and only one of several possible outcomes which is not likely to be known until its completion.
2. Experiment: Refers to the process of obtaining an observed result of some phenomenon, tossing a coin.
3. Outcome: A possible result of an experiment (e.g., getting a 4 on a die roll).
Only one of the possible outcomes will occur at any given trial of the experiment.
4. Sample Space S : Refers to a set of all possible outcomes of an experiment and denoted by S . (e.g., for a die, $S = \{1, 2, 3, 4, 5, 6\}$).
5. Event E : A subset of a sample space is called an event. (e.g., rolling an even number, $E = \{2, 4, 6\}$).
 - (a) Events $A_1, A_2, A_3, \dots, A_n$ are said to be *mutually exclusive*, if $A_j \cap A_i = \emptyset$ then $i \neq j$.
 - (b) Two or more events are said to be *equally likely* if each has an equal chance to occur.
 - (c) Two events are said to be *independent* if the occurrence of one does not affect the other.

Example 1.2.2.

1. Tossing of two coins and observe the side which is facing up. What is the sample space?

$$S = \{HH, HT, TH, TT\}$$

H = head up and T = Tail up.

2. Tossing two coins and observe the number of heads. What is the sample space?

$$S = \{0, 1, 2\}$$

3. A lot of N items where there are D defectives ($D \leq N$). The items are selected one by one (without replacement) until the last defective is selected (found) and observe the number of selected items. What is the sample space?

$$S = \{D, D + 1, D + 2, \dots, N\}$$

4. Light bulb is put on life test and aged to failure (Observe the time it takes fail)

$$S = \{t : t \geq 0\}, \quad t = \text{time}$$

1.3. Definitions and Axioms of Probability

Definition 1.3.1. If an experiment gives rise to a sample space S which has finite number of elements (N) of equally likely outcomes, then the probability of an event E denoted by $P(E)$ and is given by

$$P(E) = \frac{n(E)}{N},$$

while $n(E)$ is the number of elements in the set E . $P(E)$ is the ratio of number of elements in the set E with the number of elements in S .

Example 1.3.2. In an experiment of tossing two coins and observing the face up

- (i). Let the events A be a head up.
- (ii). Let the events B be exactly one tail.

Find the probabilities of (i) and (ii).

Solution. The sample space,

$$S = \{HH, HT, TH, TT\}, \quad N = 4.$$

Event is

$$A = \{HH, HT, TH\}, \quad n(A) = 3$$

then

$$P(A) = \frac{n(A)}{N} = \frac{3}{4}.$$

And

$$B = \{HT, TH\}, \quad n(B) = 2$$

hence

$$P(B) = \frac{2}{4} = \frac{1}{2}.$$

□

Another approach to probability involves the conceptual repetitions of an experiment and counting the number of trials and the number of times the event in question occurs.

Definition 1.3.3. If an experiment is repeated in times and let $M(E)$ be the number of times events A occurs in M trials $\frac{M(E)}{M}$ relative frequency of event E and $\frac{M(E)}{M}$ tends to stabilize as M becomes very large

$$P(E) = \lim_{M \rightarrow \infty} \frac{M(E)}{M}$$

If an experiment has sample space S and an event E defined on S i.e $E \subset S$ then $P(E)$ is a real number called the probability of event E . $P(\cdot)$ is a function whose domain in S and range is $[0, 1]$

$$P : S \longrightarrow [0, 1]$$

1.3.1 Axioms of probability

(i). For any event E , $0 \leq P(E) \leq 1$.

(ii). $P(S) = 1$.

(iii). For any finite number of mutually exclusives events $E_1, E_2, \dots, E_k$ then

$$P\left(\bigcup_{j=1}^k E_j\right) = \sum_{j=1}^k P(E_j).$$

Note 1.3.4.

$$\bigcup_{j=1}^k E_j = E_1 \cup E_2 \cup E_3 \cup \dots \cup E_k$$

$$\sum_{j=1}^k P(E_j) = P(E_1) + P(E_2) + \dots + P(E_k)$$

Theorem 1.3.5. Let A and B be any set events and S be the sample space of an experiment then

(i). $P(\emptyset) = 0$ where $\emptyset$ is empty set.

(ii). $P(A') = 1 - P(A)$

(iii). $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

(iv). if $A \subset B \implies P(A) \leq P(B)$

Proof. We prove (iii), we express $A \cup B$, A and B as

$$A \cup B = (A \cap B') \cup (A \cap B) \cup (A' \cap B) \quad (1.1)$$

$$A = (A \cap B) \cup (A \cap B') \quad (1.2)$$

$$B = (A \cap B) \cup (A' \cap B) \quad (1.3)$$

From the three equations we get

$$P(A \cup B) = P(A \cap B') + P(A \cap B) + P(A' \cap B)$$

$$P(A) = P(A \cap B) + P(A \cap B')$$

$$P(B) = P(A \cap B) + P(A' \cap B)$$

Therefore

$$\begin{aligned} P(A \cup B) &= P(A) - P(A \cap B) + P(A \cap B) + P(B) - P(A \cap B) \\ &= P(A) + P(B) - P(A \cap B). \end{aligned}$$

□

Example 1.3.6. If event A and B are such that $P(A) = 0.3$, $P(B) = 0.6$ and $P(A \cup B) = 0.8$, evaluate the following

(i). $P(A')$

Solution. $P(A') = 1 - P(A) = 1 - 0.3 = 0.7$

□

(ii). $P(A \cap B)$

Solution. $P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.3 + 0.6 - 0.8 = 0.1$

□

(iii). $P(A' \cup B')$

Solution. $P(A' \cup B') = P((A \cap B)') = 1 - P(A \cap B) = 1 - 0.1 = 0.9$

□

(iv). $P(A' \cap B')$

Solution. $P(A' \cap B') = P((A \cup B)') = 1 - P(A \cup B) = 1 - 0.8 = 0.2$

□

Example 1.3.7. The Samsung cell phones require replacement of possible defects of the LCD and in the charging system defects. Samsung have being informed that 2% of the phones have defective LCD only and 4% have defective charging system only. If 90% of these phones have neither defective, what percentage of these have both defects?

Solution. Let A be the event of a defective LCD and B be the event of a defective charging system. We are given:

- $P(A \cap B') = 0.02$ (Defective LCD *only*)
- $P(B \cap A') = 0.04$ (Defective charging *only*)
- $P((A \cup B)') = 0.90$ (Neither defective)

First, find the probability that a phone has at least one defect:

$$P(A \cup B) = 1 - P((A \cup B)') = 1 - 0.90 = 0.10$$

We know that $P(A \cup B)$ is the sum of the disjoint regions:

$$P(A \cup B) = P(A \cap B') + P(B \cap A') + P(A \cap B)$$

$$0.10 = 0.02 + 0.04 + P(A \cap B)$$

$$P(A \cap B) = 0.10 - 0.06 = 0.04$$

Conclusion: 4% of the phones have both defects. □

1.4. Independent Events, Condition Probability and Bayes Theorem

1.4.1 Independent Events

Definition 1.4.1. Events A and B are said to be independent if and only if $P(A \cap B) = P(A)P(B)$.

Lemma 1.4.2. Two events A and B are independent if and only if $P(A \cap B) = P(A)P(B)$.

Theorem 1.4.3. If A and B are independent then the following are also independent

- (i). A and B'

Proof. We know that $P(A \cap B) = P(A)P(B)$, since A and B are independent.

$$A = (A \cap B') \cup (A \cap B)$$

$$P(A) = P(A \cap B') + P(A \cap B)$$

Therefore

$$\begin{aligned} P(A \cap B') &= P(A) - P(A \cap B) \\ &= P(A) - P(A)P(B) \\ &= P(A)(1 - P(B)) \\ &= P(A)P(B'). \end{aligned}$$

i.e $P(A \cap B') = P(A)P(B')$ implies A and B' are independent. □

(ii). A' and B

Proof. Write $B = (A' \cap B) \cup (A \cap B)$, then $P(B) = P(A' \cap B) + P(A \cap B)$

$$\begin{aligned} P(A' \cap B) &= P(B) - P(A \cap B) \\ &= P(B) - P(A)P(B) \\ &= P(B)(1 - P(A)) \\ &= P(B)P(A'). \end{aligned}$$

i.e A' and B are independent. □

(iii). A' and B'

Proof. Write $A' = (A' \cap B) \cup (A' \cap B')$, then

$$P(A') = P(A' \cap B) + P(A' \cap B')$$

$$\begin{aligned} P(A' \cap B') &= P(A') - P(A' \cap B) \\ &= P(A')(1 - P(B)) \\ &= P(A')P(B'). \end{aligned}$$

i.e A' and B' are independent. □

1.4.2 Conditional Probabilities

The probability properties discussed so far relate to the entire sample space. We have used the symbol $P(A)$ and will continue to do so however we should have been more precise and used $P(A/S)$ to read "probability of A given the sample space S ".

The following definition justifies why $P(A/S)$ is simply $P(A)$ if S is the sample space.

Definition 1.4.4. Given a sample space S , and events A and B , then the conditional probability of A given B is denoted by $P(A/B)$ and is given by

$$P(A/B) = \begin{cases} \frac{P(A \cap B)}{P(B)} & \text{if } P(B) > 0 \\ 0 & \text{if } P(B) = 0 \end{cases}$$

Theorem 1.4.5. If A and B are independent events, then

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)P(B)}{P(B)} = P(A).$$

Remark 1.4.6. We have $P(A \cap B) = P(B)P(A/B)$ this is known as multiplicative rule of probability.

Definition 1.4.7. If $B_1, B_2, \dots, B_k$ are mutually exclusive events of the sample space S and if

$$\bigcup_{j=1}^k B_j = S,$$

then $B_1, B_2, \dots, B_k$ are said to be a partition of the sample space S .

Theorem 1.4.8 (Total Probability). If $B_1, B_2, \dots, B_k$ is a partition of the sample space S and A is an arbitrary event on S then the probability of A is given by

$$\begin{aligned} P(A) &= \sum_{j=1}^k P(B_j)P(A/B_j) \\ &= P(B_1)P(A/B_1) + P(B_2)P(A/B_2) + \dots + P(B_k)P(A/B_k). \end{aligned}$$

Proof. Let S be a sample space and A be an event in S . Then for a partition $B_1, B_2, \dots, B_k$

of S . We have

$$A = (A \cap B_1) \cup (A \cap B_2) \cup (A \cap B_3) \cup \cdots \cup (A \cap B_k) = \bigcup_{j=1}^k (A \cap B_j)$$

Therefore

$$\begin{aligned} P(A) &= \sum_{j=1}^k P(A \cap B_j) \\ &= \sum_{j=1}^k P(B_j)P(A/B_j). \end{aligned}$$

□

1.4.3 Bayes theorem

There are many situations where the ultimate outcome of an experiment depends on what happens in various intermediate stages. This issue is resolved by the Bayes' Theorem.

Definition 1.4.9 (Bayes theorem). If $B_1, B_2, \dots, B_k$ is a partition of the sample space S and A is an arbitrary event on S , then

$$P(B_r/A) = \frac{P(B_r)P(A/B_r)}{\sum_{j=1}^k P(B_j)P(A/B_j)}, \quad r = 1, 2, \dots, k.$$

Proof. From the definitions of conditional probability and total probability, we obtain

$$P(B_r/A) = \frac{P(B_r \cap A)}{P(A)} = \frac{P(B_r)P(A/B_r)}{\sum_{j=1}^k P(B_j)P(A/B_j)}.$$

□

Note 1.4.10. In the Bayes Theorem, the probability $P(B_j)$ is called prior probability. The Probability $P(B_j/A)$ is called posterior probability.

Example 1.4.11. In a factory for manufacturing of car batteries, machines A , B and C manufacture respectively 25%, 35% and 40% of the total produce. The Percentage of defective batteries from machine A , B and C are 5, 4 and 2 respectively.

- (i). A car battery is selected at random. What is the probability that it is defective?

Solution. Machines are the partitions: A , B , C . Let D represent the battery is defective. Given

$$P(A) = 0.25, \quad P(B) = 0.35, \quad P(C) = 0.40,$$

$$P(D/A) = 0.05, \quad P(D/B) = 0.04, \quad P(D/C) = 0.02$$

Then

$$\begin{aligned} P(D) &= P(A) P(D/A) + P(B) P(D/B) + P(C) P(D/C) \\ &= 0.25 \times 0.05 + 0.35 \times 0.04 + 0.40 \times 0.02 \\ &= 0.0125 + 0.0140 + 0.0080 \\ &= 0.0345 \end{aligned}$$

□

- (ii). A car battery is selected at random and found to be defective. What is the probability that it was manufactured by machine B ?

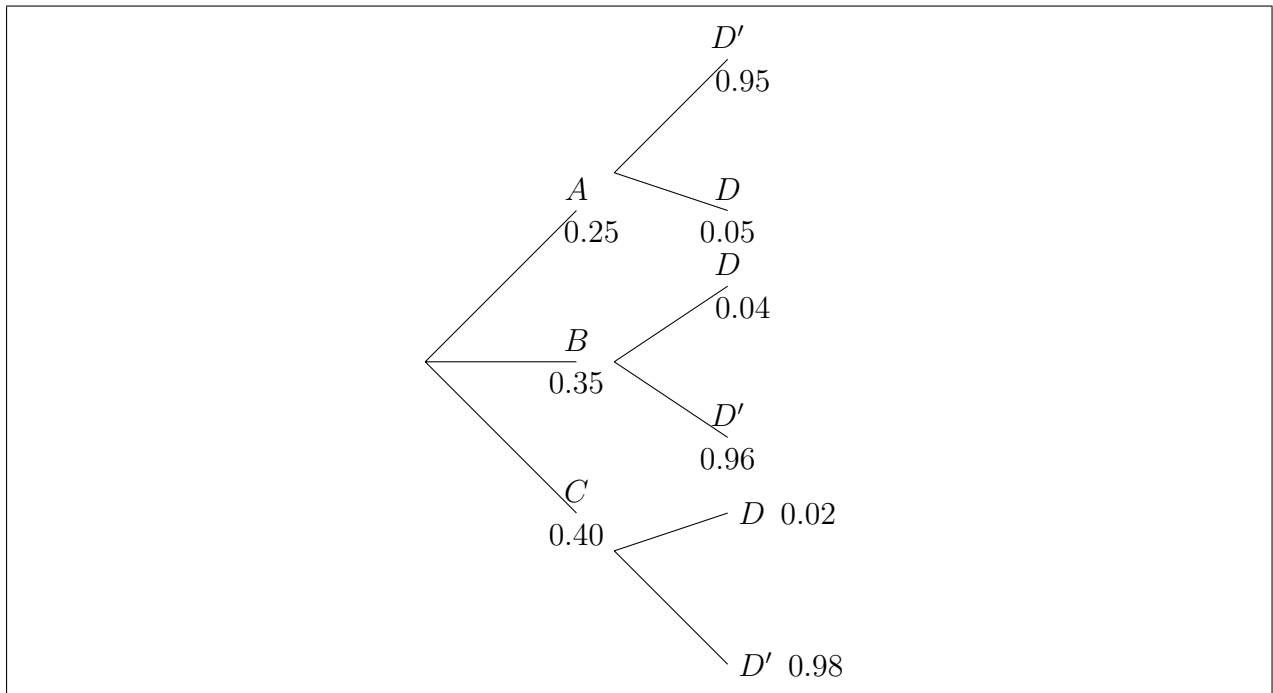
Solution.

$$\begin{aligned} P(B/D) &= \frac{P(B)P(D/B)}{P(D)} \\ &= \frac{(0.35 \times 0.04)}{0.0345} \\ &= \frac{0.0140}{0.0345} \\ &= \frac{28}{69} \end{aligned}$$

□

- (iii). Battery is selected at random and found to be non-defective. What is the probability that it was manufactured by machine A ?

Solution. You can use the tree diagram to find $P(D')$



Or you can use the fact that $P(D') = 1 - P(D)$.

$$\begin{aligned}
 P(A/D') &= \frac{P(A) P(D'/A)}{P(D')} \\
 &= \frac{0.25 \times 0.95}{1 - 0.0345} \\
 &= \frac{475}{1931} \\
 &= 0.246
 \end{aligned}$$

□

1.4.4 Contingency Tables

A Contingency Table is a tabular method used to classify sample observations according to two or more identifiable categories. In the case of two variables, the table rows represent the levels of one variable, and the columns represent the levels of the other. The intersections (cells) represent the joint probability of those specific events occurring simultaneously.

Purpose of Contingency Tables:

1. To organize complex joint event data into a readable format.

2. The row and column totals (margins) provide the probabilities of a single event occurring, regardless of the other variable.
3. They provide the raw data necessary to determine if two categorical variables are associated.

Example 1.4.12. To investigate the relationship between smoking and health outcomes, let A be the event that an individual smokes (A^c for non-smokers) and D be the event that an individual develops lung cancer (D^c for no cancer). If the probability that

- (i). an individual smokes and develops lung cancer is 0.15.
- (ii). an individual smoke and doesn't develops lung cancer is 0.25.
- (iii). an individual doesn't smoke and develops lung cancer is 0.10.
- (iv). and individual doesn't smoke and doesn't develop lung cancer is 0.50.

How can this information be used to examine, the relationship between smoking and lung cancer?

Solution. We are given the following joint probabilities:

$$P(A \cap D) = 0.15, \quad P(A \cap D^c) = 0.25, \quad P(A^c \cap D) = 0.10, \quad P(A^c \cap D^c) = 0.50$$

Step 1: Construct the probability table.

By summing the rows and columns, we obtain the marginal probabilities $P(A)$ and $P(D)$.

Smoking Status	Cancer (D)	No Cancer (D^c)	Marginal Totals
Smoker (A)	0.15	0.25	$P(A) = 0.40$
Non-Smoker (A^c)	0.10	0.50	$P(A^c) = 0.60$
Marginal Totals	$P(D) = 0.25$	$P(D^c) = 0.75$	1.00

Table 1.1: Contingency table of Smoking and Lung Cancer probabilities.

Step 2: Comparative Analysis (Relative Risk)

To see how smoking affects cancer risk, we compare the conditional probabilities:

Risk for Smokers

$$P(D|A) = \frac{P(A \cap D)}{P(A)} = \frac{0.15}{0.40} = 0.375$$

Risk for Non-Smokers

$$P(D|A^c) = \frac{P(A^c \cap D)}{P(A^c)} = \frac{0.10}{0.60} \approx 0.167$$

The Relative Risk (RR):

$$RR = \frac{P(D|A)}{P(D|A^c)} = \frac{0.375}{0.1667} = 2.25$$

An individual who smokes is 2.25 times more likely to develop lung cancer than a non-smoker.

Step 3: The Odds Ratio (OR)

In epidemiology, we often use the Odds Ratio (or Cross-Product Ratio) to estimate the strength of association. For a table with cells a, b, c, d :

$$OR = \frac{a \times d}{b \times c} = \frac{0.15 \times 0.50}{0.25 \times 0.10} = \frac{0.075}{0.025} = 3$$

While the Relative Risk (2.25) tells us how much the probability increases, the Odds Ratio (3.0) provides a different mathematical perspective on the strength of the link. Since $OR \neq 1$, we can conclude that smoking and lung cancer are not independent events. $\square$

1.5. Counting Techniques

In the classical definition of probability, $P(E) = \frac{n(E)}{N}$, we must be able to count the number of outcomes in the event space and the sample space. When the number of outcomes is too large to list manually, we use *counting techniques* or *combinatorial analysis*.

1.5.1 Multiplication Principle

If an operation can be performed in n_1 ways, and for each of these a second operation can be performed in n_2 ways, then the two operations can be performed together in $n_1 \times n_2$ ways.

If there are k operations, the total number of ways is $n_1 \times n_2 \times \cdots \times n_k$.

Example 1.5.1. Find the number of possible outcomes of the rolling of a die and then a

tossing a coin.

Solution. Here $n_1 = 6$ and $n_2 = 2$. By the multiplication principle, the number of possible outcomes is $6 \times 2 = 12$.

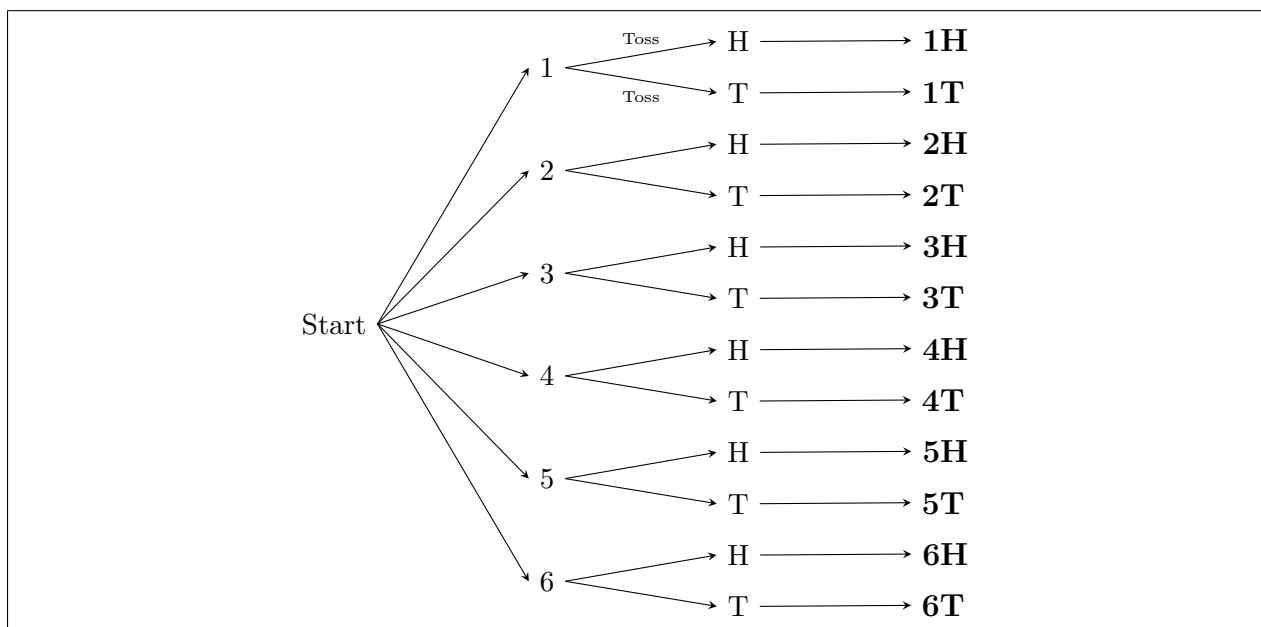


Figure 1.1: Tree Diagram for Rolling a Die and Tossing a Coin

□

Example 1.5.2. How many different license plates are possible if Lusaka uses three letters followed by two digits.

Solution.

$$(26)^3(10)^2 = 1,757,600$$

□

1.5.2 Permutations

A permutation is an arrangement of all or part of a set of objects in a specific order. The order of selection matters and one permutation differs from another permutation.

Definition 1.5.3. The number of permutations of n distinct objects taken k at a time is denoted by ${}_nP_k$ or $P(n, k)$ or P_k^n and is given by:

$${}_nP_k = \frac{n!}{(n-k)!}$$

where $n!$ (n-factorial) is $n \times (n - 1) \times \cdots \times 1$. By definition, $0! = 1$.

Example 1.5.4. Suppose that there are 4 objects: p, q, r, s – then the possible arrangements of these objects if 3 are taken at a time are: $n = 4$ and $k = 3$

$${}_nP_k = \frac{n!}{(n - k)!} = \frac{4!}{(4 - 3)!} = \frac{4 \times 3 \times 2 \times 1!}{1!} = 24.$$

There are 24 arrangements; permutations of four things taken three at a time.

pqr	pqs	prs	qrs
prq	psq	psr	qsr
qrp	qsp	rsp	rsq
qpr	qps	rps	rqs
rqp	sqp	srp	srq
rpq	spq	spr	sqr

Definition 1.5.5. Suppose the objects there are n objects, where k_1 are of one type, k_2 are of another type, $\cdots$, k_m are the m^{th} type, where $k_1 + k_2 + \cdots + k_m = n$.

The number of distinct permutations of n objects is denoted by $P_{k_1, k_2, \dots, k_m}^n$ and is given by

$$P_{k_1, k_2, \dots, k_m}^n = \frac{n!}{k_1! k_2! \cdots k_m!}.$$

Example 1.5.6. (a). How many different ways can 3 red, 4 yellow and 2 blue light bulbs be arranged on line of tree lights with nine sockets.

Solution. We have $k_1 = 3$, $k_2 = 4$, $k_3 = 2$ and $n = 3 + 4 + 2$

$$P_{k_1, k_2, k_3}^n = \frac{n!}{k_1! k_2! k_3!}$$

$$P_{3, 4, 2}^9 = \frac{9!}{3! 4! 2!} = \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4!}{3 \times 2 \times 1 \times 2 \times 1 \times 4!} = 1260.$$

□

(b). How many different ways can seven female students on a field trip course be assigned to one room and two double rooms at a quest house.

Solution. The seven students are being partitioned into three distinct group: one group of 3 (triple room) and two group of 2 (double rooms) We have $k_1 = 3$, $k_2 = 2$, $k_3 = 2$ and $n = 7$

$$P_{k_1, k_2, k_3}^n = \frac{n!}{k_1! k_2! k_3!}$$

$$P_{1,2}^7 = \frac{7!}{3!2!2!} = \frac{7 \times 6 \times 5 \times 4 \times 3!}{3! \times 2 \times 2} = 210.$$

□

1.5.3 Combinations

A combination is a selection of objects where the order does not matter. We are only interested in which objects are chosen, not the sequence in which they are picked.

Definition 1.5.7. The number of combinations of n distinct objects taken k at a time is denoted by nC_k , $\binom{n}{k}$, or $C(n, k)$ and is given by:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Remark 1.5.8.

1. Express the permutation as a combination

$$P_k^n = k! \binom{n}{k}.$$

2. Let $n \in \mathbb{N}$ and $k = 0, 1, 2, \dots, n$. Then

$$\binom{n}{k} = \binom{n}{n-k}.$$

3. For any positive integer n and $k = 1, 2, 3, \dots, n$, we have

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}.$$

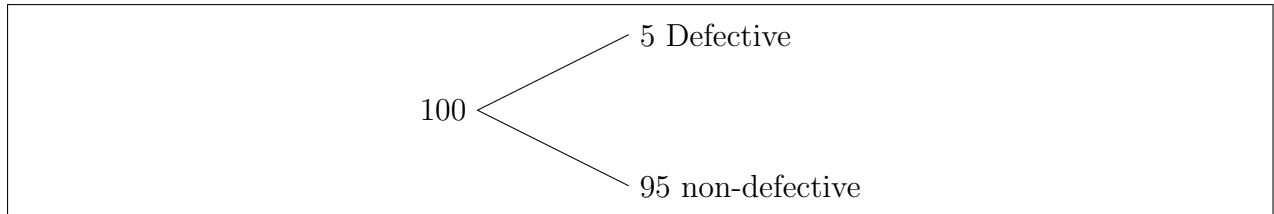
4. Let m and n be positive integers. Then

$$\sum_{r=0}^k \binom{m}{r} \binom{n}{k-r} = \binom{m+n}{k}.$$

Example 1.5.9. A production lot of size 100 has 5% defective items. A random sample of size 10 is selected (without replacement). Let A be the event that there are exactly 2 defective, B be the event that there are no defectives and C be the event that there are at most 2 defectives. D event that there is at least 2 defectives. Find the probabilities of A, B and C, D .

Solution. Let $N = 100$, with $D = 5$ defectives and $n - D = 95$ non-defectives. We select $n = 10$.

Possible ways of the samples $\binom{100}{10}$.



A: Exactly 2 defectives

Ways to pick 2 defectives from 5

$$\binom{5}{2} = \frac{5!}{2!(5-2)!} = \frac{5 \times 4 \times 3!}{2 \times 1 \times 3!} = 10.$$

Event space $n(A)$

$$n(A) = \binom{5}{2} \binom{95}{8}.$$

The probability

$$P(A) = \frac{\binom{5}{2} \binom{95}{8}}{\binom{100}{10}} = \frac{1335}{19012} \approx 0.07022$$

B: No defectives

Ways to pick 0 defectives from 5

$$\binom{5}{0} = \frac{5!}{0!(5-0)!} = 1.$$

Event space $n(B)$

$$n(B) = \binom{5}{0} \binom{95}{10}.$$

The probability

$$P(B) = \frac{\binom{5}{0} \binom{95}{10}}{\binom{100}{10}} \approx 0.584$$

C: At most 2 defectives

Ways to pick at most 2 defectives

$$\binom{5}{0} + \binom{5}{1} + \binom{4}{2}$$

Event space

$$n(C) = \binom{5}{0} \binom{95}{10} + \binom{5}{1} \binom{95}{9} + \binom{5}{2} \binom{95}{8}.$$

The probability

$$P(C) = \frac{\binom{5}{0} \binom{95}{10}}{\binom{100}{10}} + \frac{\binom{5}{1} \binom{95}{9}}{\binom{100}{10}} + \frac{\binom{5}{2} \binom{95}{8}}{\binom{100}{10}} \approx 0.9934$$

D: At least 2 defectives

$$P(D) = 1 - P(X < 2)$$

$$P(D) = 1 - \left[\frac{\binom{5}{0} \binom{95}{10}}{\binom{100}{10}} + \frac{\binom{5}{1} \binom{95}{9}}{\binom{100}{10}} \right] = 0.0769$$

□

CHAPTER 2 : RANDOM VARIABLES AND PROBABILITY DISTRIBUTIONS

In many experiments, we are not interested in the specific outcome itself, but rather in a numerical value associated with that outcome. For example, if we test three electronic components, we may not care which specific ones fail, but only the total number of failures

2.1. Random Variables

Definition 2.1.1. A random variable denoted by X is a function whose value is a real number determined by each element in the sample space.

$$X : S \longrightarrow \mathbb{R}$$

S is the domain of X , $\mathbb{R}_X$ is the range (space of the random variable X) which is the subset of $\mathbb{R}$.

Note: A random variable is neither random nor variable, it is simply a function. The values it takes on are both random and variable.

Example 2.1.2. Consider tossing a fair coin three times. The sample space S contains 8 outcomes. If we define X as the number of heads, the random variable maps each sequence to an integer:

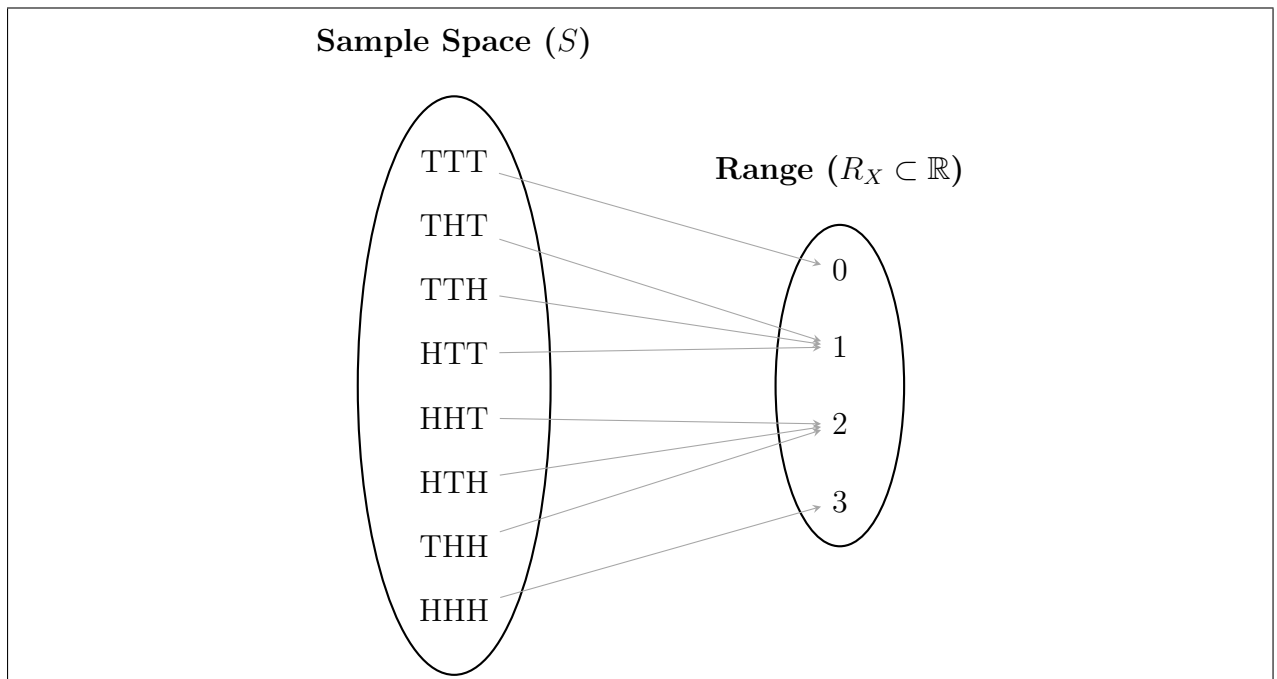


Figure 2.1: Mapping from the Sample Space to the Real Line for X = number of heads.

Example 2.1.3. In industrial monitoring, such as that performed by the Environmental Council of Zambia (ECZ), the nature of the data determines the classification of the variable.

1. Pesticide Concentration (X): If we measure the mass of pesticide in milligrams per liter, X can take any value in a continuous interval. Then the range is

$$R_X = \{x \in \mathbb{R} : x \geq 0\},$$

and the data type is continuous (uncountably infinite outcomes).

2. Monthly Compliance (X): If we count the number of months until a company fails an inspection, X can be any whole number. The range is

$$R_X = \{0, 1, 2, 3, \dots\},$$

and the data type is discrete (countably infinite outcomes).

Note 2.1.4. In practice, we use X (uppercase) to refer to the name of the variable and x (lowercase) to refer to a specific value it might take.

2.2. Discrete Random Variables

Definition 2.2.1. A random variable X is said to be discrete if its range R_X is either finite or countably infinite. This means the possible values can be listed in a sequence $x_1, x_2, x_3, \dots$.

Definition 2.2.2 (Probability Mass Function (PMF)). The probability distribution of a discrete random variable is described by a probability mass function, denoted as $f_X(x)$ or $P(X = x)$. This function assigns a probability to each possible value in the range of X such that:

1. $f_X(x) \geq 0$ for all x .
2. $\sum_{x \in R_X} f(x) = 1$.

Example 2.2.3. Let the random variable X represent the number of heads obtained in three independent tosses of a fair coin.

1. **Tabular Representation.** The distribution can be summarized in a probability table, which provides a clear view of the entire probability mass:

x	0	1	2	3
$P(X = x)$	1/8	3/8	3/8	1/8

Table 2.1: Probability distribution for the number of heads in 3 tosses.

2. **Functional Representation (The Formula).** For a balanced coin, the probability of heads $p = 1/2$. The probability of observing exactly x heads in 3 trials follows the Binomial logic:

$$\begin{aligned} P(X = x) &= \binom{3}{x} \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{3-x} \\ &= \binom{3}{x} \left(\frac{1}{2}\right)^3 \\ &= \frac{\binom{3}{x}}{8}, \quad x = 0, 1, 2, 3 \end{aligned}$$

There are special discrete probability functions which do arise in many experiments and will discuss them later. There are Bernoulli, Binomial, Poisson, Geometric, Hypergeometric, discrete uniform e.t.c

2.3. Continuous Random Variables

Definition 2.3.1. A random variable X is continuous if its range R_X is an uncountable interval (or a collection of intervals) of real numbers. X can take on any value within a given range.

Definition 2.3.2. For a continuous random variable, we define a probability density function, $f_X(x)$, such that:

1. $f_X(x) \geq 0$ for all $x \in \mathbb{R}$.
2. $\int_{-\infty}^{\infty} f_X(x) dx = 1$ (The total area under the curve is 1).

Remark 2.3.3. 1. For a continuous random variable, the probability of assuming any exact value is zero: $P(X = x) = 0$.

2. Because $P(X = x) = 0$, the inclusion or exclusion of endpoints in an interval does not change the probability:

$$P(a < X < b) = P(a \leq X \leq b) = P(a \leq X < b) = P(a < X \leq b)$$

3. Probability is found by integrating the p.d.f over the desired interval:

$$P(a < X < b) = \int_a^b f_X(x) dx.$$

Example 2.3.4. Let X be a random variable with the following p.d.f

$$f_X(x) = \begin{cases} \frac{x+1}{8}, & 2 < x < 4 \\ 0, & \text{elsewhere} \end{cases}$$

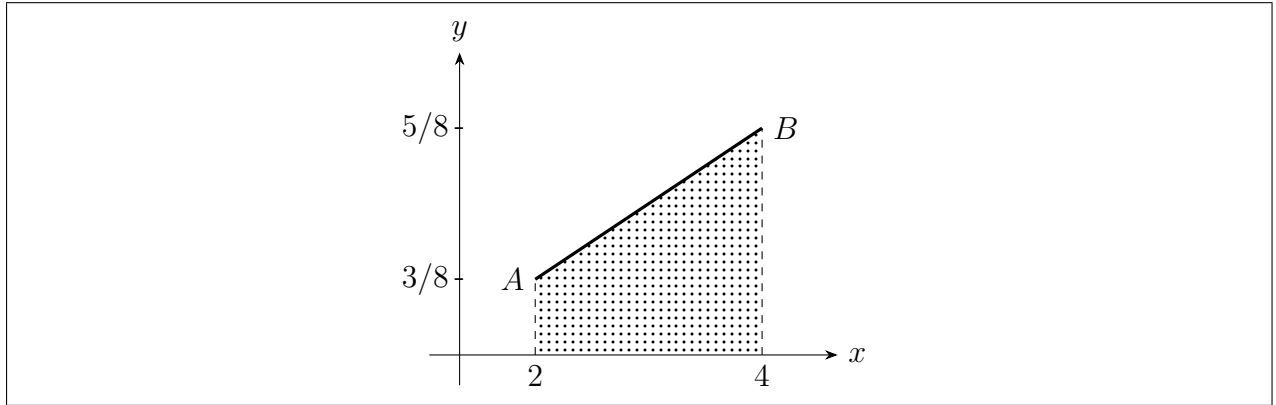


Figure 2.2:

(i). Show that $f(x)$ is probability density function.

Solution.

$$\text{Area} = \int_2^4 \frac{x+1}{8} dx = \left[\frac{(x+1)^2}{16} \right]_2^4 = \frac{(4+1)^2 - (2+1)^2}{16} = \frac{25 - 9}{16} = 1$$

□

(ii). Find $P(X > 3.5)$

Solution.

$$P(X > 3.5) = \int_{3.5}^4 \frac{x+1}{8} dx = \left[\frac{(x+1)^2}{16} \right]_{3.5}^4 = \frac{5^2 - 4.5^2}{16} = \frac{25 - 20.25}{16} = \frac{4.75}{16}.$$

□

(iii). Find $P(2.4 < X < 3.5)$

Solution.

$$P(2.4 < X < 3.5) = \int_{2.4}^{3.5} \frac{x+1}{8} dx = \left[\frac{(x+1)^2}{16} \right]_{2.4}^{3.5} = \frac{4.5^2 - 3.4^2}{16} \approx 0.5431$$

□

Property	Discrete Random Variable	Continuous Random Variable
Support (R_X)	Finite or Countably Infinite	Uncountable Interval(s)
Function Type	Probability Mass Function (PMF)	Probability Density Function (PDF)
Notation	$f(x) = P(X = x)$	$f(x)$ (where $P(X = x) = 0$)
Conditions	$0 \leq f(x) \leq 1$	$f(x) \geq 0$
Total Value	$\sum_{x \in R_X} f(x) = 1$	$\int_{-\infty}^{\infty} f(x) dx = 1$
Interval Prob.	$P(a \leq X \leq b) = \sum_{x=a}^b f(x)$	$P(a \leq X \leq b) = \int_a^b f(x) dx$

Table 2.2: Side-by-side comparison of Discrete and Continuous frameworks.

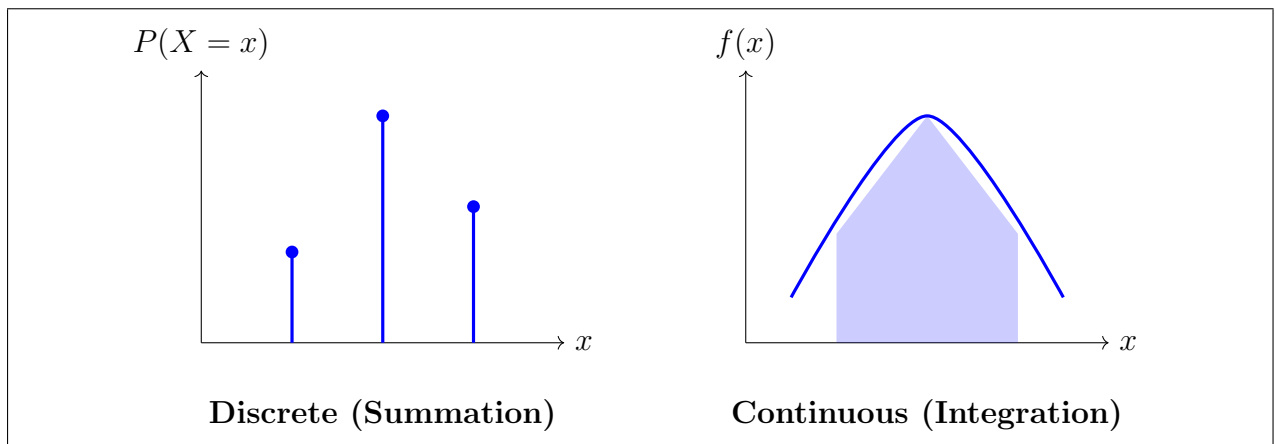


Figure 2.3: Visualizing the transition from individual points of mass to a density of area.

2.4. Mathematical Expectations and Generating Functions of Random Variables

While the p.m.f and p.d.f provide a complete description of a random variable, we often require concise numerical descriptors to summarize a distribution's location and spread. This section moves from basic Expectations to the powerful "transform" methods—Generating Functions—which serve as essential tools for identifying distributions and simplifying the calculation of higher-order moments.

2.4.1 Expectations and Moments

Definition 2.4.1. Let X be a random variable the mean or expectation of the random variable X denoted by μ or $E(X)$ and it's given by

$$E(X) = \begin{cases} \sum_{\text{all } x} x f_X(x) = \sum_{\text{all } x} x P(X = x), & \text{if } X \text{ is discrete.} \\ \int_{\text{all } x} x f_X(x) dx, & \text{if } X \text{ is continuous} \end{cases}$$

Example 2.4.2. Suppose we toss a fair coin three times. Let the random variable X represent the number of heads obtained. The probability mass function (PMF) of X is given by:

$$f_X(x) = P(X = x) = \frac{\binom{3}{x}}{8}, \quad \text{for } x \in \{0, 1, 2, 3\}$$

Find the expected value of X .

Solution. The expected value is

$$\begin{aligned} E(X) &= \sum_x x \cdot f_X(x) \\ &= \sum_{x=0}^3 x \cdot \frac{\binom{3}{x}}{8} \\ &= \frac{1}{8} \left[\sum_{x=0}^3 x \cdot \binom{3}{x} \right] \\ &= \frac{1}{8} \left[0 \cdot \binom{3}{0} + 1 \cdot \binom{3}{1} + 2 \cdot \binom{3}{2} + 3 \cdot \binom{3}{3} \right] \\ &= \frac{1}{8} [0(1) + 1(3) + 2(3) + 3(1)] \\ &= \frac{3 + 6 + 3}{8} = \frac{12}{8} \\ \mathbf{E(X) = 1.5} \end{aligned}$$

Interpretation: If we were to repeat this experiment many times, we would expect to see an average of 1.5 heads per three tosses. Note that while the outcome 1.5 is not possible in a single trial, it represents the long-run theoretical mean. $\square$

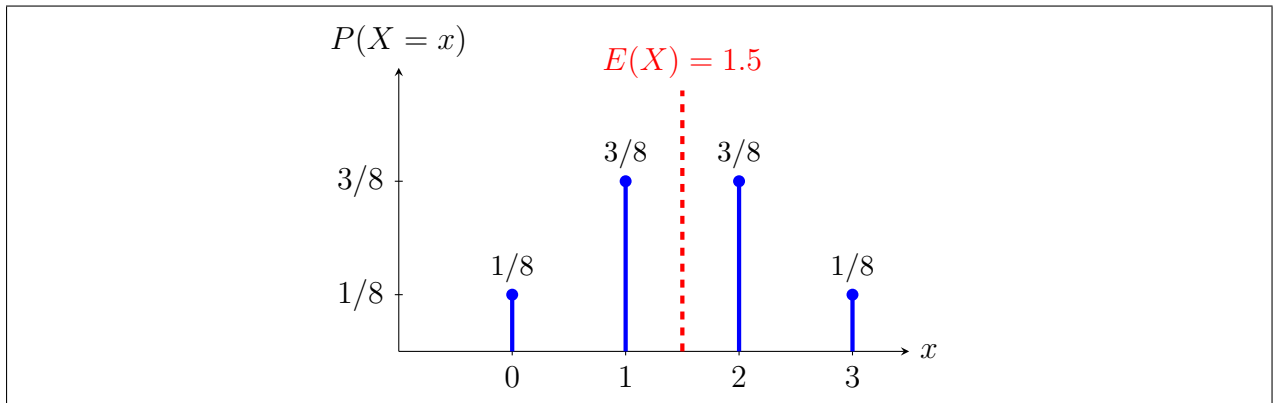


Figure 2.4: PMF for the number of heads in 3 tosses, showing the mean at the center of symmetry.

Example 2.4.3. Let X be a continuous random variable with the probability density function (pdf) defined by:

$$f_X(x) = \begin{cases} \frac{x+1}{8}, & 2 < x < 4 \\ 0, & \text{otherwise} \end{cases}$$

Calculating the Expected Value $E(X)$.

Solution.

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} x \cdot f_X(x) dx \\ &= \int_2^4 x \left(\frac{x+1}{8} \right) dx \\ &= \frac{1}{8} \int_2^4 (x^2 + x) dx \\ &= \frac{1}{8} \left[\frac{x^3}{3} + \frac{x^2}{2} \right]_2^4 \\ &= \frac{1}{8} \left[\left(\frac{64}{3} + 8 \right) - \left(\frac{8}{3} + 2 \right) \right] \\ &= \frac{1}{8} \left[\frac{56}{3} + 6 \right] = \frac{1}{8} \left[\frac{56 + 18}{3} \right] \\ &= \frac{74}{24} \\ &= \frac{37}{12} \end{aligned}$$

□

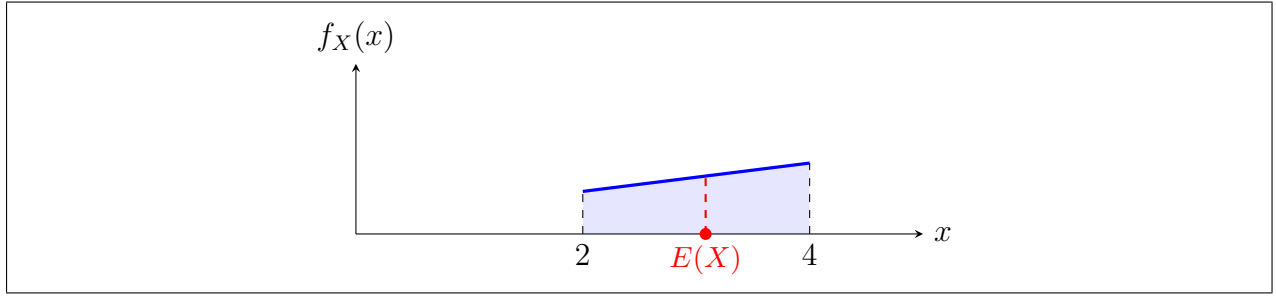


Figure 2.5: The PDF of X over the interval $(2, 4)$. The mean is located at $x \approx 3.083$.

Theorem 2.4.4. Let X be a random variable with probability function $f_X(x)$, and $g(x)$ be any function then the expectation of $h(x)$ is given by

$$E(g(x)) = \begin{cases} \sum_x h(x)f_X(x), & \text{if } X \text{ is discrete} \\ \int_x h(x)f_X(x), & \text{if } X \text{ is continuous} \end{cases}$$

Remark 2.4.5. If $h(x) = x$ which is the identity function $E[h(x)] = E(x) = \mu$.

Theorem 2.4.6 (Properties of Expectations). Let X be a random variable, b and c be constants, $g(x)$ and $h(x)$ be functions, then

$$(i). E[b g(X) + c h(X)] = b E[g(X)] + c E[h(X)]$$

$$(ii). E[g(X) + b] = E[g(X)] + b$$

Proof. Suppose that X is a continuous random variable, then

$$\begin{aligned} E[g(X) + b] &= \int_{-\infty}^{\infty} [g(X) + b] f_X(x) dx \\ &= \int_{-\infty}^{\infty} g(X) f_X(x) dx + \int_{-\infty}^{\infty} b f_X(x) dx \\ &= E[g(X)] + b \int_{-\infty}^{\infty} f_X(x) dx \\ &= E[g(X)] + b. \end{aligned}$$

This completes the proof. □

$$(iii). \text{ If } g(X) \leq h(X), \text{ then } E(g(X)) \leq E(h(X))$$

Proof. Let $g(x) = a$ and $h(x) = b$, where $a < b$.

$$E[g(X)] = E(a) = a$$

$$E[h(X)] = E(b) = b$$

Therefore, $a < b \implies E[g(X)] < E[h(X)]$. □

(iv). $E[E(X)] = E(X)$

Remark 2.4.7 (Special Expectations). If X is a random variable, then

1. The k^{th} moment (about the origin):

$$E(X^k)$$

2. The k^{th} moment about the mean:

$$E[(X - \mu)^k].$$

3. The k^{th} factorial moment:

$$E[X^{(k)}] = E[X(X-1)(X-2) \cdots (X-k+1)]$$

Definition 2.4.8. Let X be a random variable with mean μ , then the variance of X denoted by σ^2 or $\text{Var}(X)$ is given by

$$\sigma^2 = E(X - \mu)^2,$$

where $\mu = E(X)$, $\sigma^2 > 0$.

Computation formula of σ^2 is given by

$$\begin{aligned} \sigma^2 &= E(X - \mu)^2 \\ &= E[X - E(X)]^2 \\ &= E[X^2 - 2XE(X) + E(X)^2] \\ &= E(X^2) - E[2XE(X)] + E[E(X)]^2 \\ &= E(X^2) - 2E(X)E(X) + [E(X)]^2 \\ &= E(X^2) - 2[E(X)]^2 + [E(X)]^2 \\ &= E(X^2) - [E(X)]^2. \end{aligned}$$

Example 2.4.9. Using *Example 2.4.2*, we found the mean $E(X) = 1.5$. Now we find the variance

$$\text{Var}(X) = E(X^2) - E[E(X)]^2.$$

First we find $E(X^2)$.

$$\begin{aligned} E(X^2) &= \sum_x x^2 \cdot f_X(x) \\ &= \sum_{x=0}^3 x^2 \cdot \frac{\binom{3}{x}}{8} \\ &= \frac{1}{8} \left[0^2 \binom{3}{0} + 1^2 \binom{3}{1} + 2^2 \binom{3}{2} + 3^2 \binom{3}{3} \right] \\ &= \frac{1}{8} [0(1) + 1(3) + 4(3) + 9(1)] \\ &= \frac{24}{8} \\ &= 3. \end{aligned}$$

Now, substitute $E(X^2) = 3$ and $E(X) = 1.5$ into the variance formula:

$$\begin{aligned} \text{Var}(X) &= E(X^2) - [E(X)]^2 \\ &= 3 - (1.5)^2 \\ &= 3 - 2.25 \\ &= 0.75 \end{aligned}$$

Example 2.4.10. In *Example 2.4.3* we calculated the mean to be $E(X) = \frac{37}{12}$. Now, we find $E(X^2)$ and subsequently the variance.

$$\begin{aligned} E(X^2) &= \int_2^4 x^2 \left(\frac{x+1}{8} \right) dx \\ &= \frac{1}{8} \int_2^4 (x^3 + x^2) dx \\ &= \frac{1}{8} \left[\frac{x^4}{4} + \frac{x^3}{3} \right]_2^4 \\ &= \frac{1}{8} \left[\left(\frac{256}{4} + \frac{64}{3} \right) - \left(\frac{16}{4} + \frac{8}{3} \right) \right] \\ &= \frac{236}{24} = \frac{59}{6} \end{aligned}$$

Now, we substitute $E(X^2) = \frac{59}{6}$ and $E(X) = \frac{37}{12}$ into the variance formula:

$$\begin{aligned}
 \text{Var}(X) &= E(X^2) - [E(X)]^2 \\
 &= \frac{59}{6} - \left(\frac{37}{12}\right)^2 \\
 &= \frac{59}{6} - \frac{1369}{144} \\
 &= \frac{59 \times 24}{144} - \frac{1369}{144} \\
 &= \frac{1416 - 1369}{144} \\
 &= \frac{47}{144}.
 \end{aligned}$$

Theorem 2.4.11. If X is a random variable with mean μ and variance σ^2 , then

$$\text{Var}(aX + b) = a^2 \text{Var}(X),$$

where a and b are any real constants.

Proof.

$$\begin{aligned}
 \text{Var}(aX + b) &= E([(aX + b) - \mu_{aX+b}]^2) \\
 &= E([(aX + b) - E(aX + b)]^2) \\
 &= E([aX + b - aE(X) - b]^2) \\
 &= E([aX - aE(X)]^2) \\
 &= E(a^2[X - E(X)]^2) \\
 &= a^2 E([X - \mu]^2) \\
 &= a^2 \text{Var}(X).
 \end{aligned}$$

□

Remark 2.4.12. 1. The mean is the balance point (the center of gravity) where the seesaw stays level. while, the variance describes how the weight is spread out. In physics, this is called the moment of inertia.

2. The square root of the $\text{Var}(X)$ is called the *standard deviation* of X , that is

$$\sigma_X = \sqrt{\text{Var}(X)}.$$

Exercise 2.4.13. Let X have the p.d.f

$$f_X(x) = \begin{cases} \frac{2x}{k^2}, & 0 \leq x \leq k \\ 0, & \text{otherwise.} \end{cases}$$

For what value of k is the variance of X equal to 2?

2.4.2 Probability Generating Functions (PGFs)

For discrete random variables taking non-negative integer values, the PGF acts as a powerful "storage device" where the individual probabilities of the distribution are encoded as the coefficients of a power series.

Definition 2.4.14. For a discrete random variable X , the PGF, denoted by $G_X(t)$, is defined as

$$G_X(t) = E(t^X) = \sum_{x=0}^{\infty} t^x P(X = x)$$

for all values of $t \in \mathbb{R}$ for which the power series converges.

Remark 2.4.15. The three key properties of PGF include:

1. Evaluating the PGF at $t = 1$ must always equal 1, as it represents the sum of all probabilities in the sample space:

$$G_X(1) = \sum_{x=0}^{\infty} P(X = x) = 1$$

2. The individual probabilities can be recovered by taking the k -th derivative of the PGF and evaluating it at $t = 0$:

$$P(X = k) = \frac{G_X^{(k)}(0)}{k!}$$

3. The derivatives of the PGF evaluated at $t = 1$ generate what is called *factorial moments*. The first derivative gives the mean:

$$G_X'(1) = E(X)$$

The second derivative provides the second factorial moment, which is used to calculate variance:

$$G_X''(1) = E[X(X - 1)] = E(X^2) - E(X).$$

Therefore

$$\text{Var}(X) = G_X''(1) + G_X'(1) - [G_X'(1)]^2.$$

Example 2.4.16. Consider the following discrete probability distribution

x	2	4	5	10
$P(X = x)$	0.1	0.2	0.3	0.4

Find the Probability Generating Function, $G_X(t)$.

Solution. By definition, the PGF is the expected value of t^X

$$\begin{aligned} G_X(t) &= \sum_x t^x P(X = x) \\ &= t^2 P(X = 2) + t^4 P(X = 4) + t^5 P(X = 5) + t^{10} P(X = 10) \\ &= 0.1t^2 + 0.2t^4 + 0.3t^5 + 0.4t^{10} \end{aligned}$$

□

Example 2.4.17. In a game, the probability that A wins on her x th go can be described as a discrete random variable X , with probability function

$$P(X = x) = \frac{1}{2^x}, \quad \text{for } x = 1, 2, 3, 4, \dots$$

1. Find the probability generating function.

Solution. Note that

$$P(X = 1) = \frac{1}{2}, \quad P(X = 2) = \frac{1}{2^2}, \quad P(X = 3) = \frac{1}{2^3}, \quad P(X = 4) = \frac{1}{2^4}, \quad \dots$$

$$\begin{aligned} G_X(t) &= \sum_x t^x P(X = x) \\ &= \frac{1}{2}t + \frac{1}{2^2}t^2 + \frac{1}{2^3}t^3 + \frac{1}{2^4}t^4 + \dots \\ &= \sum_{x=1}^{\infty} \left(\frac{t}{2}\right)^x \end{aligned}$$

Not that “Standard Geometric Series” is usually presented as:

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$$

However, if you start at $n = 1$, you are simply missing the first term ($r^0 = 1$):

$$\sum_{n=1}^{\infty} r^n = \left(\sum_{n=0}^{\infty} r^n \right) - 1 = \frac{1}{1-r} - 1 = \frac{r}{1-r}.$$

Thus, the closed-form PGF is

$$G_X(t) = \frac{\frac{t}{2}}{1 - \frac{t}{2}} = \frac{t}{2-t}, \quad \text{for } |t| < 2.$$

□

2. Find $E(X)$ and $\text{Var}(X)$.

Solution. The mean $E(X)$,

$$G'_X(t) = \frac{(2-t)(1) - t(-1)}{(2-t)^2} = \frac{2-t+t}{(2-t)^2} = \frac{2}{(2-t)^2}$$

Evaluating at $t = 1$,

$$E(X) = G'_X(1) = \frac{2}{(2-1)^2} = 2.$$

The variance $\text{Var}(X)$. $G''_X(1)$:

$$G''_X(t) = \frac{d}{dt}[2(2-t)^{-2}] = 2(-2)(2-t)^{-3}(-1) = \frac{4}{(2-t)^3}$$

Evaluating at $t = 1$,

$$G''_X(1) = \frac{4}{(2-1)^3} = 4$$

Now, use the variance formula for PGFs.

$$\text{Var}(X) = G''_X(1) + G'_X(1) - [G'_X(1)]^2$$

$$\text{Var}(X) = 4 + 2 - (2)^2 = 6 - 4$$

$$\text{Var}(X) = 2.$$

□

2.4.3 Moment Generating Functions (MGFs)

There are some distributions whose moments are difficult to compute from the definition. A moment generating function is a real valued function from which one can generate all the moments of a given random variable.

Definition 2.4.18 (Moment Generating function (m.g.f)). Let X be a random variable with probability function $f_X(x)$ then the moment generating function of X is denoted by $M_X(t)$ and given by $M_X(t) = E(e^{tx})$.

$$M_X(t) = \begin{cases} \sum_x e^{tx} P(X = x), & \text{if } X \text{ is discrete} \\ \int_x e^{tx} f_X(x) dx & \text{if } X \text{ is continuous} \end{cases}$$

The exponential series

$$e^y = \sum_{k=0}^{\infty} \frac{y^k}{k!} = 1 + y + \frac{y^2}{2!} + \frac{y^3}{3!} + \dots$$

If $y = 1$,

$$e = 1 + 1 + \frac{1}{2} + \frac{1}{3!} + \frac{1}{4!} + \dots$$

Remark 2.4.19. (a). We only need $M_X(t)$ to be defined in the neighbourhood of zero i.e $t \in (-\varepsilon, \varepsilon)$ for any $\varepsilon > 0$.

(b). Not every random variable has a moment generating function. But if the moment generating function of a random variable exists, then it is unique.

c). The m.g.f is used to generate moments i.e $E(X^k)$ is known as the k^{th} raw moment.

$$E(X) = \left. \frac{d}{dt} M_X(t) \right|_{t=0}$$

$$E(X^2) = \left. \frac{d}{dt} \left[\frac{d}{dt} M_X(t) \right] \right|_{t=0} = \left. \frac{d^2}{dt^2} M_X(t) \right|_{t=0}$$

$$\sigma^2 = E(X^2) - (E(X))^2.$$

Example 2.4.20. What is the moment generating function of the random variable X whose p.d.f is given by

$$f_X(x) = e^{-x} \quad x > 0.$$

What are the mean and variance of X ?

Solution. The moment generating function of X is

$$\begin{aligned} M_X(t) &= E(e^{tX}) \\ &= \int_0^\infty e^{tx} e^{-x} dx \\ &= \int_0^\infty e^{-(1-t)x} dx \\ &= \frac{1}{1-t} [-e^{-(1-t)x}]_0^\infty \\ &= \frac{1}{1-t}, \quad 1-t > 0. \end{aligned}$$

The expected value of X is

$$\begin{aligned} E(X) &= \left. \frac{d}{dt} M_X(t) \right|_{t=0} \\ &= (-1)(-1)(1-t)^{-2} \Big|_{t=0} \\ &= \left. \frac{1}{(1-t)^2} \right|_{t=0} \\ &= 1. \end{aligned}$$

Similarly

$$\begin{aligned} E(X^2) &= \left. \frac{d^2}{dt^2} M(t) \right|_{t=0} \\ &= \left. \frac{(-1)(-2)}{(1-t)^3} \right|_{t=0} \\ &= \left. \frac{2}{(1-t)^3} \right|_{t=0} \\ &= 2. \end{aligned}$$

Hence, the variance of X is

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = 2 - 1 = 1.$$

□

Theorem 2.4.21. Let $M_X(t)$ be the moment generating function of the random variable X . If

$$M_X(t) = a_0 + a_1t + a_2t^2 + \cdots a_nt^n + \cdots \quad (2.1)$$

is the Taylor series expansion of $M_X(t)$, then

$$E(X^n) = (n!) a_n$$

for $n \in \mathbb{R}$.

Proof. Let $M_X(t)$ be the m.g.f. of the random variable X . The Taylor series expansion of $M_X(t)$ about zero is given by

$$M_X(t) = M(0) + \frac{M'(0)}{1!}t + \frac{M''(0)}{2!}t^2 + \frac{M'''(0)}{3!}t^3 + \cdots + \frac{M^{(n)}(0)}{n!}t^n + \cdots$$

Since $E(X^n) = M^{(n)}(0)$ for $n \geq 1$ and $M(0) = 1$, we have

$$M(t) = 1 + \frac{E(X)}{1!}t + \frac{E(X^2)}{2!}t^2 + \frac{E(X^3)}{3!}t^3 + \cdots + \frac{E(X^n)}{n!}t^n + \cdots \quad (2.2)$$

From (2.1) and (2.2), equating the coefficients of the like powers of t , we obtain

$$a_n = \frac{E(X^n)}{n!} \implies E(X^n) = (n!) a_n.$$

Hence the proof. □

Example 2.4.22. Find the 99th moment of X about the origin, if the moment generating function of X is $M_X(t) = \frac{1}{1+t}$?

Solution. The Taylor series expansion of $M_X(t)$ is

$$\begin{aligned} M_X(t) &= \frac{1}{1+t} = \frac{1}{1-(-t)} \\ &= 1 + (-t) + (-t)^2 + (-t)^3 + \cdots + (-t)^n + \cdots \\ &= 1 - t + t^2 - t^3 + \cdots + (-1)^n t^n + \cdots \end{aligned}$$

Therefore, $a_n = (-1)^n$ and we get $a_{99} = -1$. Thus,

$$E(X^{99}) = (99!)a_{99} = -99!$$

□

2.4.4 Characteristic Functions

Definition 2.4.23. If a random variable does not have a well-defined moment generating function m.g.f, we can use the *characteristic function* as

$$\phi_X(t) = E[e^{itX}],$$

where $i = \sqrt{-1}$ and t is a real number.

Theorem 2.4.24. Some of the properties of the characteristic function include

1. $\phi_X(t)$ is defined for all $t \in \mathbb{R}$.
2. $|\phi_X(t)| \leq \phi_X(0) = 1$.

Proof. Since $|e^{itX}| = 1$, then

$$|\phi_X(t)| = |E[e^{itX}]| \leq E[|e^{itX}|] \leq 1.$$

□

3. If the n^{th} moment $E(X^n)$ exists, it can be found by differentiating the function k times at the origin:

$$E(X^n) = \frac{1}{i^n} \left[\frac{d^n}{dt^n} \phi_X(t) \right]_{t=0}.$$

4. If you shift or scale a random variable (e.g., $Y = aX + b$), then

$$\phi_{aX+b}(t) = e^{itb} \phi_X(at).$$

Example 2.4.25. Let X be a random variable with a p.d.f

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0.$$

Use the characteristic function to find the mean.

Solution. Derive the characteristic function

$$\begin{aligned}
 \phi_X(t) &= E[e^{itX}] \\
 &= \int_0^\infty e^{itx} \lambda e^{-\lambda x} dx \\
 &= \lambda \int_0^\infty e^{-(\lambda - it)x} dx \\
 &= \lambda \left[-\frac{e^{-(\lambda - it)x}}{(\lambda - it)} \right]_0^\infty \\
 &= \frac{\lambda}{\lambda - it}.
 \end{aligned}$$

Then first derivative ($n = 1$)

$$\phi'_X(t) = \lambda(-1)(-i)(\lambda - it)^{-2} = \frac{i\lambda}{(\lambda - it)^2}.$$

The mean

$$E(X) = \frac{1}{i} \phi'_X(0) = \frac{1}{i} \left(\frac{i\lambda}{\lambda^2} \right) = \frac{1}{\lambda}.$$

□

2.5. Some Important Discrete Probability Mass Function

we focus on specific families of discrete random variables that appear frequently in scientific research, engineering, and economics. Each distribution is defined by one or more parameters that determine its shape and characteristics. By identifying a situation as fitting one of these models, we can use pre-derived formulas for their mean, variance, and generating functions.

2.5.1 The Bernoulli Distribution

The Bernoulli trial is the simplest form of a random experiment, characterized by exactly two possible outcomes: "success" and "failure." By convention, we map the outcome "success" to the value 1 and "failure" to the value 0. Let p denote the probability of success, where $0 \leq p \leq 1$. It follows that the probability of failure is $1 - p$, often denoted as q .

Definition 2.5.1. A random variable X is said to follow a Bernoulli distribution, denoted

$X \sim \text{Ber}(p)$, if its probability mass function (PMF) is:

$$f(x; p) = p^x(1 - p)^{1-x}, \quad x \in 0, 1$$

where $p = P(X = 1)$.

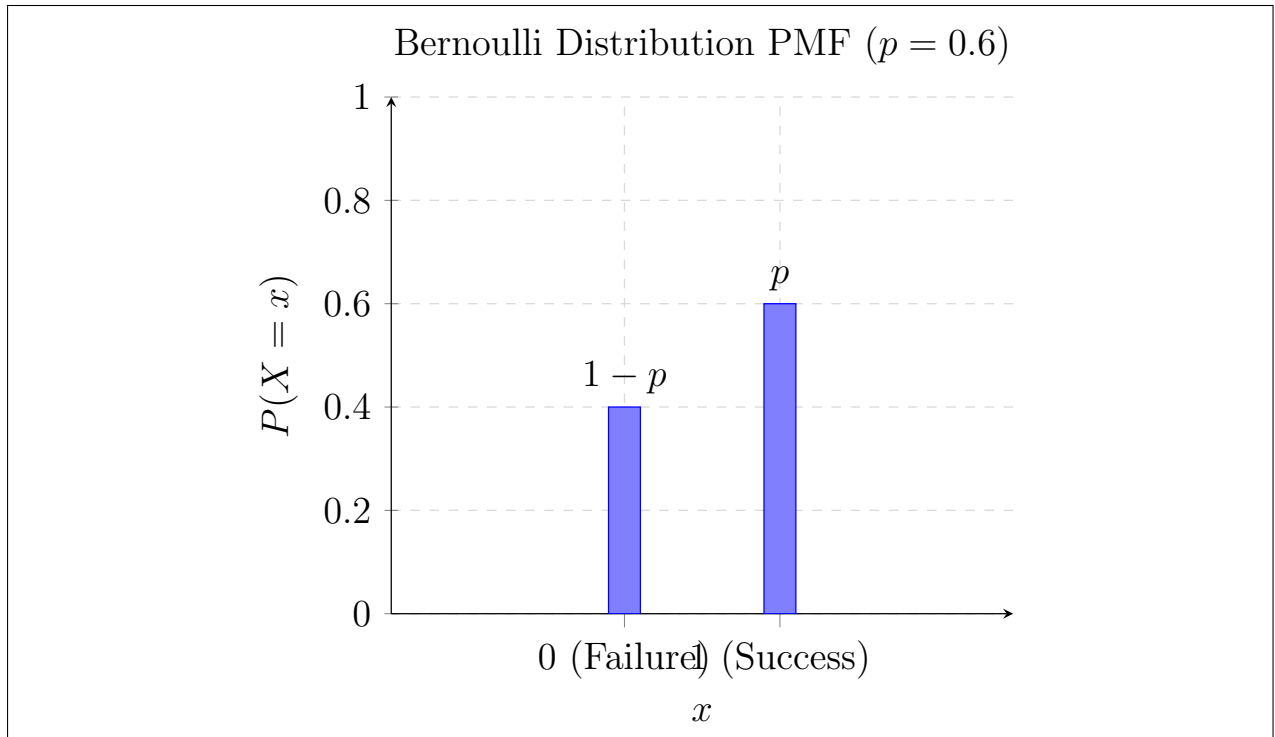


Figure 2.6: Probability Mass Function of a Bernoulli trial where $p = 0.6$.

Example 2.5.2. Consider the experiment of rolling a fair six-sided die. What is the probability of obtaining a score of at least 5?

Solution. The sample space is $S = \{1, 2, 3, 4, 5, 6\}$. We can define a Bernoulli trial by partitioning this set into:

- **Success ($X = 1$):** The outcome is in the set $\{5, 6\}$.
- **Failure ($X = 0$):** The outcome is in the set $\{1, 2, 3, 4\}$.

Since the die is fair, the probability of success is:

$$p = P(X = 1) = \frac{2}{6} = \frac{1}{3}$$

Thus, the probability of rolling a score of at least 5 is $1/3$. □

Theorem 2.5.3. If $X \sim \text{Ber}(p)$, then:

1. **Mean:** $E(X) = p$.

Proof. By the definition of expectation:

$$E(X) = \sum_{x=0}^1 x \cdot f(x; p) = (0)(1-p) + (1)(p) = p.$$

□

2. **Variance:** $\text{Var}(X) = p(1-p)$.

Proof. First, we calculate the second raw moment:

$$E(X^2) = \sum_{x=0}^1 x^2 \cdot f(x; p) = (0^2)(1-p) + (1^2)(p) = p.$$

Using the identity $\text{Var}(X) = E(X^2) - [E(X)]^2$:

$$\text{Var}(X) = p - p^2 = p(1-p).$$

□

Theorem 2.5.4. The generating functions for $X \sim \text{Ber}(p)$ are given by:

1. Probability Generating Function (PGF):

$$G_X(t) = E[t^X] = t^0(1-p) + t^1(p) = (1-p) + pt.$$

2. Moment Generating Function (MGF):

$$M_X(t) = E[e^{tX}] = e^0(1-p) + e^t(p) = (1-p) + pe^t, \quad t \in \mathbb{R}.$$

3. Characteristic Function:

$$\phi_X(t) = E[e^{itX}] = (1-p) + pe^{it}, \quad t \in \mathbb{R}.$$

2.5.2 Discrete Uniform

The Discrete Uniform distribution describes a situation where a finite number of outcomes are all equally likely to occur. A classic example is the roll of a fair N -sided die, where each face has an identical probability of appearing.

Definition 2.5.5. A random variable X is said to follow a Discrete Uniform distribution on the set $\{1, 2, \dots, N\}$ if its probability mass function (PMF) is given by:

$$f_X(x) = P(X = x) = \frac{1}{N}, \quad x = 1, 2, 3, \dots, N$$

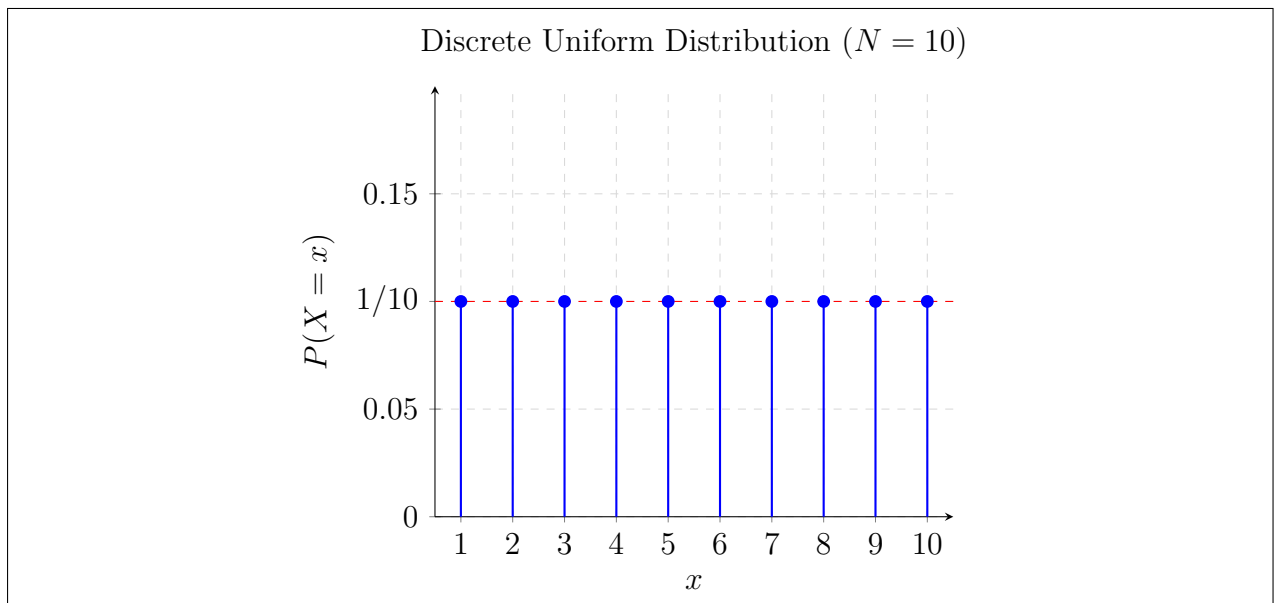


Figure 2.7: PMF of a Discrete Uniform Distribution where $P(X = x) = \frac{1}{10}$ for all x .

Remark 2.5.6. To derive the mean and variance, we utilize the standard power sum formulas:

$$\sum_{i=1}^N i = \frac{N(N+1)}{2} \quad \text{and} \quad \sum_{i=1}^N i^2 = \frac{N(N+1)(2N+1)}{6}.$$

Theorem 2.5.7. Moments of the distribution

1. Mean (Expectation):

$$\begin{aligned}
 E(X) &= \sum_{x=1}^N x \cdot P(X = x) \\
 &= \sum_{x=1}^N x \cdot \frac{1}{N} \\
 &= \frac{1}{N} \sum_{x=1}^N x \\
 &= \frac{1}{N} \left[\frac{N(N+1)}{2} \right] \\
 &= \frac{N+1}{2}.
 \end{aligned}$$

2. Variance: First, we calculate the second raw moment $E(X^2)$:

$$\begin{aligned}
 E(X^2) &= \sum_{x=1}^N x^2 \cdot \frac{1}{N} \\
 &= \frac{1}{N} \left[\frac{N(N+1)(2N+1)}{6} \right] \\
 &= \frac{(N+1)(2N+1)}{6}.
 \end{aligned}$$

Using the identity $\text{Var}(X) = E(X^2) - [E(X)]^2$:

$$\begin{aligned}
 \text{Var}(X) &= \frac{(N+1)(2N+1)}{6} - \left(\frac{N+1}{2} \right)^2 \\
 &= \frac{N+1}{2} \left[\frac{2N+1}{3} - \frac{N+1}{2} \right] \\
 &= \frac{N+1}{2} \left[\frac{4N+2-3N-3}{6} \right] \\
 &= \frac{(N+1)(N-1)}{12} \\
 &= \frac{N^2-1}{12}.
 \end{aligned}$$

Theorem 2.5.8. The transform functions for the Discrete Uniform distribution are based on the sum of a geometric progression.

1. Probability Generating Function (PGF): For $t \neq 1$:

$$G_X(t) = E[t^X] = \sum_{x=1}^N t^x \frac{1}{N} = \frac{t(1 - t^N)}{N(1 - t)}.$$

2. Moment Generating Function (MGF): For $t \neq 0$:

$$M_X(t) = E[e^{tX}] = \sum_{x=1}^N e^{tx} \frac{1}{N} = \frac{e^t(1 - e^{Nt})}{N(1 - e^t)}.$$

3. Characteristic Function:

$$\phi_X(t) = E[e^{itX}] = \frac{e^{it}(1 - e^{iNt})}{N(1 - e^{it})}.$$

Remark 2.5.9. To derive the mean $E(X)$ using the PGF, it is mathematically more convenient to differentiate the series expansion $\sum \frac{1}{N}t^x$ rather than the closed-form quotient. If the quotient form is used, one must apply L'Hôpital's Rule twice (for the second derivative) to resolve the indeterminate forms at $t = 1$.

Proof. 1. Instead of using the "shortcut" formula, it is often easier to differentiate the summation form directly. This avoids the denominator issue entirely:

$$G'_X(t) = \frac{d}{dt} \left[\sum_{x=1}^N \frac{1}{N} t^x \right] = \sum_{x=1}^N \frac{x}{N} t^{x-1}$$

When we evaluate at $t = 1$:

$$E(X) = G'_X(1) = \sum_{x=1}^N \frac{x}{N} (1)^{x-1} = \frac{1}{N} \sum_{x=1}^N x = \frac{N+1}{2}$$

2. The L'Hôpital's Rule Method (Using the Closed Form) If you insist on using the closed-form $G_X(t) = \frac{t(1-t^N)}{N(1-t)}$, you must use limits.

□

2.5.3 The Binomial Distribution

The Binomial distribution models the number of successes in a sequence of n independent trials, each with the same probability of success p . It is essentially the sum of n independent

and identically distributed (i.i.d.) Bernoulli random variables.

Definition 2.5.10. A random variable X follows a Binomial distribution, denoted $X \sim \text{Bin}(n, p)$, if its probability mass function (PMF) is:

$$f(x; n, p) = P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}, \quad x = 0, 1, 2, \dots, n$$

where n is the number of trials and p is the probability of success.

Remark 2.5.11. If $X_1, X_2, \dots, X_n$ are i.i.d. Bernoulli random variables with parameter p , then the random variable $X = \sum_{i=1}^n X_i$ follows a Binomial distribution with parameters n and p .

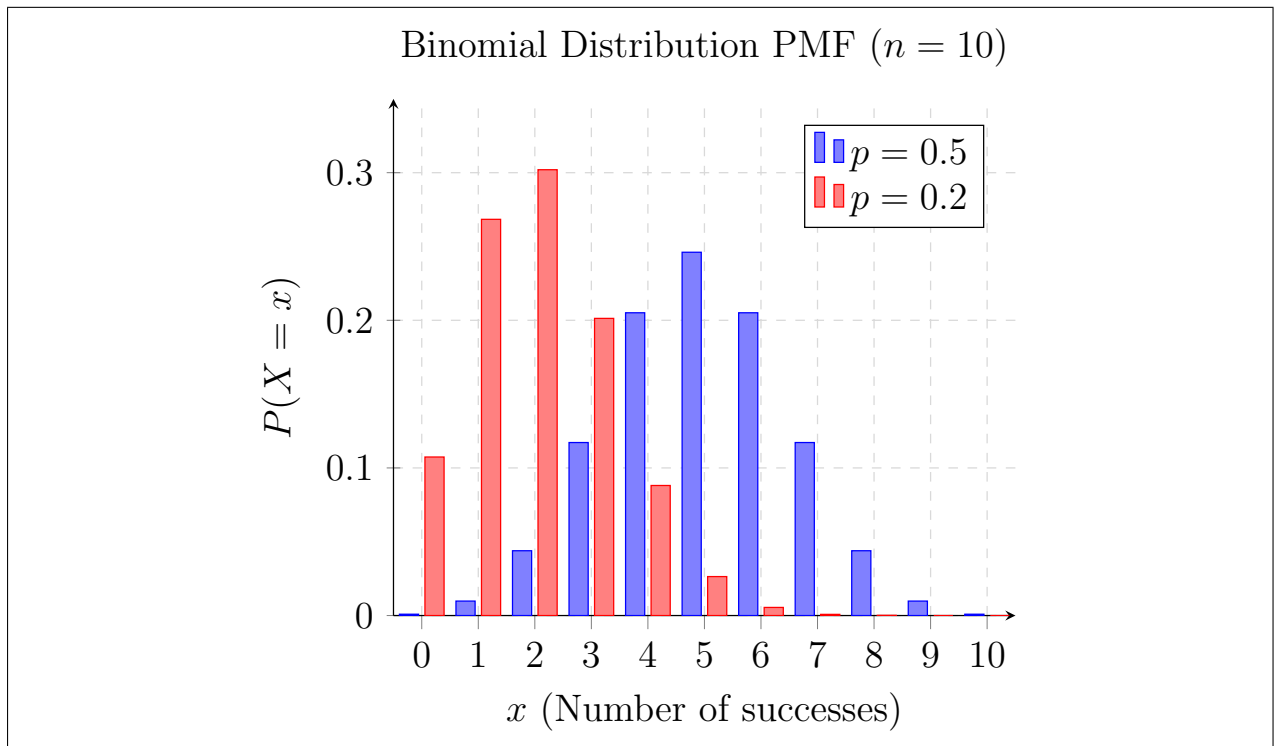


Figure 2.8: PMF of the Binomial distribution for $n = 10$. Note the symmetry when $p = 0.5$ and the right skew when $p = 0.2$.

Example 2.5.12. A textile firm finds that only 20% of applicants are qualified. If 5 people are interviewed, find the probability that at least two are qualified.

Solution. Here $n = 5$ and $p = 0.2$ (so $q = 0.8$). We want $P(X \geq 2)$:

$$\begin{aligned}
 P(X \geq 2) &= 1 - P(X < 2) \\
 &= 1 - [P(X = 0) + P(X = 1)] \\
 &= 1 - \left[\binom{5}{0} (0.2)^0 (0.8)^5 + \binom{5}{1} (0.2)^1 (0.8)^4 \right] \\
 &= 1 - [0.32768 + 0.4096] \\
 &= 0.26272.
 \end{aligned}$$

□

Remark 2.5.13. Some identities

1. $x^{(k)} \binom{n}{x} = n^{(k)} \binom{n-k}{x-k}$
2. $x^{(k)} = x(x-1)(x-2) \cdots (x-k+1)$
3. $\binom{x+k-1}{x} = \binom{x+k-1}{k-1} = (-1)^x \binom{-k}{x}$

Theorem 2.5.14. If $X \sim \text{Bin}(n, p)$, the mean and variance are given by:

1. $E(X) = np$

Proof. Mean

$$\begin{aligned}
 E(X) &= \sum_{x=0}^n x \binom{n}{x} p^x (1-p)^{n-x} \\
 &= \sum_{x=1}^n \frac{x \cdot n!}{x!(n-x)!} p^x (1-p)^{n-x} \\
 &= np \sum_{x=1}^n \frac{(n-1)!}{(x-1)!(n-x)!} p^{x-1} (1-p)^{n-x} \\
 &= np \sum_{y=0}^{n-1} \binom{n-1}{y} p^y (1-p)^{n-1-y} \quad (\text{letting } y = x-1) \\
 &= np(p + (1-p))^{n-1} \\
 &= np(1)^{n-1} \\
 &= np.
 \end{aligned}$$

□

2. $\text{Var}(X) = np(1 - p)$

Proof. To find $\text{Var}(X)$, we first find $E[X(X - 1)]$:

$$\begin{aligned} E[X(X - 1)] &= \sum_{x=2}^n x(x - 1) \binom{n}{x} p^x (1 - p)^{n-x} \\ &= \sum_{x=2}^n x^{(2)} \binom{n}{x} p^x (1 - p)^{n-x} \\ &= n(n - 1)p^2 \sum_{x=2}^n \binom{n-2}{x-2} p^{x-2} (1 - p)^{n-x} \\ &= n(n - 1)p^2. \end{aligned}$$

Since $E(X^2) = E[X(X - 1)] + E(X) = n(n - 1)p^2 + np$:

$$\begin{aligned} \text{Var}(X) &= n(n - 1)p^2 + np - (np)^2 \\ &= n^2p^2 - np^2 + np - n^2p^2 \\ &= np(1 - p). \end{aligned}$$

□

Theorem 2.5.15. The generating functions for $X \sim \text{Bin}(n, p)$ are:

1. Probability Generating Function (PGF):

$$\begin{aligned} G_X(t) &= E[t^X] \\ &= \sum_{x=0}^n t^x \binom{n}{x} p^x (1 - p)^{n-x} \\ &= \sum_{x=0}^n \binom{n}{x} (pt)^x (1 - p)^{n-x} \\ &= (pt + 1 - p)^n. \end{aligned}$$

2. Moment Generating Function (MGF):

$$\begin{aligned}
 M_X(t) &= E[e^{tx}] \\
 &= \sum_{x=0}^n e^{tx} P(X = x) \\
 &= \sum_{x=0}^n e^{tx} \binom{n}{x} p^x (1-p)^{n-x} \\
 &= \sum_{x=0}^n \binom{n}{x} [pe^t]^x (1-p)^{n-x} \\
 &= (pe^t + 1 - p)^n.
 \end{aligned}$$

3. Characteristic Function (CF):

$$\phi_X(t) = E[e^{itX}] = (pe^{it} + 1 - p)^n.$$

2.5.4 The Hypergeometric Distribution

Consider a population of N items, where k items possess a certain characteristic (successes) and $N - k$ do not (failures). If a sample of n items is drawn at random without replacement, let X represent the number of successes in the sample.

Definition 2.5.16. A random variable X follows a Hypergeometric distribution, denoted $X \sim \text{Hyp}(N, n, k)$, if its probability mass function (PMF) is:

$$P(X = x) = \frac{\binom{k}{x} \binom{N-k}{n-x}}{\binom{N}{n}}, \quad x = \max(0, n - (N - k)), \dots, \min(n, k)$$

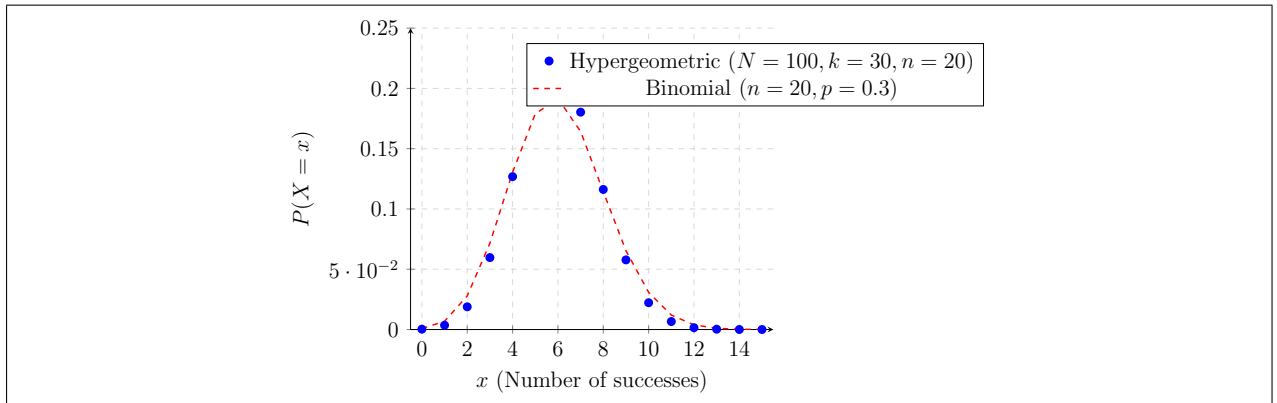


Figure 2.9: The Hypergeometric distribution (dots) compared to the Binomial distribution (dashed line). As $N \rightarrow \infty$, these two distributions become identical.

Example 2.5.17. A shipment contains 100 items, of which k are defective. A sample of size $n = 2$ is selected. The shipment is accepted only if both items are non-defective ($X = 0$).

Solution. The PMF for $N = 100, n = 2$ is

$$P(X = x) = \frac{\binom{k}{x} \binom{100-k}{2-x}}{\binom{100}{2}}.$$

(i) If 10% are defective ($k = 10$):

$$P(X = 0) = \frac{\binom{10}{0} \binom{90}{2}}{\binom{100}{2}} = \frac{1 \cdot 4005}{4950} \approx 0.8091.$$

(ii) If 20% are defective ($k = 20$):

$$P(X = 0) = \frac{\binom{20}{0} \binom{80}{2}}{\binom{100}{2}} = \frac{1 \cdot 3160}{4950} \approx 0.6384.$$

□

Remark 2.5.18. Vandermonde's Identity and Useful Relations:

$$\sum_{x=0}^n \binom{k}{x} \binom{N-k}{n-x} = \binom{N}{n} \quad (\text{Ensures } \sum P(X = x) = 1) \quad (2.3)$$

$$x \binom{k}{x} = k \binom{k-1}{x-1} \quad (2.4)$$

$$n \binom{N}{n} = N \binom{N-1}{n-1} \quad (2.5)$$

Lemma 2.5.19. If $X \sim \text{Hyp}(N, n, k)$, then:

$$\begin{aligned}
E[X^r] &= \sum_{x=0}^n x^r P(X=x) \\
&= \sum_{x=1}^n \frac{x^r \binom{k}{x} \binom{N-k}{k-x}}{\binom{N}{n}} \\
&= n \sum_{x=1}^n \frac{x^{r-1} k \binom{k-1}{x-1} \binom{N-k}{n-x}}{N \binom{N-1}{n-1}} \quad (\text{by equ 2.4 and 2.5}) \\
&= \frac{nk}{N} \sum_{x=1}^n \frac{x^{r-1} \binom{k-1}{x-1} \binom{N-k}{n-x}}{\binom{N-1}{n-1}} \\
&= \frac{nk}{N} \sum_{y=0}^{n-1} \frac{(y+1)^{r-1} \binom{k-1}{y} \binom{N-k}{n-1-y}}{\binom{N-1}{n-1}} \\
&= \frac{nk}{N} E[(Y+1)^{r-1}]
\end{aligned}$$

where $Y \sim \text{Hyp}(N-1, n-1, k-1)$

Theorem 2.5.20. If $X \sim \text{Hyp}(N, n, k)$, then:

$$E(X) = \frac{nk}{N} \quad \text{and} \quad \text{Var}(X) = \frac{nk}{N} \left(\frac{N-k}{N} \right) \left(\frac{N-n}{N-1} \right)$$

Proof. Using the r -th moment relation derived: $E[X^r] = \frac{nk}{N} E[(Y+1)^{r-1}]$, where $Y \sim \text{Hyp}(N-1, n-1, k-1)$.

1. **Mean** ($r=1$):

$$E[X] = \frac{nk}{N} E[1] = \frac{nk}{N}.$$

2. **Variance:** First find $E[X(X-1)]$. By similar logic to the mean:

$$E[X(X-1)] = \frac{nk}{N} \cdot \frac{(n-1)(k-1)}{N-1}.$$

Then, $\text{Var}(X) = E[X(X-1)] + E(X) - [E(X)]^2$:

$$\begin{aligned}
\text{Var}(X) &= \frac{nk(n-1)(k-1)}{N(N-1)} + \frac{nk}{N} - \left(\frac{nk}{N} \right)^2 \\
&= \frac{nk}{N} \left[\frac{(n-1)(k-1)}{N-1} + 1 - \frac{nk}{N} \right] \\
&= \frac{nk(N-k)(N-n)}{N^2(N-1)}.
\end{aligned}$$

□

Remark 2.5.21. The Finite Population Correction: The variance of the Hypergeometric distribution is the same as the Binomial variance (npq , where $q = \frac{N-k}{N}$), multiplied by the factor $\frac{N-n}{N-1}$. As $N \rightarrow \infty$, this factor approaches 1, and the Hypergeometric converges to the Binomial.

Remark 2.5.22. The hypergeometric has no closed form of MGF.

$$\sum_{x=0}^{\min(n,k)} e^{tx} \frac{\binom{k}{x} \binom{N-k}{n-x}}{\binom{N}{n}}.$$

2.5.5 The Geometric Distribution

The Geometric distribution models the number of independent Bernoulli trials required to achieve the first success. Unlike the Binomial distribution, which has a fixed number of trials n , the Geometric distribution has a fixed number of successes (one) and a variable number of trials.

Definition 2.5.23. A random variable X follows a Geometric distribution, denoted $X \sim \text{Geo}(p)$, if its probability mass function (PMF) is given by:

$$f(x; p) = P(X = x) = q^{x-1}p, \quad x = 1, 2, 3, \dots$$

where p is the probability of success and $q = 1 - p$ is the probability of failure.

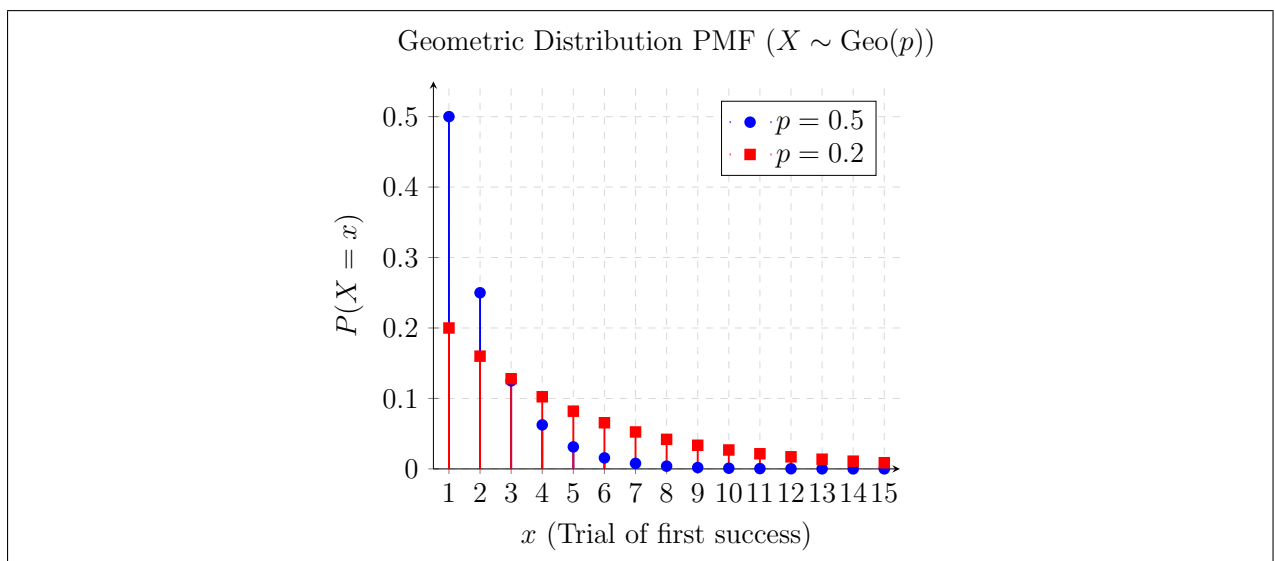


Figure 2.10: PMF of the Geometric distribution. The probability decreases as the number of trials increases, illustrating that the first success is most likely to occur early.

Useful Results: Geometric Series

To derive the moments of this distribution, we rely on the properties of the geometric series for $|t| < 1$:

1. $\sum_{x=0}^{\infty} t^x = \frac{1}{1-t}$
2. $\sum_{x=1}^{\infty} x t^{x-1} = \frac{d}{dt} \left(\sum_{x=0}^{\infty} t^x \right) = \frac{1}{(1-t)^2}$
3. $\sum_{x=2}^{\infty} x(x-1) t^{x-2} = \frac{d^2}{dt^2} \left(\sum_{x=0}^{\infty} t^x \right) = \frac{2}{(1-t)^3}$

Example 2.5.24. Show that the Geometric distribution is indeed a PMF

Proof.

$$\begin{aligned}
 \sum_{x=1}^{\infty} P(X = x) &= \sum_{x=1}^{\infty} p(1-p)^{x-1} \\
 &= p \sum_{x=1}^{\infty} (1-p)^{x-1} \\
 &= p \sum_{k=0}^{\infty} (1-p)^k \quad (\text{letting } k = x-1) \\
 &= p \frac{1}{1 - (1-p)} = \frac{p}{p} \\
 &= 1
 \end{aligned}$$

□

Theorem 2.5.25 (Moments of the Geometric Distribution). If $X \sim \text{Geo}(p)$, then:

1. Mean:

$$E(X) = \frac{1}{p}$$

2. Variance:

$$\text{Var}(X) = \frac{1-p}{p^2}$$

Proof. 1. Mean: Using the second series result above:

$$E(X) = \sum_{x=1}^{\infty} x \cdot p q^{x-1} = p \sum_{x=1}^{\infty} x q^{x-1} = p \left(\frac{1}{(1-q)^2} \right) = p \left(\frac{1}{p^2} \right) = \frac{1}{p}.$$

2. Variance: First, we find the second factorial moment using the third series result:

$$\begin{aligned}
 E[X(X-1)] &= \sum_{x=1}^{\infty} x(x-1)pq^{x-1} \\
 &= pq \sum_{x=2}^{\infty} x(x-1)q^{x-2} \\
 &= pq \left(\frac{2}{(1-q)^3} \right) \\
 &= \frac{2pq}{p^3} \\
 &= \frac{2q}{p^2}.
 \end{aligned}$$

Using the identity $\text{Var}(X) = E[X(X-1)] + E(X) - [E(X)]^2$:

$$\text{Var}(X) = \frac{2q}{p^2} + \frac{1}{p} - \frac{1}{p^2} = \frac{2q + p - 1}{p^2} = \frac{2(1-p) + p - 1}{p^2} = \frac{1-p}{p^2}.$$

□

Theorem 2.5.26. Transform Functions:

1. Probability Generating Function (PGF):

$$G_X(t) = E[t^X] = \sum_{x=1}^{\infty} t^x pq^{x-1} = pt \sum_{x=1}^{\infty} (qt)^{x-1} = \frac{pt}{1-qt}, \quad |t| < \frac{1}{q}.$$

2. Moment Generating Function (MGF):

$$M_X(t) = E[e^{tX}] = \frac{pe^t}{1-qe^t}, \quad t < -\ln(q).$$

Theorem 2.5.27 (Memoryless Property). The Geometric distribution is the only discrete distribution with the memoryless property:

$$P(X > i + j \mid X > i) = P(X > j)$$

This means that the probability of needing j more trials to get a success does not depend on how many failed trials (i) have already occurred.

Example 2.5.28. In a large production batch, 5% of items are defective. A quality control inspector tests items one by one until the first defective item is found.

- (i) Find the probability that the first defective item is the 10th item tested.
- (ii) Find the expected number of items the inspector must test.

Solution. Let X be the number of items tested until the first defective. Here, $p = 0.05$ and $q = 0.95$.

(i) $P(X = 10) = q^{10-1}p = (0.95)^9(0.05) \approx 0.0315$.

(ii) $E(X) = \frac{1}{p} = \frac{1}{0.05} = 20$ items.

□

2.5.6 The Negative Binomial Distribution

The Negative Binomial distribution models the number of independent Bernoulli trials required to achieve a fixed number of successes, denoted by r . It is often referred to as the Pascal distribution or the waiting time distribution.

Definition 2.5.29. A random variable X follows a Negative Binomial distribution, denoted $X \sim \text{NB}(r, p)$, if its probability mass function (PMF) is:

$$f_X(x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}, \quad x = r, r+1, r+2, \dots$$

where p is the probability of success in each trial.

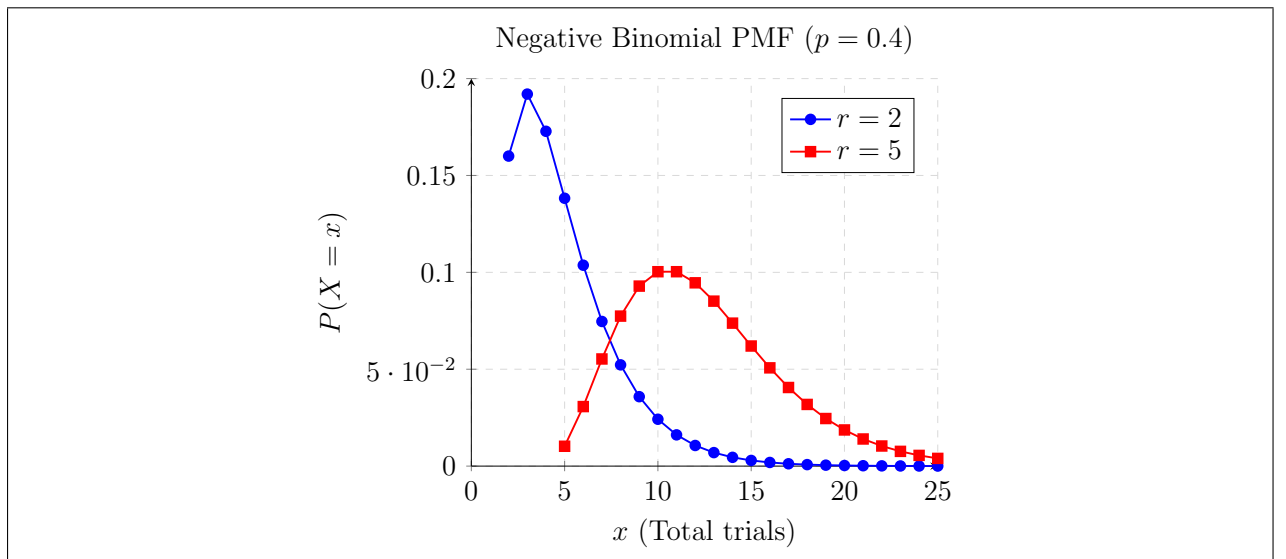


Figure 2.11: Negative Binomial Distribution for $r = 2$ and $r = 5$. Note how the distribution starts at $x = r$.

The Two Versions of the Distribution

In probability theory, there are two common ways to define this random variable:

1. **Number of trials (X):** The total number of trials required to get r successes ($x \in \{r, r+1, \dots\}$).
2. **Number of failures (Y):** The number of failures encountered before the r^{th} success occurs ($y \in \{0, 1, \dots\}$).

The relationship is simply $Y = X - r$. The PMF for the number of failures Y is:

$$f_Y(y) = \binom{y+r-1}{r-1} p^r (1-p)^y, \quad y = 0, 1, 2, \dots$$

Theorem 2.5.30. If $X \sim \text{NB}(r, p)$, then:

1. Mean:

$$E(X) = \frac{r}{p}$$

2. Variance:

$$\text{Var}(X) = \frac{r(1-p)}{p^2}$$

Proof. Recall that X can be viewed as the sum of r independent Geometric random variables, $X = X_1 + X_2 + \dots + X_r$, where each $X_i \sim \text{Geo}(p)$. Using the linearity of expectation:

$$E(X) = E\left(\sum_{i=1}^r X_i\right) = \sum_{i=1}^r E(X_i) = \sum_{i=1}^r \frac{1}{p} = \frac{r}{p}.$$

Since the trials are independent:

$$\text{Var}(X) = \text{Var}\left(\sum_{i=1}^r X_i\right) = \sum_{i=1}^r \text{Var}(X_i) = \sum_{i=1}^r \frac{1-p}{p^2} = \frac{r(1-p)}{p^2}.$$

□

Theorem 2.5.31 (Generating Functions). For the version Y (number of failures):

1. Probability Generating Function (PGF):

$$G_Y(t) = \left(\frac{p}{1-qt}\right)^r$$

2. Moment Generating Function (MGF):

$$M_Y(t) = \left(\frac{p}{1 - qe^t} \right)^r, \quad t < -\ln(q)$$

Example 2.5.32. A scientist inoculates mice one at a time until 3 contract a disease. If the probability of contracting the disease is $1/6$, what is the probability that exactly 8 mice are required?

Solution. Here $r = 3$ (target successes) and $p = 1/6$ (success probability). We want the probability that the 3^{rd} success occurs on the 8^{th} trial ($x = 8$).

$$P(X = 8) = \binom{8-1}{3-1} \left(\frac{1}{6} \right)^3 \left(\frac{5}{6} \right)^{8-3} = \binom{7}{2} \left(\frac{1}{216} \right) \left(\frac{3125}{7776} \right)$$

$$P(X = 8) = 21 \cdot \frac{3125}{1679616} \approx 0.0391.$$

□

2.5.7 The Poisson Distribution

The Poisson distribution is a discrete probability distribution that models the number of independent events occurring within a fixed interval of time or space. It is uniquely characterized by a single parameter, λ (lambda), which represents the constant average rate of occurrence.

Typical applications include:

- The number of calls received by a telecommunications switchboard per minute.
- The number of radioactive particles decayed from a source within a second.
- The number of typographical errors found in a single page of a manuscript.

Definition 2.5.33. A discrete random variable X is said to follow a *Poisson distribution* with parameter $\lambda > 0$, denoted as $X \sim \text{Po}(\lambda)$, if its probability mass function (PMF) is:

$$f(x; \lambda) = P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots$$

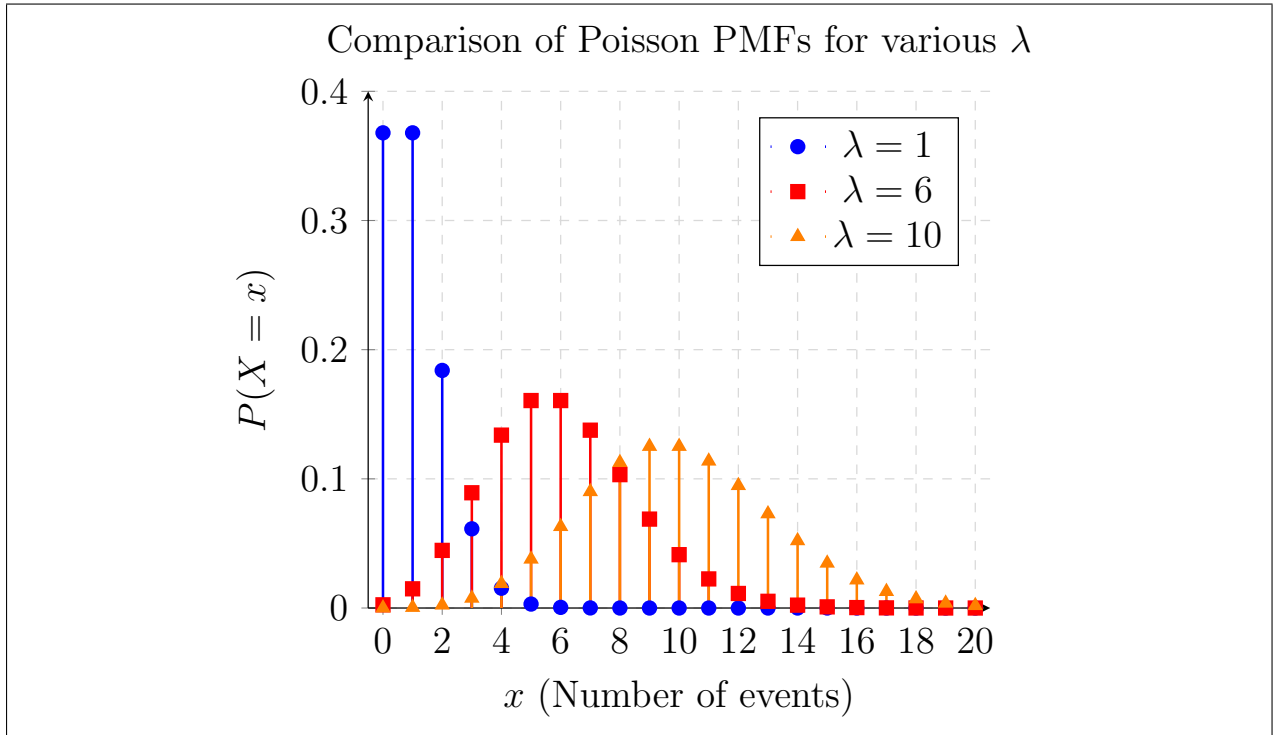


Figure 2.12: Probability Mass Function of the Poisson Distribution showing the effect of the rate parameter λ .

Example 2.5.34. Verify that $f(x; \lambda)$ is a valid probability mass function.

Proof. A valid PMF must satisfy $\sum_x f(x) = 1$. Using the Taylor series expansion for the exponential function $e^\lambda = \sum_{x=0}^{\infty} \frac{\lambda^x}{x!}$:

$$\begin{aligned}
 \sum_{x=0}^{\infty} f_X(x) &= \sum_{x=0}^{\infty} \frac{e^{-\lambda} \lambda^x}{x!} \\
 &= e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} \\
 &= e^{-\lambda} e^{\lambda} = 1.
 \end{aligned}$$

□

Theorem 2.5.35. If $X \sim \text{PO}(\lambda)$, then

1. mean is $E(X) = \lambda$.

Proof. Using the definition of expectations

$$\begin{aligned}
 E(X) &= \sum_{x=0}^{\infty} x, \frac{e^{-\lambda} \lambda^x}{x!} \\
 &= e^{-\lambda} \sum_{x=1}^{\infty} \frac{x \lambda^x}{x(x-1)!} \\
 &= \lambda e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!} \\
 &= \lambda e^{-\lambda} \sum_{y=0}^{\infty} \frac{\lambda^y}{y!}, \quad \text{letting } y = x - 1 \\
 &= \lambda e^{-\lambda} e^{\lambda} \\
 &= \lambda.
 \end{aligned}$$

□

2. variance is $\text{Var}(X) = \lambda$.

Proof. We first compute $E(X^2)$:

$$\begin{aligned}
 E(X^2) &= \sum_{x=0}^{\infty} \frac{x^2 e^{-\lambda} \lambda^x}{x!} \\
 &= \lambda e^{-\lambda} \sum_{x=1}^{\infty} \frac{x \cdot x \lambda^{x-1}}{x(x-1)!} \\
 &= \lambda e^{-\lambda} \sum_{x=1}^{\infty} \frac{x \lambda^{x-1}}{(x-1)!} \\
 &= \lambda e^{-\lambda} \sum_{y=0}^{\infty} \frac{(y+1) \lambda^y}{y!}, \quad \text{letting } y = x - 1 \\
 &= \lambda \left[\sum_{y=0}^{\infty} \frac{y e^{-\lambda} \lambda^y}{y!} + \sum_{y=0}^{\infty} \frac{e^{-\lambda} \lambda^y}{y!} \right] \\
 &= \lambda(\lambda + 1).
 \end{aligned}$$

Therefore, the variance

$$\begin{aligned}
 \text{Var}(X) &= E(X^2) - [E(X)]^2 \\
 &= \lambda^2 + \lambda - [\lambda]^2 \\
 &= \lambda.
 \end{aligned}$$

□

Theorem 2.5.36. If $X \sim \text{POI}(\lambda)$, then

1. Probability Generating Function (PGF):

$$\begin{aligned} G_X(t) &= E[t^X] = \sum_{x=0}^{\infty} t^x \frac{e^{-\lambda} \lambda^x}{x!} \\ &= e^{-\lambda} \sum_{x=0}^{\infty} \frac{(\lambda t)^x}{x!} \\ &= e^{-\lambda} e^{\lambda t} \\ &= e^{\lambda(t-1)}. \end{aligned}$$

2. Moment Generating Function (MGF):

$$\begin{aligned} M_X(t) &= E[e^{tX}] = \sum_{x=0}^{\infty} e^{tx} \frac{e^{-\lambda} \lambda^x}{x!} \\ &= e^{-\lambda} \sum_{x=0}^{\infty} \frac{(\lambda e^t)^x}{x!} \\ &= e^{-\lambda} e^{\lambda e^t} \\ &= e^{\lambda(e^t-1)}. \end{aligned}$$

3. Characteristic Function:

$$\phi_X(t) = E[e^{itX}] = e^{\lambda(e^{it}-1)}.$$

Remark 2.5.37 (The Poisson Process). A Poisson experiment satisfies the following properties:

- Independence:** The number of outcomes in one time interval is independent of the number in any other disjoint interval.
- Proportionality:** The probability of a single outcome occurring in a very short interval is proportional to the length of the interval.
- Negligibility:** The probability of more than one outcome occurring in a sufficiently short interval is negligible.

Example 2.5.38. The average number of calls coming into a switchboard is 4 per minute. Assuming a Poisson distribution, find the probability that:

- (i) Exactly 5 calls are received.
- (ii) More than one call is received.

Solution. Here, $\lambda = 4$.

(i) $P(X = 5) = \frac{4^5 e^{-4}}{5!} \approx 0.1563$.

(ii) $P(X > 1) = 1 - P(X \leq 1) = 1 - [P(0) + P(1)]$.

$$\begin{aligned} P(X > 1) &= 1 - \left[\frac{4^0 e^{-4}}{0!} + \frac{4^1 e^{-4}}{1!} \right] \\ &= 1 - [e^{-4} + 4e^{-4}] \\ &= 1 - 5e^{-4} \\ &\approx 1 - 0.09158 = 0.9084. \end{aligned}$$

□

Theorem 2.5.39 (Poisson Approximation to the Binomial). If n is large ($n \rightarrow \infty$) and p is small ($p \rightarrow 0$), the Binomial distribution $B(n, p)$ can be approximated by a Poisson distribution with $\lambda = np$.

Example 2.5.40. Suppose that on average one person in every 1, 000 has an eye problem. Find the probability that a random sample of 8, 000 people will have fewer than 7 people with eye problem.

Solution. Given

X = number with eye problem

p = probabilities of having an eye problem

$$p = 0.001$$

n = number of independent trials

$$n = 8000$$

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

$$P(X < 7) = \sum_{x=0}^6 \binom{8000}{x} (0.001)^x (0.999)^{8000-x}.$$

Which is tedious to complete. We use Poisson approximation to Binomial

$$\lambda = np = 8000 \times 0.001 = 8.$$

$$\begin{aligned} P(X < 7) &= P(X \leq 6) \\ &= \sum_{x=0}^6 \frac{8^x e^{-8}}{x!} \\ &= e^{-8} \left(\frac{8^0}{0!} + \frac{8}{1!} + \frac{8^2}{2!} + \frac{8^3}{3!} + \frac{8^4}{4!} + \frac{8^5}{5!} + \frac{8^6}{6!} \right) \\ &= 0.3134 \end{aligned}$$

□

2.5.8 The Zeta (or Zipf) Distribution

The Zeta distribution, frequently referred to as the Zipf distribution, is a power-law distribution typically utilized to model the frequency of occurrences of data in diverse fields such as linguistics, economics, and social sciences. It effectively accounts for "long-tail" phenomena where a small number of members in a population account for a high proportion of the total frequency (popularity), while the majority remain in relative obscurity. Notable applications include the distribution of family incomes, the frequency of words in a language, and the population sizes of cities.

Definition 2.5.41. A discrete random variable X is said to have a Zeta distribution with parameter α , denoted $X \sim \text{Zeta}(\alpha)$, if its probability mass function is given by:

$$f(x; \alpha) = \frac{x^{-\alpha}}{\zeta(\alpha)}, \quad x = 1, 2, 3, \dots$$

for $\alpha > 1$, where $\zeta(\alpha)$ is the **Riemann Zeta Function** defined by the infinite series:

$$\zeta(\alpha) = \sum_{n=1}^{\infty} \frac{1}{n^{\alpha}}.$$

Note 2.5.42. Observe that the support of X starts at $x = 1$. If the summation in the denominator were to start at $x = 0$, the term $0^{-\alpha}$ would be undefined for $\alpha > 1$.

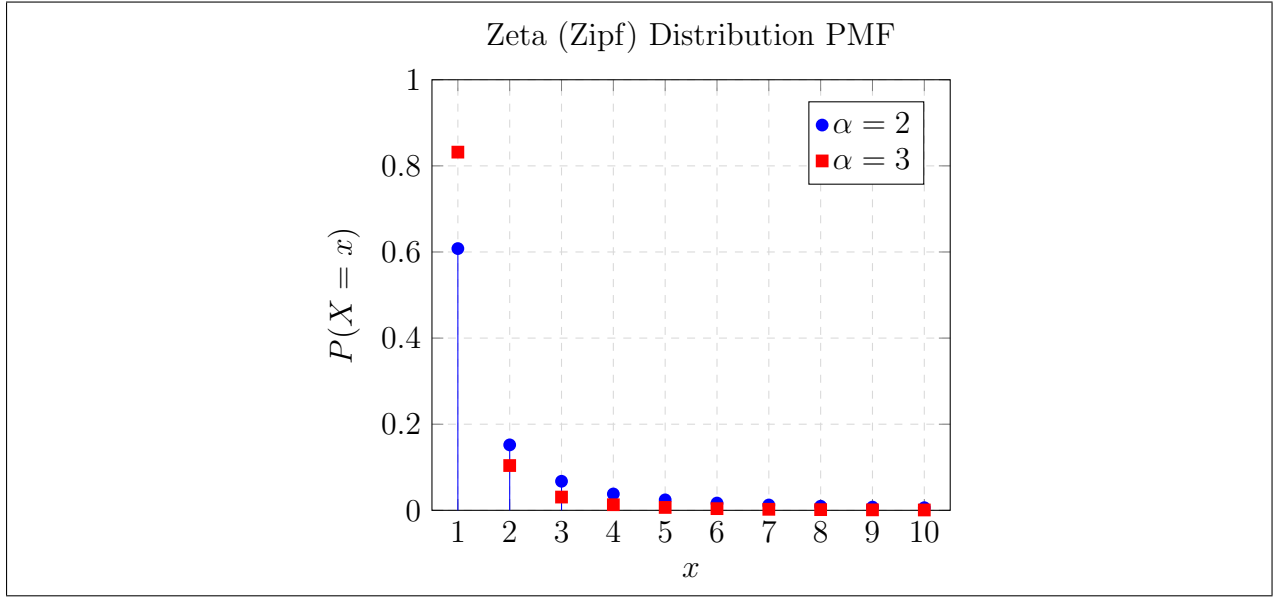


Figure 2.13: Probability Mass Function of the Zeta Distribution for $\alpha = 2$ and $\alpha = 3$.

Theorem 2.5.43. If $X \sim \text{Zeta}(\alpha)$, the moments of the distribution are defined as follows:

1. Mean

$$E(X) = \frac{\zeta(\alpha - 1)}{\zeta(\alpha)},$$

provided $\alpha > 2$.

Proof. By the definition of expectation

$$\begin{aligned} E(X) &= \sum_{x=1}^{\infty} x \cdot f(x; \alpha) \\ &= \frac{1}{\zeta(\alpha)} \sum_{x=1}^{\infty} x \cdot \frac{1}{x^{\alpha}} \\ &= \frac{1}{\zeta(\alpha)} \sum_{x=1}^{\infty} \frac{1}{x^{\alpha-1}} \\ &= \frac{\zeta(\alpha - 1)}{\zeta(\alpha)}. \end{aligned}$$

For the series $\zeta(\alpha - 1)$ to converge, we require $\alpha - 1 > 1$, thus $\alpha > 2$. □

2. Variance

$$\text{Var}(X) = \frac{\zeta(\alpha)\zeta(\alpha - 2) - \zeta(\alpha - 1)^2}{\zeta(\alpha)^2},$$

provided $\alpha > 3$.

Proof. First, we determine the second raw moment:

$$E(X^2) = \sum_{x=1}^{\infty} x^2 \frac{1}{x^\alpha \zeta(\alpha)} = \frac{1}{\zeta(\alpha)} \sum_{x=1}^{\infty} \frac{1}{x^{\alpha-2}} = \frac{\zeta(\alpha-2)}{\zeta(\alpha)}.$$

Applying the variance formula $\text{Var}(X) = E(X^2) - [E(X)]^2$:

$$\begin{aligned} \text{Var}(X) &= \frac{\zeta(\alpha-2)}{\zeta(\alpha)} - \left(\frac{\zeta(\alpha-1)}{\zeta(\alpha)} \right)^2 \\ &= \frac{\zeta(\alpha)\zeta(\alpha-2) - \zeta(\alpha-1)^2}{\zeta(\alpha)^2}. \end{aligned}$$

The term $\zeta(\alpha-2)$ converges only if $\alpha-2 > 1$, hence $\alpha > 3$. □

Theorem 2.5.44. The transform functions for $X \sim \text{Zeta}(\alpha)$ are characterized by the following series:

1. Probability Generating Function:

$$G_X(t) = E[t^X] = \frac{1}{\zeta(\alpha)} \sum_{x=1}^{\infty} \frac{t^x}{x^\alpha}, \quad |t| \leq 1.$$

2. Moment Generating Function:

$$M_X(t) = E[e^{tX}] = \frac{1}{\zeta(\alpha)} \sum_{x=1}^{\infty} \frac{e^{tx}}{x^\alpha}, \quad t \leq 0.$$

3. Characteristic Function:

$$\phi_X(t) = E[e^{itX}] = \frac{1}{\zeta(\alpha)} \sum_{x=1}^{\infty} \frac{e^{itx}}{x^\alpha}.$$

2.6. Some Important Continuous Probability Density Functions

In this section, we describe models frequently used in continuous random phenomena. These distributions are the building blocks of statistical inference and engineering applications.

2.6.1 The Uniform Distribution

The uniform distribution provides a probability model for selecting points at random from an $[\alpha, \beta]$. An important application of the uniform distribution lies in random number generation.

Definition 2.6.1 (Uniform Distribution $X \sim \text{UNIF}(\alpha, \beta)$). A random variable X is said to be uniform on the interval $[\alpha, \beta]$ if its p.d.f. is of the form

$$f_X(x) = \begin{cases} \frac{1}{\beta - \alpha}, & \alpha \leq x \leq \beta \\ 0, & \text{otherwise} \end{cases}$$

Diagrams of the Uniform Distribution.

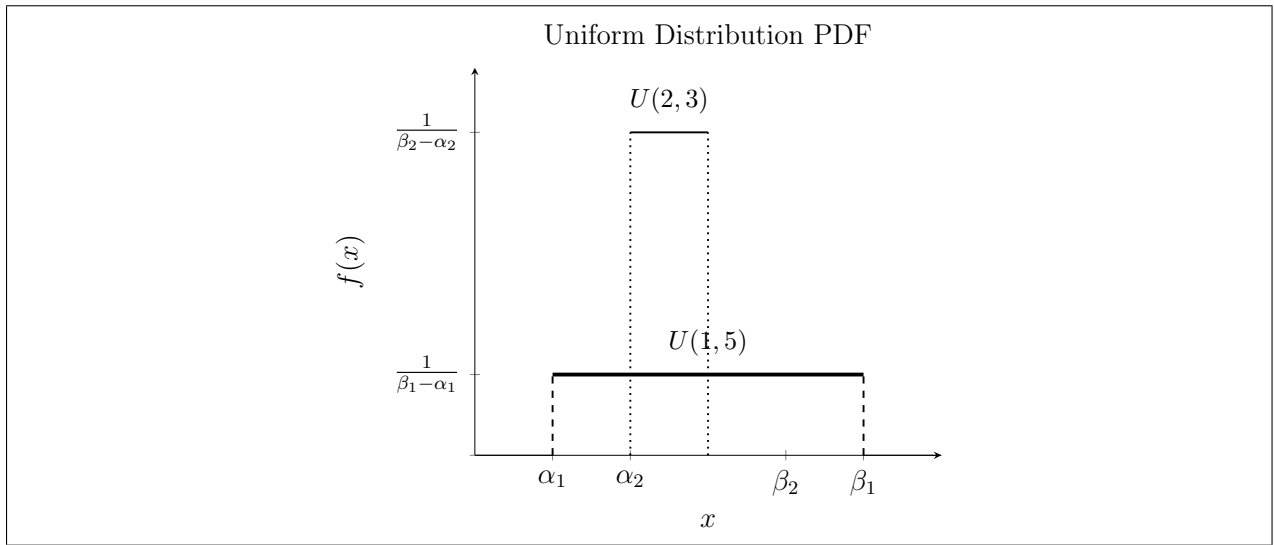


Figure 2.14: The PDF of the Uniform distribution. Note that the total area under each rectangle is exactly 1.

The mean of X is

$$\begin{aligned} E(X) &= \int_{\alpha}^{\beta} x \frac{1}{\beta - \alpha} dx \\ &= \frac{1}{\beta - \alpha} \int_{\alpha}^{\beta} x dx \\ &= \frac{\alpha^2}{2(\beta - \alpha)} \Big|_{\alpha}^{\beta} \\ &= \frac{\beta^2 - \alpha^2}{2(\beta - \alpha)} \\ &= \frac{(\beta + \alpha)(\beta - \alpha)}{2(\beta - \alpha)} \\ &= \frac{\beta + \alpha}{2}. \end{aligned}$$

Also

$$\begin{aligned}
 E(X^2) &= \frac{1}{\beta - \alpha} \int_{\alpha}^{\beta} x^2 dx \\
 &= \frac{1}{\beta - \alpha} \left[\frac{x^3}{3} \right]_{\alpha}^{\beta} \\
 &= \frac{1}{\beta - \alpha} \frac{\beta^3 - \alpha^3}{3} \\
 &= \frac{(\beta - \alpha)(\beta^2 + \beta\alpha + \alpha^2)}{3(\beta - \alpha)} \\
 &= \frac{\beta^2 + \beta\alpha + \alpha^2}{3}.
 \end{aligned}$$

Hence, variance is

$$\begin{aligned}
 \text{Var}(X) &= \frac{\beta^2 + \beta\alpha + \alpha^2}{3} - \frac{(\beta + \alpha)^2}{4} \\
 &= \frac{\beta^2 - 2\beta\alpha + \alpha^2}{12} \\
 &= \frac{(\beta - \alpha)^2}{12}.
 \end{aligned}$$

The m.g.f of X is

$$\begin{aligned}
 M_X(t) &= E(e^{tx}) \\
 &= \int_{\alpha}^{\beta} e^{tx} dx \\
 &= \frac{1}{\beta - \alpha} \int_{\alpha}^{\beta} e^{tx} dx \\
 &= \frac{1}{\beta - \alpha} \left. \frac{e^{tx}}{t} \right|_{\alpha}^{\beta} \\
 &= \frac{e^{\beta t} - e^{\alpha t}}{t(\beta - \alpha)}.
 \end{aligned}$$

Thus

$$M_X(t) = \begin{cases} \frac{e^{\beta t} - e^{\alpha t}}{t(\beta - \alpha)}, & \text{if } t \neq 0 \\ 1, & \text{if } t = 0. \end{cases}$$

Remark 2.6.2. Foundation of Simulation: In computer science and computational statistics, the "Random" function in almost all programming languages generates a $U(0, 1)$ value. Other distributions (like Normal or Exponential) are then generated by transforming these

uniform values using the Inverse Transform Method.

2.6.2 Exponential Distribution

The exponential is the distribution of the amount of time until some specific event occurs (i.e waiting time). For instance, how long one waits before a new component "dies"?

Definition 2.6.3. A continuous random variable X whose p.d.f is given, for some $\lambda > 0$, by

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x > 0, \quad \lambda > 0 \\ 0, & \text{otherwise} \end{cases}$$

is said to be an exponential random variable.

Remark 2.6.4. An exponential has the mean and variable

$$E(X) = \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2}.$$

Note 2.6.5. The p.d.f of an exponential random variable can also be presented in this form

$$f_X(x) = \begin{cases} \frac{1}{\lambda} e^{-\frac{x}{\lambda}}, & x > 0, \quad \lambda > 0 \\ 0, & \text{otherwise} \end{cases}$$

where the mean and variance are

$$E(X) = \lambda, \quad \text{Var}(X) = \lambda^2.$$

Graphs of an Exponential

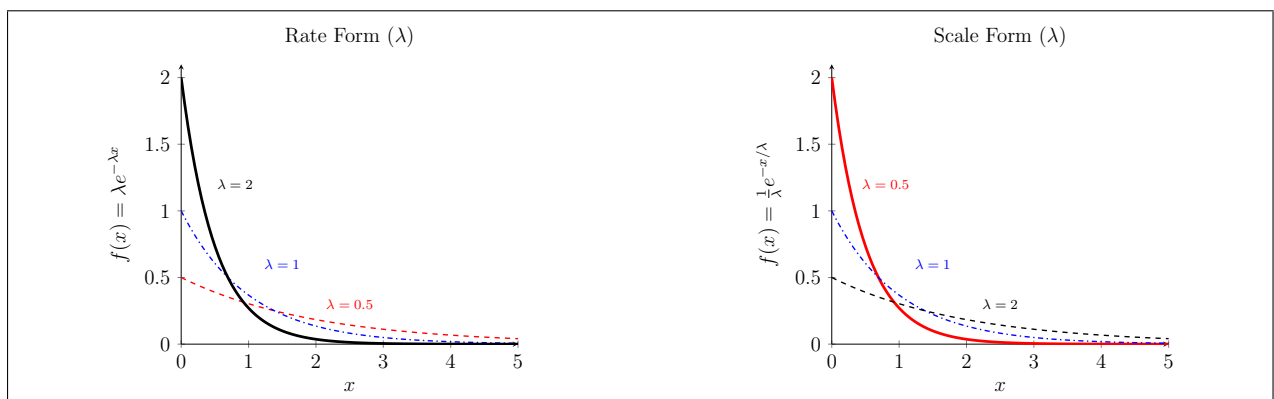


Figure 2.15: Comparison of Exponential Distribution parameterizations. Note how increasing λ makes the Rate Form (left) "taller," but makes the Scale Form (right) "flatter."

Note 2.6.6 (Important).

1. *Rate Parameterization* (λ): Used most commonly in Poisson processes. Here, $E(X) = \frac{1}{\lambda}$. As λ increases, the mean decreases, and the curve becomes steep.
2. *Scale Parameterization* (θ or λ): Often used in engineering and reliability (Mean time to failure). Here, $E(X) = \lambda$. As λ increases, the mean increases, the curve spreads out further along the x -axis.

The cumulative distribution function is

$$\begin{aligned} F_X(x) &= P[X \leq x] \\ &= \int_0^x \lambda e^{-\lambda x} dx \\ &= -e^{-\lambda x} \Big|_0^x \\ &= 1 - e^{-\lambda x}, \quad x \geq 0. \end{aligned}$$

The moment generating function is

$$\begin{aligned} M_X(t) &= E[e^{tX}] \\ &= \int_0^\infty e^{tx} \lambda e^{-\lambda x} dx \\ &= \lambda \int_0^\infty e^{-(\lambda-t)x} dx \\ &= \frac{\lambda}{\lambda - t}, \quad t < \lambda. \end{aligned}$$

Then differentiating $M'_X(t)$ we get

$$\begin{aligned} M'_X(t) &= \frac{\lambda(-1)(-1)}{(\lambda - t)^2} = \frac{\lambda}{(\lambda - t)^2} \\ M''_X(t) &= \frac{\lambda(-1)(-2)}{(\lambda - t)^3} = \frac{2\lambda}{(\lambda - t)^3} \end{aligned}$$

Then the mean is

$$E(X) = M'_X(0) = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda}$$

and

$$E(X^2) = M''_X(0) = \frac{2\lambda}{\lambda^3} = \frac{2}{\lambda^2}$$

then variance

$$\text{Var}(X) = \frac{2}{\lambda^2} - \left(\frac{1}{\lambda^2}\right) = \frac{1}{\lambda^2}.$$

Example 2.6.7. A manufacture of microwave ovens is trying to determine the length of warranty period it should attach to its magneto tube, the most critical component in the microwave oven. Preliminary testing has shown that the life of the tube has an exponential distribution with mean 6.25 years.

- (i). What is the warranty period if the manufacturer wants to replace only 10% of the magneto tube.

Solution. Let the warranty period be α . $X \sim \text{Exp}(\lambda)$, $E(X) = 6.25$

$$f_X(x) = \frac{1}{6.25} e^{-\frac{x}{6.25}}$$

$$P(X \leq \alpha) = 0.1$$

$$\int_0^\alpha \frac{1}{6.25} e^{-\frac{x}{6.25}} dx = 0.1$$

$$\left. -e^{-\frac{x}{6.25}} \right|_0^\alpha = 0.1$$

$$1 - e^{-\frac{\alpha}{6.25}} = 0.1$$

$$e^{-\frac{\alpha}{6.25}} = 0.90 \quad \implies \quad \frac{-\alpha}{6.25} = \ln(0.90)$$

$$\alpha = -6.25 \ln(0.90).$$

Therefore, $\alpha = 0.6585$ years ≈ 7.9 months ≈ 8.0 months. □

- (ii). If the warranty is 4 years, what percentage of the tubes will be replaced?

Solution.

$$\begin{aligned} P(X \leq 4) &= \int_0^4 \frac{1}{6.25} e^{-\frac{x}{6.25}} dx \\ &= \left. e^{-\frac{x}{6.25}} \right|_0^4 \\ &= 1 - e^{-\frac{4}{6.25}} \\ &= 0.4727 = 47.27\% \end{aligned}$$

□

2.6.3 Gamma Distribution

Definition 2.6.8 (Gamma Function). The gamma function, denoted by $\Gamma(\alpha)$ for all $\alpha > 0$, is given by

$$\Gamma(\alpha) = \int_0^{\infty} t^{\alpha-1} e^{-t} dt.$$

Properties of Γ

- (i). $\Gamma(\alpha + 1) = \alpha \Gamma(\alpha)$
- (ii). $\Gamma(n) = (n - 1)!, n = 1, 2, \dots$
- (iii). $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

Definition 2.6.9 (Gamma Distribution $X \sim \text{GAM}(\alpha, \lambda)$). A gamma random variable X with parameters $\alpha > 0$, $\lambda > 0$, has probability density function

$$f_X(x) = \frac{x^{\alpha-1} e^{-\frac{x}{\lambda}}}{\lambda^{\alpha} \Gamma(\alpha)} \quad \text{or} \quad f_X(x) = \frac{\lambda^{\alpha} x^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)}, \quad x > 0$$

Note 2.6.10. When $\alpha = 1$, then the gamma reduces to the exponential distribution.

$$f_X(x) = \frac{x^{1-1} e^{-\frac{x}{\lambda}}}{\lambda^1 \Gamma(1)} = \frac{1}{\lambda} e^{-\frac{x}{\lambda}}, \quad x > 0, \quad \lambda > 0$$

$$f_X(x) = \frac{\lambda^1 x^{1-1} e^{-\lambda x}}{\Gamma(1)} = \lambda e^{-\lambda x}, \quad x > 0, \quad \lambda > 0$$

The moment generating function of X is

$$\begin{aligned} M_X(t) &= E[e^{tx}] = \int_0^{\infty} e^{tx} \frac{\lambda^{\alpha} x^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)} dx \\ &= \frac{\lambda^{\alpha}}{\Gamma(\alpha)} \int_0^{\infty} x^{\alpha-1} e^{-(\lambda-t)x} dx \\ &= \frac{\lambda^{\alpha}}{\Gamma(\alpha)} \int_0^{\infty} \left(\frac{y}{\lambda-t}\right)^{\alpha-1} e^{-y} \frac{dy}{\lambda-t}, \quad \text{letting } y = (\lambda-t)x \\ &= \frac{\lambda^{\alpha}}{\Gamma(\alpha)} \frac{1}{(\lambda-t)^{\alpha-1}} \frac{1}{(\lambda-t)} \int_0^{\infty} y^{\alpha-1} e^{-y} dy \\ &= \frac{\lambda^{\alpha}}{\Gamma(\alpha)} \left(\frac{1}{\lambda-t}\right)^{\alpha} \Gamma(\alpha) \\ &= \left(\frac{\lambda}{\lambda-t}\right)^{\alpha}. \end{aligned}$$

Differentiating $M_X(t)$

$$M'_X(t) = \lambda^\alpha(-\alpha)(\lambda - t)^{-\alpha-1}(-1) = \alpha\lambda^\alpha(\lambda - t)^{-\alpha-1} = \frac{\alpha\lambda^\alpha}{(\lambda - t)^{\alpha+1}}$$
$$M''_X(t) = \frac{\alpha[-(\alpha+1)](-1)\lambda^\alpha}{(\lambda - t)^{\alpha+2}} = \frac{\alpha(\alpha+1)\lambda^\alpha}{(\lambda - t)^{\alpha+2}}$$

Then the mean is

$$E(X) = M'_X(0) = \frac{\alpha\lambda^\alpha}{\lambda^{\alpha+1}} = \frac{\alpha}{\lambda}$$

and for the variance

$$E(X^2) = M''_X(0) = \frac{\alpha(\alpha+1)\lambda^\alpha}{\lambda^{\alpha+2}} = \frac{\alpha(\alpha+1)}{\lambda^2}$$

therefore

$$\text{Var}(X) = \frac{\alpha^2 + \alpha}{\lambda^2} - \left(\frac{\alpha}{\lambda}\right)^2 = \frac{\alpha}{\lambda^2}.$$

Remark 2.6.11.

1. We can present the m.g.f of the gamma as

$$M_X(t) = \left(\frac{\lambda}{\lambda - t}\right)^\alpha = \left(\frac{1}{1 - t/\lambda}\right)^\alpha = \left(1 - \frac{t}{\lambda}\right)^{-\alpha}.$$

2. The "Shape" parameter (α) dictates the physical appearance of the curve (as seen in plots below), while the "Rate" parameter (λ) determines the scale or "spread" of the data along the x -axis.
3. While the Exponential distribution models the time until the first event occurs in a Poisson process, the Gamma distribution models the time required for α independent events to occur.
4. The gamma is a sum of independent exponentials. Exponential is $X \sim \text{GAM}(1, \lambda)$.

Example 2.6.12. The lifetime of a battery is exponentially distributed with rate λ . If a stereo cassette requires one battery to operate, then the total playing time one can obtain from the total of n batteries is a gamma random variable with parameters (n, λ) .

Graphs of the Gamma

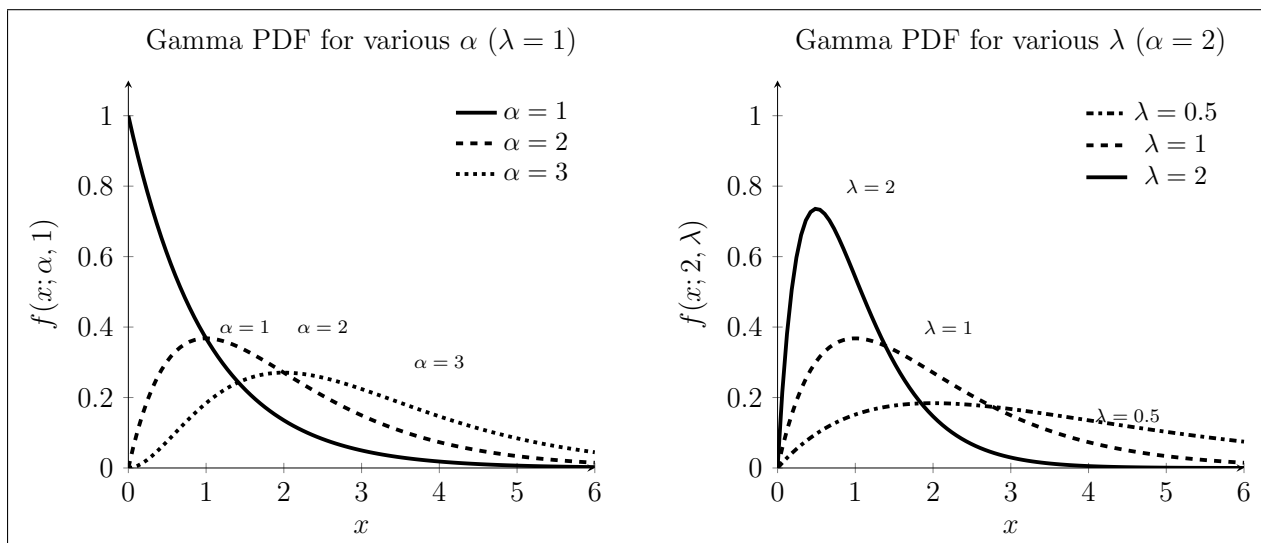


Figure 2.16: Comparison of Gamma distribution parameters. Left: Changing shape (α) shifts the peak and changes the skew. Right: Changing rate (λ) affects the spread and height.

2.6.4 Beta Distribution

Definition 2.6.13 (Beta Function.). We define the *Beta* function of α and β , by

$$B(\alpha, \beta) = \int_0^1 t^{\alpha-1} (1-t)^{\beta-1} dt$$

for $\alpha > 0$ and $\beta > 0$.

Properties of B :

1. $B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$
2. $B(\alpha, \beta) = B(\beta, \alpha)$
3. $B(\alpha + 1, \beta) = \frac{\alpha}{\alpha + \beta} B(\alpha, \beta)$
4. $B(\alpha, \beta + 1) = \frac{\beta}{\alpha + \beta} B(\alpha, \beta)$

Definition 2.6.14 (Beta Distribution $X \sim \text{Beta}(\alpha, \beta)$). A random variable is said to

have a *beta distribution* if its density p.d.f is given by

$$f_X(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}, \quad \alpha > 0, \quad \beta > 0, \quad 0 < x < 1.$$

The k^{th} expectation of X is

$$\begin{aligned} E(x^k) &= \int_x x^k f_X(x) dx \\ &= \int_0^1 \frac{x^k x^{\alpha-1} (1-x)^{\beta-1}}{B(\alpha, \beta)} dx \\ &= \frac{1}{B(\alpha, \beta)} \int_0^1 x^{k+\alpha-1} (1-x)^{\beta-1} dx \\ &= \frac{1}{B(\alpha, \beta)} B(k+\alpha, \beta) \\ &= \frac{\Gamma(k+\alpha)\Gamma(\beta)}{\Gamma(k+\alpha+\beta)} = \frac{\Gamma(k+\alpha)\Gamma(\beta)}{\Gamma(\alpha)\Gamma(\beta)} \times \frac{\Gamma(\alpha+\beta)}{\Gamma(k+\alpha+\beta)} \\ &= \frac{\Gamma(k+\alpha)\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(k+\alpha+\beta)}. \end{aligned}$$

When $k = 1$, we get the mean of X

$$E(x) = \frac{\Gamma(\alpha+1)\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\alpha+\beta+1)} = \frac{\alpha\Gamma(\alpha)\Gamma(\alpha+\beta)}{\Gamma(\alpha)(\alpha+\beta)\Gamma(\alpha+\beta)} = \frac{\alpha}{\alpha+\beta}.$$

When $k = 2$, we get

$$\begin{aligned} E(x^2) &= \frac{\Gamma(\alpha+2)\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\alpha+\beta+2)} \\ &= \frac{(\alpha+1)\alpha\Gamma(\alpha)\Gamma(\alpha+\beta)}{\Gamma(\alpha)(\alpha+\beta+1)(\alpha+\beta)\Gamma(\alpha+\beta)} \\ &= \frac{\alpha(\alpha+1)}{(\alpha+\beta+1)(\alpha+\beta)}. \end{aligned}$$

Therefore, the variance of X is

$$\begin{aligned}
 \text{Var}(x) &= E(x^2) - (E(x))^2 \\
 &= \frac{\alpha(\alpha+1)}{(\alpha+\beta)(\alpha+\beta+1)} - \left(\frac{\alpha}{\alpha+\beta}\right)^2 \\
 &= \frac{\alpha}{\alpha+\beta} \left[\frac{\alpha+1}{\alpha+\beta+1} - \frac{\alpha}{\alpha+\beta} \right] \\
 &= \frac{\alpha}{\alpha+\beta} \left[\frac{(\alpha+1)(\alpha+\beta) - \alpha(\alpha+\beta+1)}{(\alpha+\beta)(\alpha+\beta+1)} \right] \\
 &= \frac{\alpha}{(\alpha+\beta)^2} \left[\frac{\alpha^2 + \alpha\beta + \alpha + \beta - \alpha^2 - \alpha\beta - \alpha}{\alpha + \beta + 1} \right] \\
 &= \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}.
 \end{aligned}$$

Graph of the Beta Distribution.

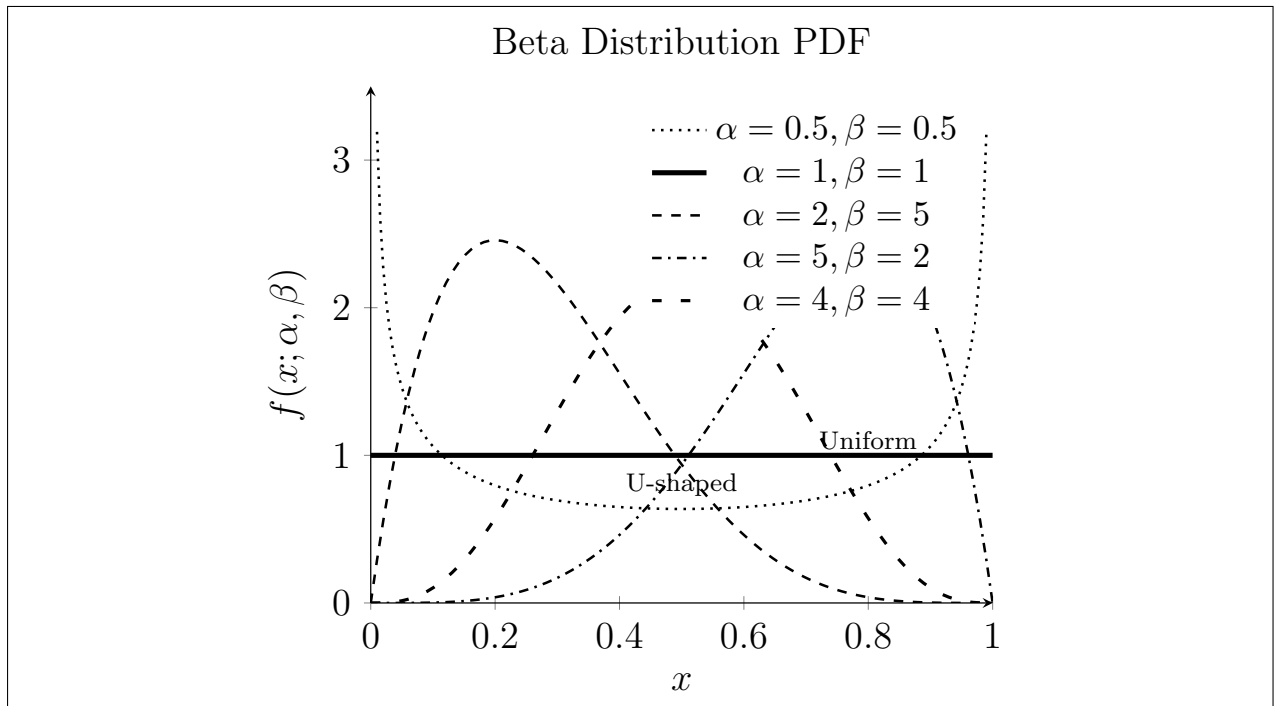


Figure 2.17: The Beta distribution takes various shapes depending on α and β , making it ideal for modeling proportions and probabilities.

Remark 2.6.15. The Beta distribution is famously known as the "Probability of a Probability." In Bayesian statistics, it is used as the conjugate prior for the Bernoulli and Binomial distributions. For example, if you are trying to estimate the probability of success of a new

crop variety in a specific Zambian province, the Beta distribution represents your "certainty" about that probability.

2.6.5 Normal Distribution

The main importance of normal distribution lies on the central limit theorem which says that the sample mean has a normal distribution if the sample size is large.

Definition 2.6.16 (Normal Distribution $X \sim N(\mu, \sigma^2)$). A random variable X is said to normally distributed, with parameter μ and σ^2 if the density of X is given by

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}, \quad x \in (-\infty, \infty).$$

This density function is a bell-shaped curve that is symmetric about μ . However, the special normal distribution is the standard normal distribution denoted by

$$Z \sim N(0, 1) \implies \mu = 0, \quad \sigma^2 = 1.$$

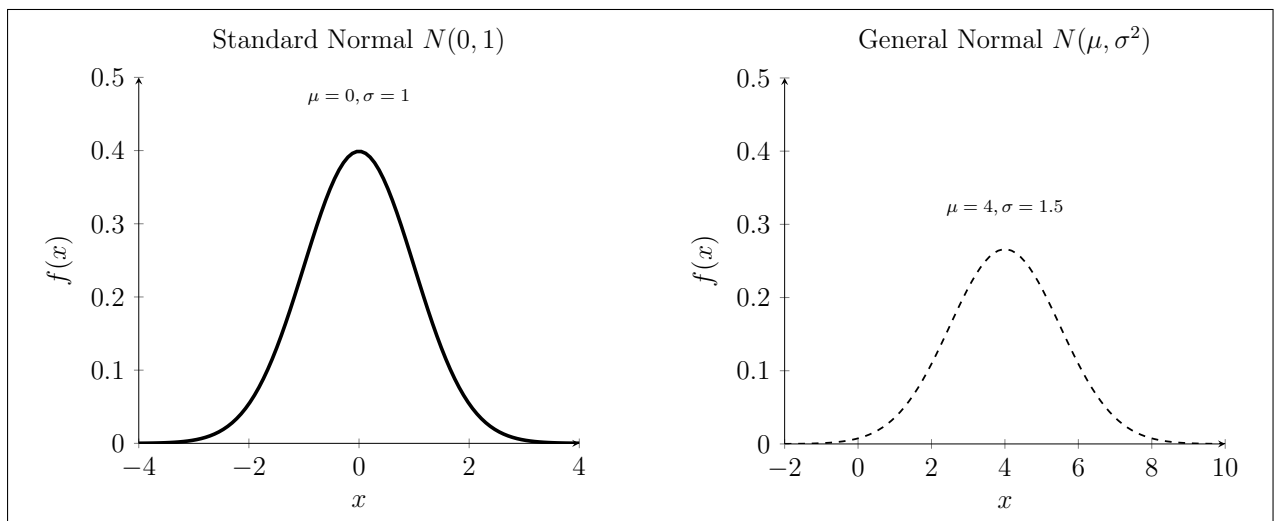


Figure 2.18: Comparison between the Standard Normal distribution and a General Normal distribution with shifted mean and increased variance.

Theorem 2.6.17. If $X \sim N(\mu, \sigma^2)$, then

$$\begin{aligned} E(X) &= \mu \\ \text{Var}(X) &= \sigma^2 \\ M_X(t) &= e^{\mu t + \frac{1}{2}\sigma^2 t^2}. \end{aligned}$$

The moment generating function

$$\begin{aligned}
M_X(t) &= E(e^{tx}) \\
&= \int_X e^{tx} f_X(x) dx \\
&= \int_{-\infty}^{\infty} e^{tx} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx \\
&= \int_{-\infty}^{\infty} e^{tx} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x^2-2\mu x+\mu^2)} dx \\
&= \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x^2-2\mu x+\mu^2-2\sigma^2 tx)} dx \\
&= \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu-\sigma^2 t)^2} e^{\mu t + \frac{1}{2}\sigma^2 t^2} dx \\
&= e^{\mu t + \frac{1}{2}\sigma^2 t^2} \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu-\sigma^2 t)^2} dx \\
&= e^{\mu t + \frac{1}{2}\sigma^2 t^2}.
\end{aligned}$$

Note: The last integral integrates to 1 because the integrand is the p.d.f of a normal, i.e $N(\mu + \sigma^2 t, \sigma^2)$.

Differentiating the moment generating function

$$\begin{aligned}
M'_X(t) &= (\mu + t\sigma^2)e^{\mu t + \frac{1}{2}t^2\sigma^2} \\
M''_X(t) &= \sigma^2 e^{\mu t + \frac{1}{2}t^2\sigma^2} + (\mu + t\sigma^2)(\mu + t\sigma^2)e^{\mu t + \frac{1}{2}t^2\sigma^2}
\end{aligned}$$

Then the mean is

$$E(x) = M'_X(0) = \mu e^0 = \mu$$

and

$$E(X^2) = M''_X(0) = \sigma^2 e^{0+0} + (\mu + 0)(\mu + 0)e^{0+0} = \sigma^2 + \mu^2$$

hence, the variance

$$\text{Var}(X) = E[x^2] - [E(x)]^2 = \sigma^2 + \mu^2 - \mu^2 = \sigma^2.$$

2.6.6 Cauchy Distribution

In physics, the Cauchy distribution is known as the Lorentz distribution. It is used to model the distribution of energy of unstable states in quantum mechanics and the resonance curves in spectroscopy.

Definition 2.6.18 (Cauchy Distribution ($X \sim \text{Cauchy}(a, \alpha)$). A random variable is said to have a Cauchy distribution with parameters a and α , if the p.d.f is give by

$$f_X(x) = \frac{1}{\alpha\pi \left[1 + \left(\frac{x-a}{\alpha}\right)^2\right]}, \quad -\infty < x < \infty$$

for $\alpha > 0$ and $-\infty < a < \infty$.

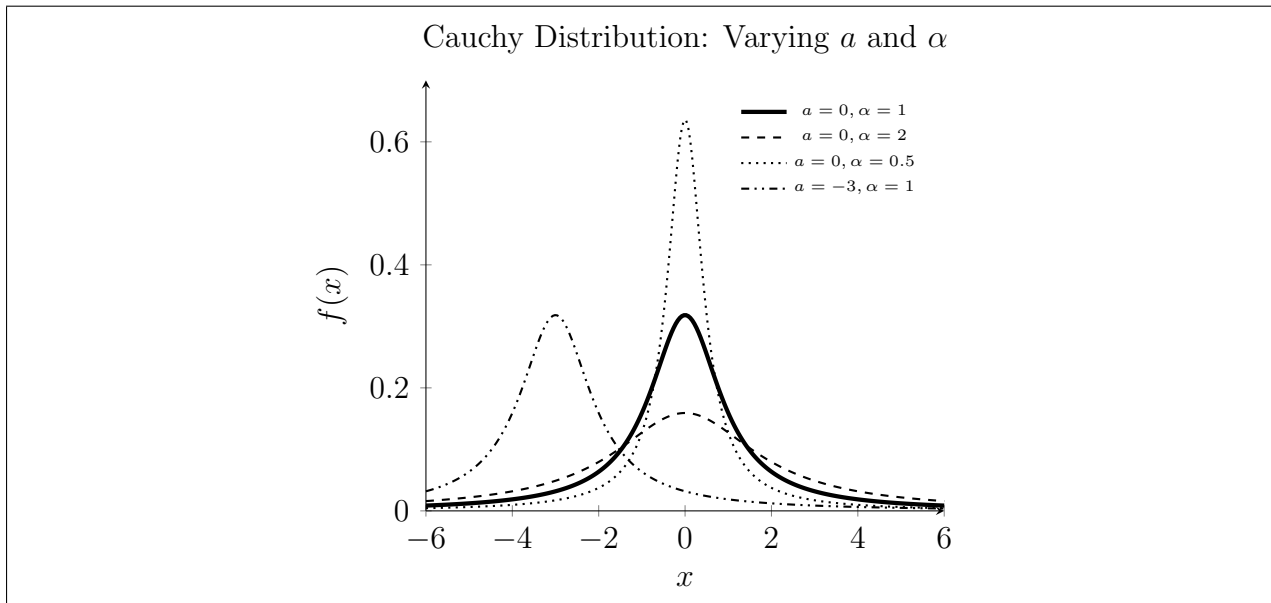


Figure 2.19: The Cauchy distribution under different parameters. Note how α controls the “scale” and a controls the location.

Remark 2.6.19. The mean does not exist

$$E(x) = \int_{-\infty}^{\infty} \frac{x}{\pi(1 + (x - \theta)^2)} dx = H(x) \Big|_{-\infty}^{\infty} \quad \text{diverges}$$

and

$$E(x^2) = \int_{-\infty}^{\infty} \frac{x^2}{\pi(1 + (x - \theta)^2)} dx \quad \text{diverges.}$$

Exercise 2.6.20. Suppose that X has Cauchy $X \sim \text{Cauchy}(1, 0)$ distribution show that $E(X)$ does not exist.

2.6.7 The Weibull Distribution

The distribution is used in reliability and survival analysis to model the lifetime of an object, the lifetime of an organism or systems that fail at a constant rate, or systems that wear out as they age.

Definition 2.6.21 (Weibull Distribution ($X \sim \text{Weibull}(\alpha, \beta)$). A Weibull random variable X has p.d.f of form

$$f_X(x) = \frac{\beta}{\alpha} x^{\beta-1} e^{(-\frac{1}{\alpha})x^\beta}, \quad x > 0,$$

with the scale parameter $\alpha > 0$ and shape parameter $\beta > 0$.

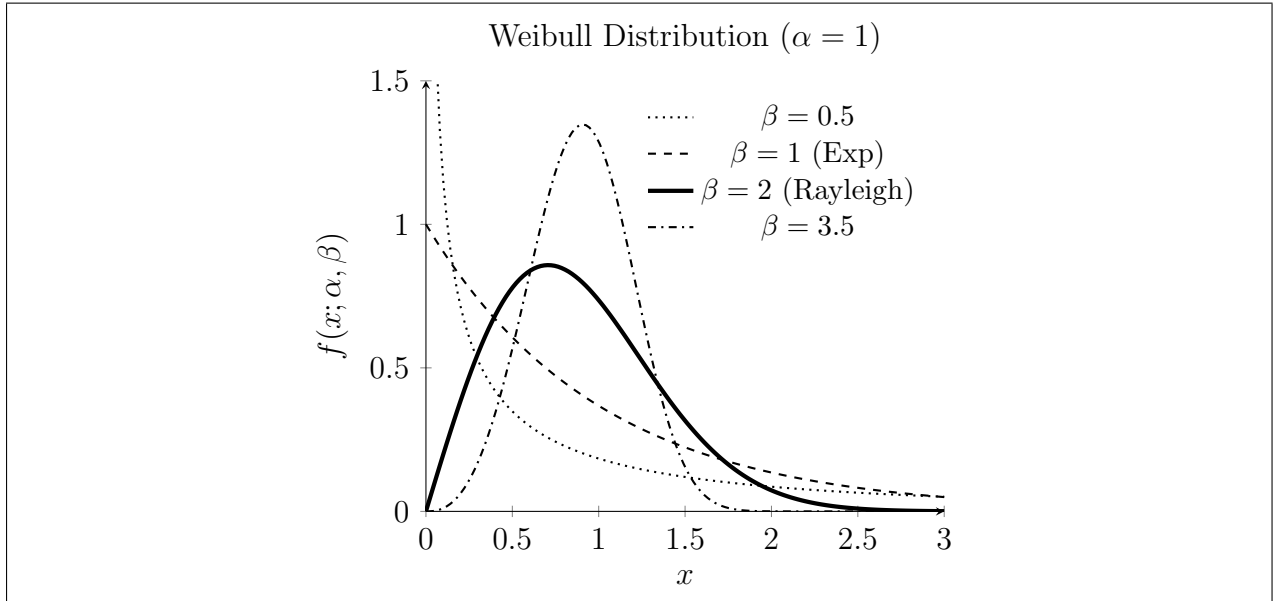


Figure 2.20: The Weibull distribution. Changing β allows the model to shift from an exponential decay to a symmetric bell shape.

Theorem 2.6.22. If $X \sim \text{Weibull}(\alpha, \beta)$, then c.d.f is

$$F_X(x) = P(X \leq x) = 1 - e^{-\frac{1}{\alpha} \beta x},$$

the mean is

$$E(X) = \frac{\alpha}{\beta} \Gamma\left(\frac{1}{\beta}\right)$$

and the variance is

$$\text{Var}(X) = \alpha^2 \left\{ \frac{2}{\beta} \Gamma\left(\frac{2}{\beta}\right) - \left[\frac{1}{\beta} \Gamma\left(\frac{1}{\beta}\right) \right]^2 \right\}.$$

2.6.8 Rayleigh Distribution

The Rayleigh distribution is a special case of the Weibull distribution (where the shape parameter is fixed), the only thing that changes here is the scale parameter α .

Definition 2.6.23 (Rayleigh Distribution ($X \sim \text{Rayleigh}(\alpha)$). A Rayleigh random

variable X with $\alpha > 0$ has p.d.f

$$f_X(x) = \frac{2x e^{-\frac{x^2}{\alpha}}}{\alpha}, \quad x > 0.$$

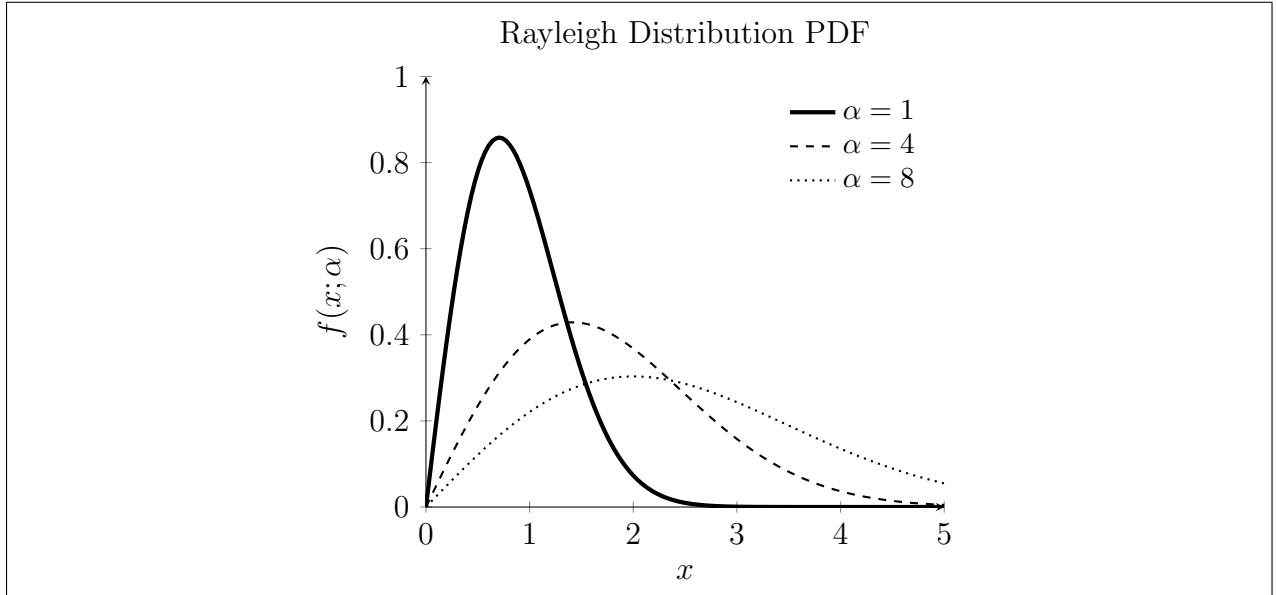


Figure 2.21: The Rayleigh distribution. As α increases, the distribution spreads out and the peak shifts to the right.

Exercise 2.6.24. Show that the mean of $X \sim \text{Rayleigh}(\alpha)$ is

$$E(X) = \frac{\sqrt{\alpha\pi}}{\pi}$$

and Variance is

$$\text{Var}(X) = \frac{\alpha(4 - \pi)}{4}.$$

2.6.9 The Pareto Distribution

The Pareto distribution is a "power-law" distribution traditionally used to model the distribution of wealth and income. In this model, the parameter λ represents the minimum value (such as a minimum wage), while κ determines the shape of the distribution. Beyond economics, it is used to model the lifetime of objects with a mandatory warranty period λ , or the duration of events with a guaranteed minimum length λ , such as a labor strike.

Definition 2.6.25 (Pareto Distribution $X \sim \text{Pareto}(\lambda, \kappa)$). A Pareto random variable

X with scale parameter $\lambda > 0$ and shape parameter $\kappa > 0$ has a p.d.f. given by:

$$f_X(x) = \frac{\kappa \lambda^\kappa}{x^{\kappa+1}}, \quad x \geq \lambda.$$

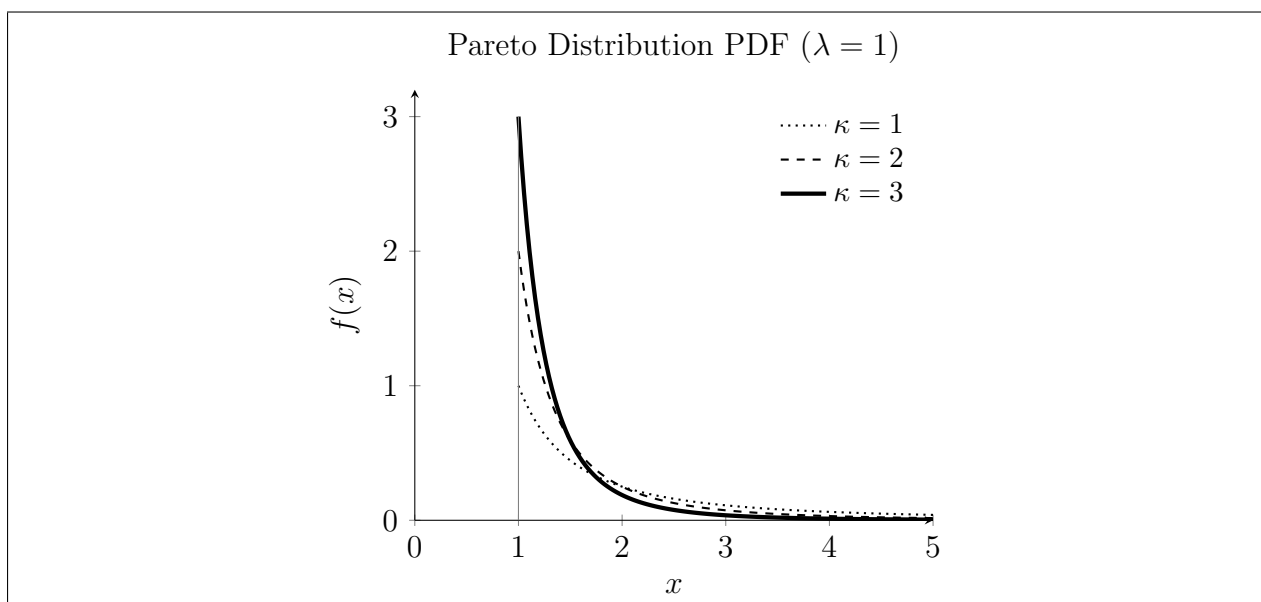


Figure 2.22: The Pareto distribution. Note that the density is zero for $x < \lambda$. Higher values of κ lead to a faster decay in the probability of extreme values.

Theorem 2.6.26. If $X \sim \text{Pareto}(\alpha)$, then

the moment generating function is

$$M_X(t) = \kappa(-\lambda t)^\kappa \Gamma(-\kappa, \lambda t),$$

the mean is

$$E(X) = \frac{\kappa \lambda}{\kappa - 1} \quad \text{for } \kappa > 1,$$

and the variance is

$$\text{Var}(X) = \frac{\kappa \lambda^2}{(\kappa - 1)^2(\kappa - 2)} \quad \text{for } \kappa > 2.$$

Remark 2.6.27.

1. The Pareto distribution is the mathematical backbone of the 80/20 rule (the Pareto Principle). For example, it often models how 80% of a country's wealth is held by 20% of the population.
2. Unlike the Normal or Gamma distributions, which can approach zero, the Pareto has a

hard “floor” at λ . In many engineering applications, this represents the “guaranteed” life of a component before any failure is even possible.

2.7. Cumulative Distribution Function (CDF)

Definition 2.7.1. Let X be a random variable with probability function $f_X(x)$, then cumulative distribution function (cdf) of X denoted by $F_X(x)$ and is given by

$$F_X(t) = P(X \leq t)$$

specifically, calculated as

$$F_X(t) = \begin{cases} \sum_{\{X: X \leq t\}} P(X = x) & \text{if } X \text{ discrete} \\ \int_{\{X: X \leq t\}} f_X(x) dx & \text{if } X \text{ is continuous} \end{cases}$$

Fundamental Properties:

1. $F_X(x)$ is a non-decreasing function.
2. $F_X(-\infty) = \lim_{x \rightarrow -\infty} F_X(x) = 0$.
3. $F_X(+\infty) = \lim_{x \rightarrow +\infty} F_X(x) = 1$.
4. $F_X(x)$ is right-continuous: $\lim_{h \rightarrow 0^+} F_X(x + h) = F_X(x)$.

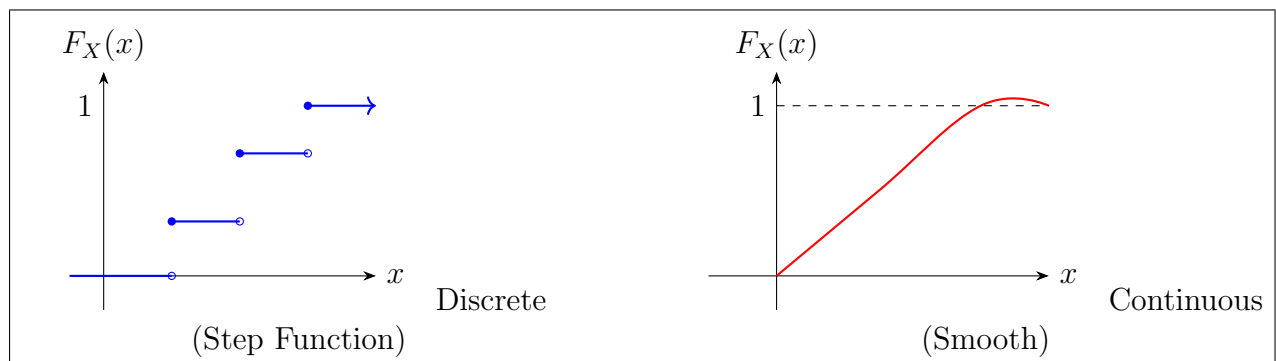


Figure 2.24: *

Example 2.7.2. Consider tossing a fair coin 3 times. Let $X \sim \text{BIN}(3, 1/2)$. The p.m.f. is:

$$f_X(x) = \binom{3}{x} \left(\frac{1}{2}\right)^3 \quad \text{for } x \in \{0, 1, 2, 3\}.$$

$$P(X = 0) = \frac{1}{8}, \quad P(X = 1) = \frac{3}{8}, \quad P(X = 2) = \frac{3}{8}, \quad P(X = 3) = \frac{1}{8}.$$

$$F_X(t) = P(X \leq t)$$

If $t < 0$

$$F_X(t) = P(X \leq t) = 0.$$

If $0 \leq t < 1$

$$F_X(t) = P(X \leq t) = P(X \leq 0) = P(X = 0) = \frac{1}{8}.$$

If $1 \leq t < 2$

$$\begin{aligned} P(1 \leq t < 2) &= P(X \leq 1) = P(X = 0, X = 1) \\ &= P(X = 0) + P(X = 1) \\ &= \frac{1}{8} + \frac{3}{8} \\ &= \frac{4}{8} \\ &= \frac{1}{2}. \end{aligned}$$

If $2 \leq t < 3$

$$\begin{aligned} P(2 \leq t < 3) &= P(X \leq 2) \\ &= P(X = 0) + P(X = 1) + P(X = 2) \\ &= \frac{1}{8} + \frac{3}{8} + \frac{3}{8} \\ &= \frac{7}{8}. \end{aligned}$$

If $t \geq 3$, $P(X \leq 3) = 1$ since

$$P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) = 1.$$

Therefore, the c.d.f

$$F_X(t) = \begin{cases} 0 & \text{if } t < 0 \\ \frac{1}{8} & \text{if } 0 \leq t < 1 \\ \frac{1}{2} & \text{if } 1 \leq t < 2 \\ \frac{7}{8} & \text{if } 2 \leq t < 3 \\ 1 & \text{if } t \geq 3 \end{cases}$$

The plot

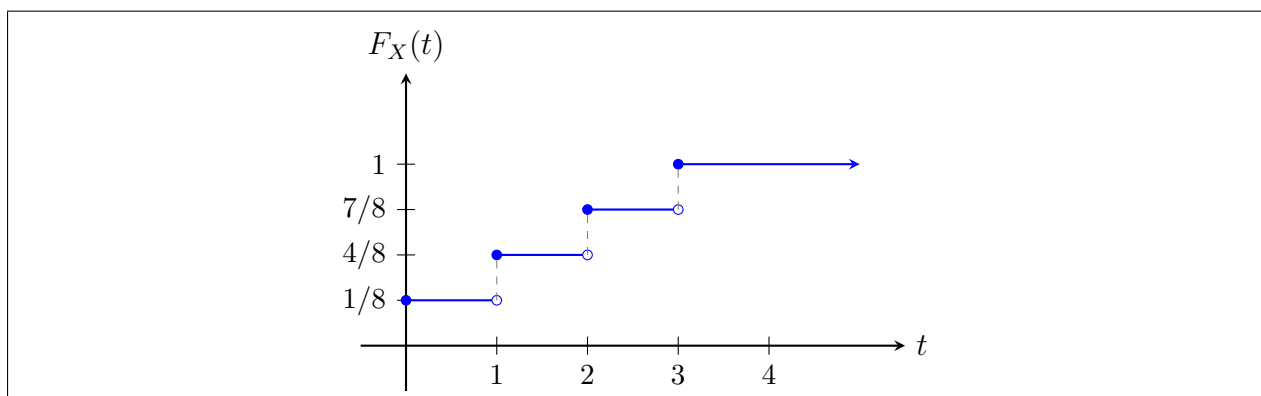


Figure 2.25: Graph of $F_X(t)$

Example 2.7.3. Let the pdf of a random variable X be

$$f_X(x) = \frac{x+1}{8}, \quad 2 < x < 4.$$

Find the c.d.f and plot it.

Solution.

$$F_X(t) = P(X \leq t) = \int_{\{X: X \leq t\}} f_X(x) dx$$

If $t \leq 2$,

$$F_X(t) = 0.$$

If t is such $2 < t < 4$

$$\begin{aligned}
 F_X(t) &= P(X \leq t) = \int_2^t f_X(x) dx \\
 &= \int_2^t \frac{x+1}{8} dx \\
 &= \left. \frac{(x+1)^2}{8} \right|_2^t \\
 &= \frac{(t+1)^2 - 9}{16}.
 \end{aligned}$$

If $t \geq 4$,

$$\begin{aligned}
 F_X(t) &= P(X \leq t) \\
 &= \int_2^t f_X(x) dx + \int_4^t f_X(x) dx \\
 &= 1 + 0 \\
 &= 1.
 \end{aligned}$$

Therefore the c.d.f

$$F_X(t) = \begin{cases} 0 & \text{if } t \leq 2 \\ \frac{(t+1)^2 - 9}{16} & \text{if } 2 < t < 4 \\ 1 & t \geq 4 \end{cases}$$

The plot

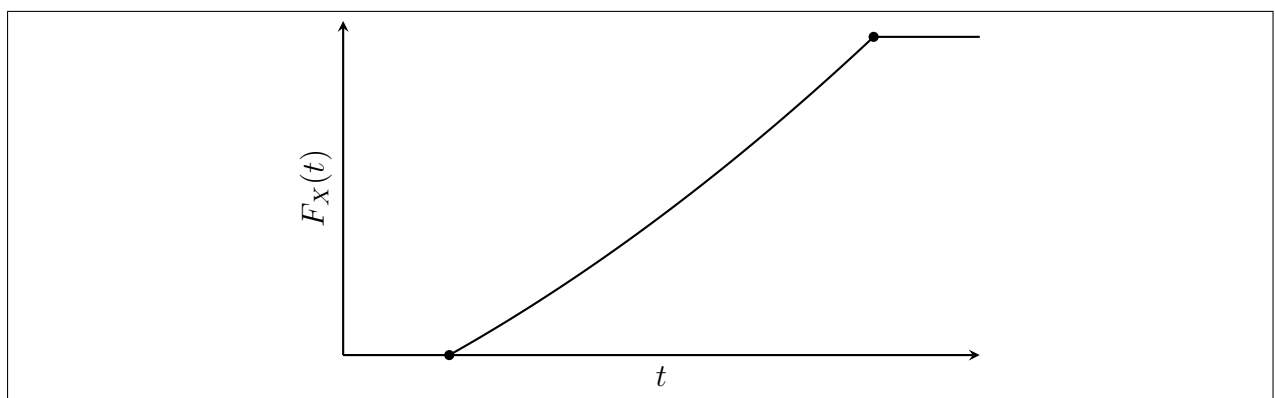


Figure 2.26: c.d.f X

□

Remark 2.7.4.

1. In the case where X is discrete with possible values $x_1, x_2, x_3, \dots$

$$f_X(x_j) = P(X = x_j) = F_X(x_j) - F_X(x_{j-1})$$

For continuous random variable

$$\begin{aligned} P(a < X < b) &= \int_b^a f_X(x) dx \\ &= \int_{-\infty}^b f_X(x) dx - \int_{-\infty}^a f_X(x) dx \\ &= F_X(b) - F_X(a) \end{aligned}$$

also, $P(a \leq X < b) = P(a \leq X \leq b) = P(a < X \leq b)$

2. From fundamental theorem of calculus, it follows that if $F_X(t)$ is differentiable then

$$f_X(x) = \frac{d}{dx} F_X(x).$$

3. For discrete random variables X with possible values in ascending order $x_1, x_2, x_3, \dots$ then

$$F_X(t) = P(X \leq t) = \sum_{j=1}^k P(X = x_j)$$

where k is such that $x_{k-1} < t \leq x_k$.

$$F_X(t) = \begin{cases} 0, & t < x_1 \\ \sum_{j=1}^k P(X = x_j) & \text{if } x_{k-1} < t \leq x_k \\ 1 & \text{if } t > x_N, \text{ where } x_N \text{ is the largest} \end{cases}$$

4. If X is continuous random variable with pdf having support in the interval (u, v) . i.e $f_X(x)$ is defined on (u, v) and 0 else where

$$F_X(t) = \begin{cases} 0 & \text{if } t < u \\ \int_u^t f_X(x) dx & \text{if } u < t < v \\ 1 & \text{if } t \geq v \end{cases}$$

Example 2.7.5. Suppose a random variable has the following probability mass functions:

$$P(X = 1) = \frac{1}{2}, \quad P(X = 2) = \frac{1}{4}, \quad P(X = 3) = \frac{1}{8}, \quad P(X = 4) = \frac{1}{8}.$$

Find the c.d.f of the random variable X and use it to find $P(X \leq 1)$, $P(1 < X < 3)$ and $P(1 \leq X \leq 3)$.

Solution. The c.d.f

$$F_X(t) = \begin{cases} 0, & t < 1 \\ \frac{1}{2}, & 1 \leq t < 2 \\ \frac{3}{4}, & 2 \leq t < 3 \\ \frac{7}{8}, & 3 \leq t < 4 \\ 1, & t \geq 4. \end{cases}$$

The $P(X \leq 1)$ is

$$P(X \leq 1) = F(1^+) = \frac{1}{2}.$$

The $P(1 < X < 3)$ is

$$\begin{aligned} P(1 < X < 3) &= F(3^-) - F(1^+) \\ &= \frac{3}{4} - \frac{1}{2} \\ &= \frac{1}{4}. \end{aligned}$$

The $P(1 \leq X \leq 3)$ is

$$\begin{aligned} P(1 \leq X \leq 3) &= F(3^+) - F(1^-) \\ &= \frac{7}{8} - 0 \\ &= \frac{7}{8}. \end{aligned}$$

□

Example 2.7.6. Use the c.d.f of

$$f_X(x) = \frac{x+1}{8}, \quad 2 < x < 4$$

to find the following probabilities: Using Cdf find:

(i). $P(1 < X < 2\frac{1}{2})$

(ii). $P(2\frac{1}{2} < X \leq 3)$

(iii). $P(3 < X < 5)$

Solution. The c.d.f is given by

$$F_X(t) = \begin{cases} 0, & t < 2 \\ \frac{(t+1)^2 - 9}{16}, & 2 \leq t < 4 \\ 1, & t \geq 4 \end{cases}$$

Then

$$\begin{aligned} P\left(1 < X < 2\frac{1}{2}\right) &= F_X\left(\frac{5}{2}\right) - F_X(1) \\ &= \frac{(2\frac{1}{2} + 1)^2 - 9}{16} - 0 \\ &= \frac{\frac{49}{4} - 9}{16} \\ &= \frac{49 - 36}{64} \\ &= \frac{13}{64}. \end{aligned}$$

Also

$$\begin{aligned} P\left(2\frac{1}{2} < X < 3\right) &= F_X(3) - F_X\left(\frac{5}{2}\right) \\ &= \frac{(3+1)^2 - 9}{16} - \frac{13}{64} \\ &= \frac{7}{16} - \frac{13}{64} \\ &= \frac{15}{64}. \end{aligned}$$

And

$$\begin{aligned}P(3 < X < 5) &= F_X(5) - F_X(3) \\&= 1 - \left(\frac{4^2 - 9}{16}\right) \\&= 1 - \frac{7}{16} \\&= \frac{9}{16}.\end{aligned}$$

□

Theorem 2.7.7. $X \sim N(\mu, \sigma^2)$ is read as X is distributed as a normal with mean μ and variance σ^2 . If

$$Z = \frac{X - \mu}{\sigma},$$

then Z is distributed as normal with mean 0 and variance 1 and is known as standard normal, notation $Z \sim N(0, 1)$.

Note 2.7.8. Normal variable is usually referred to as Gaussian Distribution.

The c.d.f for the standard normal

$$\Phi(z) = \int_{-\infty}^z \frac{e^{-\frac{1}{2}v^2}}{\sqrt{2\pi}} dv$$

$$\Phi(-1.11) = \int_{-\infty}^{-1.11} \frac{e^{-\frac{1}{2}t^2}}{\sqrt{2\pi}} dt$$

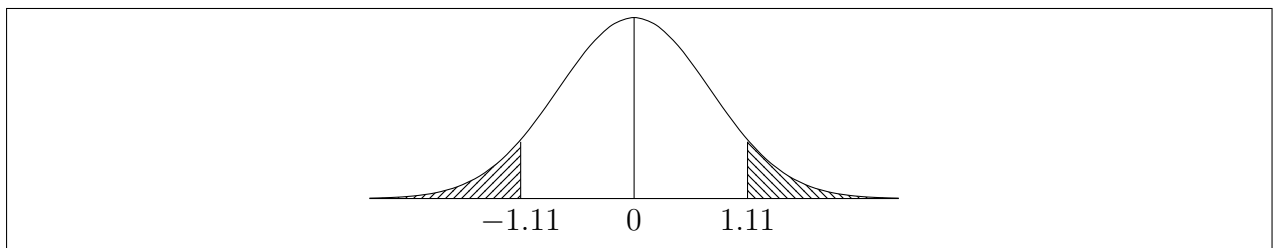


Figure 2.27:

$$\begin{aligned}
\Phi(-1.11) &= 1 - \Phi(1.11) \\
&= 1 - 0.8643 \\
&= 0.1357
\end{aligned}$$

$$P(z \geq 2) = P(z > 2)$$

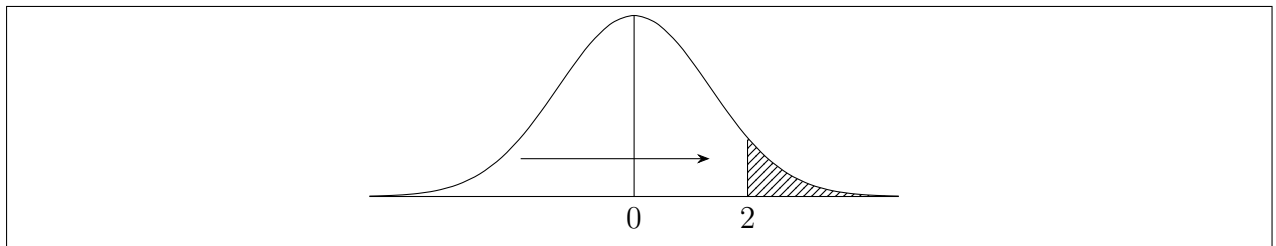


Figure 2.28:

$$\begin{aligned}
P(z \geq 2) &= 1 - P(z \leq 2) \\
&= 1 - \Phi(2) \\
&= 1 - 0.9772 \\
&= 0.0228
\end{aligned}$$

$$\begin{aligned}
P(-1.02 < z < 1.96) &= \Phi(1.96) - \Phi(-1.02) \\
&= \Phi(1.96) - (1 - \Phi(1.02)) \\
&= \Phi(1.96) + \Phi(1.02) - 1 \\
&= 0.9750 + 0.8461 - 1 \\
&= 0.8211
\end{aligned}$$

Example 2.7.9. The Rock well hardness of a metal specimen is determined by pressing the surface of the specimen with hardened point, and then measuring the depth of penetration. The hardness of an alloy is normally distributed with mean 70 units and standard deviation 3 units.

1. If a specimen is acceptable only if its hardness is between 66 and 74 units. What is

the probability that a randomly chosen specimen is acceptable?

Solution. $P(66 \leq x \leq 74) = ?$

$$\begin{aligned} P\left(\frac{66-70}{3} < Z < \frac{74-70}{3}\right) &= P\left(-\frac{4}{3} < Z < \frac{4}{3}\right) \\ &= 2P\left(0 < Z < \frac{4}{3}\right) \\ &= 2\left[\Phi\left(\frac{4}{3}\right) - \Phi(0)\right] \\ &= 2[0.9082 - 0.5000] \\ &= 0.8164 \end{aligned}$$

□

2. If the acceptable range is $70 \pm c$, for what value of c would 95% of all the specimens to be acceptable?

Solution.

$$\begin{aligned} P\left(-\frac{c}{3} < Z < \frac{c}{3}\right) &= 0.95 \\ 2P\left(0 < z < \frac{c}{3}\right) &= 0.95 \\ \Rightarrow P\left(0 < Z < \frac{c}{3}\right) &= 0.475 \end{aligned}$$

Therefore

$$\frac{c}{3} = 1.960 \Rightarrow c = 3 \times 1.960 = 5.880$$

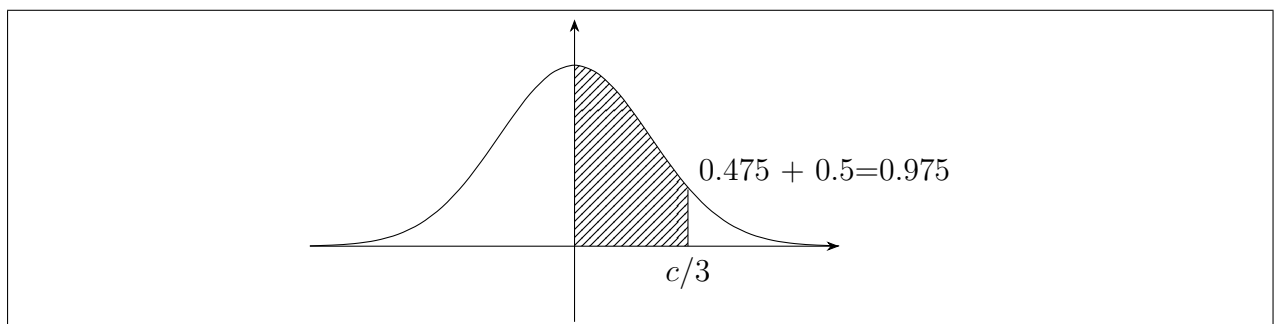


Figure 2.29:

□

2.8. Markov's and Chebyshev's Inequality

The primary value of Markov's and Chebyshev's inequalities lies in their ability to provide rigorous probability bounds when our knowledge of a distribution is limited. Specifically, these inequalities allow us to estimate the likelihood of events when we know only the mean, or both the mean and variance, of a random variable. (i.e underlying distribution is either unknown or too complex to compute directly.)

Lemma 2.8.1 (Markov's Inequality). *Let X be a non-negative random variable having a finite mean and t be a positive real number then*

$$P(X \geq t) \leq \frac{E(X)}{t}.$$

Proof. Define a random variable Y as follows

$$\begin{cases} y = 0, & \text{if } x < t \\ y = t, & \text{if } x \geq t \end{cases}$$

y is discrete. Then

$$f_Y(y) = \begin{cases} 0, & P(x < t) \\ t, & P(x \geq t) \end{cases}$$

Thus

$$X \geq Y$$

$$E(X) \geq E(Y) = 0 P(X < t) + t P(X \geq t)$$

$$E(X) \geq t P(X \geq t)$$

$$\frac{E(X)}{t} \geq P(X \geq t)$$

$$P(X \geq t) \leq \frac{E(X)}{t}.$$

□

Theorem 2.8.2 (Chebyshev's Inequality). Let X be a random variable with mean μ and finite variance σ^2 , then for any real number $t > 0$

$$P(|X - \mu| \geq t) \leq \frac{\sigma^2}{t^2}$$

or

$$P(|X - \mu| < t) > 1 - \frac{\sigma^2}{t^2}.$$

Proof. Since $(X - \mu)^2 \geq 0$, we apply the Markov's inequality to obtain

$$P((X - \mu)^2 \geq t^2) \leq \frac{E(X - \mu)^2}{t^2} = \frac{\sigma^2}{t^2}$$

That is

$$P((X - \mu) \geq t^2) \leq \frac{\sigma^2}{t^2}$$

Now, since $(X - \mu)^2 \geq t^2 \iff |X - \mu| \geq t$. Therefore,

$$P(|X - \mu| \geq t) \leq \frac{\sigma^2}{t^2}.$$

□

Remark 2.8.3.

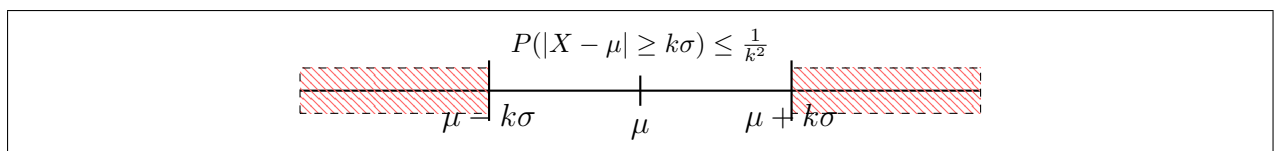
- i) **The Complementary Nature of the Bound:** The event $|X - \mu| \geq t$ represents the "tails" of the distribution—values that are far from the mean.

$$|X - \mu| \geq t \iff (X \geq \mu + t) \text{ or } (X \leq \mu - t)$$

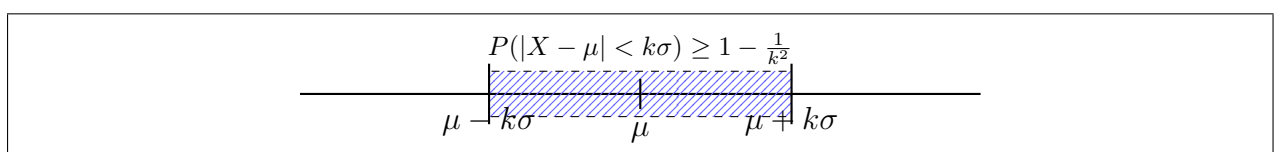
- ii) **Visualizing the Upper and Lower Bounds:** If we set $t = k\sigma$, Chebyshev's inequality tells us:

- The probability in the **shaded tails** (at most distance $k\sigma$ away) is at most $\frac{1}{k^2}$.
- The probability in the **shaded center** (within distance $k\sigma$) is at least $1 - \frac{1}{k^2}$.

Case A: Probability in the Tails (Maximum $\frac{1}{k^2}$)



Case B: Probability in the Center (Minimum $1 - \frac{1}{k^2}$)



Example 2.8.4. From a quick analysis of records a materials control manger estimates that the mean and standard deviation of the "lead time" required in ordering a small value are 8 days and 1.5 days respectively. The manager is willing to assume the mean and variance to be absolutely correct. The manager would like to determine a time interval such that the probability is at least $\frac{8}{9}$ that the order will be received during the time interval.

Solution. $\mu = 8$, $\sigma = 1.5$ and

$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2} = \frac{8}{9} \implies k = 3.$$

Then

$$(\mu - k\sigma, \mu + k\sigma) = (8 - 3(1.5), 8 + 3(1.5)) = (3.5, 12.5).$$

□

Example 2.8.5. Suppose we want the number of items produced in a factory during a week is a random variable with mean 500.

1. What can be said about the probability that the weeks production will be at least 1000?

Solution. $\mu = 500$, $P(X \geq 1000)$ by Markov's inequality we get

$$P(X \geq 1000) \leq \frac{E(X)}{1000} = \frac{500}{1000} = \frac{1}{2}.$$

□

2. If the variance of a week's production is known to be 100, what can be said about the probability that this week's production will be between 420 and 580?

Solution. $\mu = 500$, $\sigma = 10$

$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}$$

rewriting the expression

$$P(\mu - k\sigma < X < \mu + k\sigma) \geq 1 - \frac{1}{k^2}$$

but we have $P(420 < X < 580)$ therefore

$$500 - k(10) = 420 \implies k = 8$$

$$500 + k(10) = 580 \implies k = 8.$$

□

CHAPTER 3 : JOINTLY DISTRIBUTED RANDOM VARIABLES

3.1. Introduction

The study of random variables and their probability distributions so far has been restricted to one-dimensional sample spaces. There are many situations where we may want to describe several random variables. For example we might measure amount of precipitate X and the volume Y of a gas released from a controlled chemical giving rise to two-dimensional space. At this level of the course we will concentrate on two variables.

3.2. Joint Distributions

Definition 3.2.1. Let X and Y be discrete random variables, a formula or table listing all the possible values of X and Y together with the associated probabilities is called a joint probability distribution of X and Y denoted by $f_{X,Y}(x, y)$ and is given

$$f_{X,Y}(x, y) = P(X = x, Y = y)$$

and

$$\sum_x \sum_y f_{X,Y}(x, y) = \sum_x \sum_y P(X = x, Y = y).$$

Example 3.2.2. Two refills for a ball point pen are selected at random from a box that contains 3 blue refills, 2 red refills and 3 green refills. Let X be the number of blue refills and Y be the number of red refills selected. Find the joint distribution of X and Y .

Solution. Possible values of X are: 0, 1, 2 and Y are 0, 1, 2.

X	Y		
	0	1	2
0	$\frac{3}{28}$	$\frac{6}{28}$	$\frac{1}{28}$
1	$\frac{9}{28}$	$\frac{6}{28}$	0
2	$\frac{3}{28}$	0	0

$$P(X = 0, Y = 0) = \frac{\binom{3}{0} \binom{2}{0} \binom{3}{2}}{\binom{8}{2}} = \frac{3}{28}$$

$$P(X = 0, Y = 1) = \frac{\binom{3}{0} \binom{2}{1} \binom{3}{1}}{\binom{8}{2}} = \frac{2 \times 3}{28} = \frac{6}{28}$$

Therefore, the general formula is:

$$f_{X,Y}(x, y) = P(X = x, Y = y) = \frac{\binom{3}{x} \binom{2}{y} \binom{3}{2-x-y}}{\binom{8}{2}}.$$

□

Definition 3.2.3. Let X and Y be continuous random variables with joint range B in the Euclidean space, then $f_{X,Y}(x, y)$ the joint probability function has the following properties:

a). $f_{X,Y}(x, y) \geq 0$ for $x, y \in B$

b).
$$\iint_B f_{X,Y}(x, y) dx dy = \iint_B f_{X,Y}(x, y) dy dx = 1$$

Remark 3.2.4. The joint p.d.f for two continuous random variables lying in a specified region

$$P(a < X < b, c < Y < d) = \int_a^b \int_c^d f_{X,Y}(x, y) dy dx = \int_c^d \int_a^b f_{X,Y}(x, y) dx dy$$

Example 3.2.5. Suppose that the variation in two continuous random variables, X and Y , can be modeled by the joint p.d.f $f_{X,Y}(x, y) = cxy$, for $0 < y < x < 1$. Find c .

Solution. By inspection, (a) in the definition 3.1.3, $f_{X,Y}(x, y)$ will be nonnegative as long as $c \geq 0$.

$$\iint_B cxy dy dx = 1 \implies \int_0^1 \int_0^x cxy dy dx = 1$$

$$c \int_0^1 \left[\frac{y^2}{2} \right]_0^x dx = 1$$

$$c \int_0^1 \frac{x^3}{2} dx = 1$$

$$c \left[\frac{x^4}{8} \right]_0^1 = 1$$

$$\frac{c}{8} = 1.$$

Therefore, $c = 8$.

□

Example 3.2.6. Let X and Y be random variables with a joint p.d.f

$$f_{X,Y}(x, y) = x + y, \quad x \in (0, 1), \quad y \in (0, 1).$$

Find:

- (i). Show that $f_{X,Y}(x, y)$ is indeed a joint p.d.f.

Solution. We show that $\iint_B f_{X,Y}(x, y) = 1$.

$$\begin{aligned} \int_0^1 \int_0^1 (x + y) dx dy &= \int_0^1 \left[\frac{x^2}{2} + xy \right]_0^1 dy \\ &= \int_0^1 \left(\frac{1}{2} + y \right) dy \\ &= \left[\frac{1}{2}y + \frac{y^2}{2} \right]_0^1 \\ &= 1. \end{aligned}$$

□

- (ii). Find $P(x + y < 1)$.

Solution. The region of integration is the area below the line $x + y = 1$.

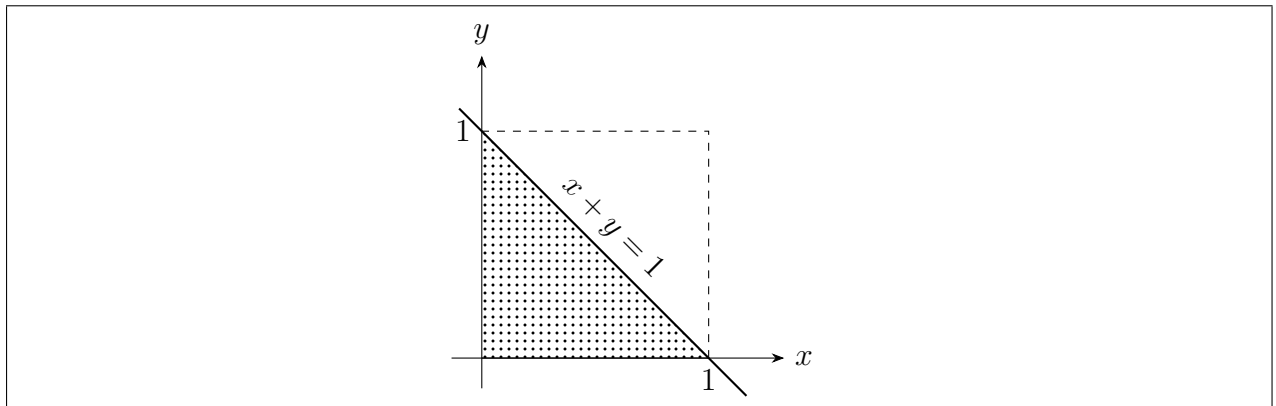


Figure 3.1:

$$\begin{aligned} P(X + Y < 1) &= \int_0^1 \int_0^{1-y} (x + y) dx dy = \int_0^1 \left[\frac{x^2}{2} + xy \right]_0^{1-y} dy \\ &= \int_0^1 \left(\frac{1 - 2y + y^2}{2} + y - y^2 \right) dy \\ &= \frac{1}{2} \int_0^1 (1 - y^2) dy \\ &= \frac{1}{3}. \end{aligned}$$

Therefore, $P(X + Y < 1) = \frac{1}{3}$. □

(iii). (*Exercise*) Find $P(X + Y > 1\frac{1}{2})$

Exercise 3.2.7. A study claims that the daily number of hours, X , a teenager watches television and the daily number of hours, Y , he works on his homework are approximated by the joint p.d.f

$$f_{X,Y}(x, y) = xye^{-(x+y)}, \quad x > 0, \quad y > 0.$$

What probability that a teenager chosen at random spends at least twice as much time watching television as he does working on this homework?

Solution. The region, B , in the xy -plane corresponding to the event “ $X \geq 2Y$ ”.

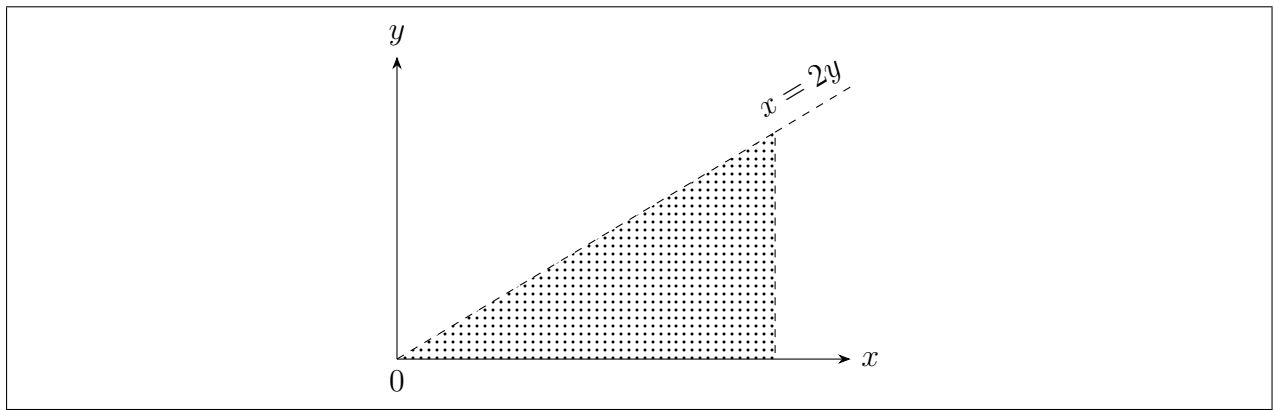


Figure 3.2:

$$P(2Y \leq X) = \int_0^\infty \int_0^{\frac{x}{2}} xye^{-(x+y)} dy dx$$

We can write

$$\begin{aligned} P(X \leq 2Y) &= \int_0^\infty xe^{-x} \left[\int_0^{x/2} ye^{-y} dy \right] dx \\ &= \int_0^\infty xe^{-x} \left[1 - \left(\frac{x}{2} + 1 \right) e^{-x/2} \right] dx \\ &= \int_0^\infty xe^{-x} dx - \frac{1}{2} \int_0^\infty e^{-3x/2} dx - \int_0^\infty xe^{-3x/2} dx \\ &= 1 - \frac{16}{54} - \frac{4}{9} \\ &= \frac{7}{27}. \end{aligned}$$

□

Definition 3.2.8. If X and Y are discrete random variables with a joint distribution function $f_{X,Y}(x,y)$ then $f_X(x)$ and $f_Y(y)$ are called marginal distribution of X and Y respectively and are given by

$$f_X(x) = \sum_{\text{all } y} f_{X,Y}(x,y)$$

and

$$f_Y(y) = \sum_{\text{all } x} f_{X,Y}(x,y).$$

Example 3.2.9. Using example 3.2.2

$$P(X = x, Y = y) = \frac{\binom{3}{x} \binom{2}{y} \binom{3}{2-x-y}}{\binom{8}{2}}.$$

Solution. We can get the marginals from the table

X	Y			$P(X = x)$
	0	1	2	
0	$\frac{3}{28}$	$\frac{6}{28}$	$\frac{1}{28}$	$\frac{16}{28} = P(X = 0)$
1	$\frac{9}{28}$	$\frac{6}{28}$	0	$\frac{15}{28} = P(X = 1)$
2	$\frac{3}{28}$	0	0	$\frac{3}{28} = P(X = 2)$
$P(Y = y)$	$\frac{15}{28}$	$\frac{12}{28}$	$\frac{1}{28}$	

The marginal of X . i.e $f_X(x) = P(X = x)$

X	0	1	2
$f_X(x)$	$\frac{10}{28}$	$\frac{15}{28}$	$\frac{3}{28}$

The marginal of Y . i.e $f_Y(y) = P(Y = y)$

Y	0	1	2
$f_Y(y)$	$\frac{15}{28}$	$\frac{12}{28}$	$\frac{1}{28}$

Using the formula

$$\begin{aligned}
 f_X(x) &= \sum_{y=0}^2 \frac{\binom{3}{x} \binom{2}{y} \binom{3}{2-x-y}}{\binom{8}{2}} \\
 &= \frac{\binom{3}{x}}{\binom{8}{2}} \sum_{y=0}^2 \binom{2}{y} \binom{3}{2-x-y} \\
 &= \frac{\binom{3}{x} \binom{5}{2-x}}{\binom{8}{2}}, \quad x = 0, 1, 2.
 \end{aligned}$$

□

Exercise 3.2.10. A supermarket has two express lines. Let X and Y denote the number of customers in the first and in the second, respectively, at any given time. During nonrush hours, the joint p.d.f of X and Y is summarized by the following table:

		X			
		0	1	2	3
Y	0	0.1	0.2	0	0
	1	0.2	0.25	0.05	0
	2	0	0.05	0.05	0.025
	3	0	0	0.025	0.05

Definition 3.2.11. Suppose X and Y are jointly continuous with joint p.d.f $f_{X,Y}(x, y)$. Then the marginal p.d.fs, $f_X(x)$ and $f_Y(y)$, are given by

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy$$

and

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx.$$

Example 3.2.12. Consider the joint p.d.f of X and Y given by

$$f_{X,Y}(x, y) = \begin{cases} y^2 e^{-y(x+1)} & x \geq 0, \quad y \geq 0 \\ 0, & \text{elsewhere} \end{cases}$$

Find the marginals of X and Y .

Solution. Marginal of X

$$f_X(x) = \int_y f_{X,Y}(x, y) dy = \int_0^{\infty} y^2 e^{-y(x+1)} dy.$$

Let $m = y(x + 1)$ implies that $\frac{dm}{x+1} = dy$

$$\begin{aligned} f_X(x) &= \int_0^\infty \left(\frac{m}{x+1} \right)^2 e^{-m} \frac{dm}{x+1} \\ &= \frac{1}{(x+1)^2} \frac{1}{x+1} \int_0^\infty m^2 e^{-m} dm \\ &= \frac{1}{(x+1)^3} 2! \\ &= \frac{2}{(x+1)^3}, \quad x \geq 0. \end{aligned}$$

Marginal of Y

$$\begin{aligned} f_Y(y) &= \int_0^\infty y^2 e^{-y(x+1)} dx \\ &= y^2 e^{-y} \int_0^\infty e^{-yx} dx \\ &= y^2 e^{-y} \left[-\frac{e^{-yx}}{y} \right]_0^\infty \\ &= y e^{-y}, \quad y \geq 0. \end{aligned}$$

□

3.2.1 Joint Cumulative Distribution Function

Definition 3.2.13. The joint c.d.f of X and Y is given by

$$F(x, y) = P(X \leq x, Y \leq y), \quad (x, y) \in \mathbb{R}^2.$$

Properties of F :

- (i). F is non-decreasing in x for fixed y
- (ii). F is non-decreasing in y for fixed x
- (iii). $\lim_{(x,y) \rightarrow (\infty, \infty)} F(x, y) = 1$.

$$\lim_{x \rightarrow -\infty} F(x, y) = \lim_{y \rightarrow -\infty} F(x, y) = \lim_{(x,y) \rightarrow (-\infty, -\infty)} F(x, y) = 0.$$

Definition 3.2.14. The marginal c.d.f. of X is given by

$$F_X(x) = P(X \leq x) = \lim_{y \rightarrow \infty} F(x, y), \quad x \in \mathbb{R}.$$

The marginal c.d.f. of Y is given by

$$F_Y(y) = P(Y \leq y) = \lim_{x \rightarrow \infty} F(x, y), \quad y \in \mathbb{R}.$$

Example 3.2.15. If X and Y are continuous random variable with joint p.d.f.

$$f_{X,Y}(x, y) = x + y, \quad 0 \leq x \leq 1, \quad 0 \leq y \leq 1.$$

Find

(i). the joint c.d.f of X and Y .

Solution. The joint c.d.f $F(x, y) = P(X \leq x, Y \leq y)$.

$$\begin{aligned} F(x, y) &= \int_{u=0}^x \int_{v=0}^y (u + v) dv du \\ &= \int_{u=0}^x \left[uy + \frac{y^2}{2} \right] du \\ &= \frac{x^2 y + xy^2}{2}, \quad 0 < x < 1, \quad 0 < y < 1. \end{aligned}$$

Therefore

$$F(x, y) = \begin{cases} 0, & 0 \leq x, \quad 0 \leq y \\ \frac{x^2 y + xy^2}{2}, & 0 < x < 1, \quad 0 < y < 1 \\ \frac{y + y^2}{2}, & 0 < y < 1, \quad x \geq 1 \\ \frac{x + x^2}{2}, & 0 < x < 1, \quad y \geq 1 \\ 1, & x \geq 1, \quad y \geq 1 \end{cases}$$

□

(ii). the marginal c.d.f of X and the marginal c.d.f of Y .

Solution. The marginal of X

$$F_X(x) = \lim_{y \rightarrow \infty} F(x, y) = \lim_{y \rightarrow 1} \frac{x^2 y + x y^2}{2} = \begin{cases} 0, & x \leq 0 \\ \frac{x^2 + x}{2}, & 0 < x < 1 \\ 1, & x \geq 1 \end{cases}.$$

The marginal of Y

$$F_Y(y) = \lim_{x \rightarrow \infty} F(x, y) = \begin{cases} 0, & y \leq 0 \\ \frac{y^2 + y}{2}, & 0 < y < 1 \\ 1, & y \geq 1 \end{cases}.$$

□

3.2.2 Expected Values

Definition 3.2.16.

1. Suppose X and Y are discrete random variables with joint p.d.f $f_{X,Y}(x, y)$ and let $g(X, Y)$ be a real-valued function of X and Y . Then the expected value of the random variable $g(X, Y)$ is given by

$$E[g(X, Y)] = \sum_{\text{all } x} \sum_{\text{all } y} g(x, y) f_{X,Y}(x, y).$$

2. Suppose X and Y are continuous random variables with joint p.d.f $f_{X,Y}(x, y)$, and let $g(X, Y)$ be a real-valued function. Then the expected value of the random variable $g(X, Y)$ is given by

$$E[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{X,Y}(x, y) dx dy.$$

Example 3.2.17. If X and Y are jointly distributed

Y	X		
	0	1	2
0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$
1	$\frac{6}{28}$	$\frac{6}{28}$	0
2	$\frac{1}{28}$	0	0

Find $E[g(X, Y)]$ where

1. $g(x, y) = xy$

Solution.

$$\begin{aligned}
 E[g(X, Y)] &= E[XY] \\
 &= \sum_{x=0}^2 \sum_{y=0}^2 xy \cdot P(X = x, Y = y) \\
 &= (1)(1) P(X = 1, Y = 1) \\
 &= \frac{6}{28} \\
 &= \frac{3}{14}.
 \end{aligned}$$

□

2. $g(x, y) = xy^2$

Solution.

$$\begin{aligned}
 E(XY^2) &= \sum_{x=0}^2 \sum_{y=0}^2 xy^2 \cdot P(X = x, Y = y) \\
 &= \sum_{x=1}^2 \sum_{y=1}^2 xy^2 \cdot P(X = x, Y = y) \\
 &= (1)(1)^2 P(X = 1, Y = 1) \\
 &= \frac{6}{28} \\
 &= \frac{3}{14}.
 \end{aligned}$$

□

3. $g(x, y) = x$

Solution.

$$\begin{aligned}
 E(X) &= \sum_{x=0}^2 x P(X = x, Y = y) \\
 &= 1 \cdot P(X = 1, Y = 0) + 1 \cdot P(X = 1, Y = 0) + 2 \cdot P(X = 2, Y = 0) \\
 &= 1 \cdot \frac{9}{28} + 1 \cdot \frac{6}{28} + 2 \cdot \frac{3}{28} \\
 &= \frac{3}{4}.
 \end{aligned}$$

□

Example 3.2.18. Suppose X and Y have a joint distribution given by

$$f_{X,Y}(x, y) = x^2 + \frac{xy}{3}, \quad 0 < x < 1, \quad 0 < y < 2$$

Find $E[g(x, y)]$

1. $g(x, y) = xy$

Solution.

$$\begin{aligned}
 E[g(X, Y)] &= E(XY) = \int_0^1 \int_0^2 xy \left(x^2 + \frac{xy}{3} \right) dy dx \\
 &= \int_0^1 \int_0^2 \left(x^3 y + \frac{x^2 y^2}{3} \right) dy dx \\
 &= \int_0^1 \left[2x^3 + \frac{8x^2}{9} \right] dx \\
 &= \left[\frac{x^4}{2} + \frac{8x^3}{27} \right]_0^1 \\
 &= \frac{1}{2} + \frac{8}{27} \\
 &= \frac{43}{54}.
 \end{aligned}$$

□

2. $g(x, y) = x^2 y$.

Solution.

$$\begin{aligned}
E(X^2Y) &= \int_0^2 \int_0^1 x^2y \left(x^2 + \frac{xy}{3} \right) dx dy \\
&= \int_0^2 \left[\frac{x^5}{5} + \frac{x^4y^2}{12} \right]_0^1 dy \\
&= \int_0^2 \left[\frac{y}{5} + \frac{y^2}{12} \right] dy \\
&= \left[\frac{y^2}{10} + \frac{y^3}{36} \right]_0^2 \\
&= \frac{2}{5} + \frac{2}{9} = \frac{18+10}{45} \\
&= \frac{28}{45}.
\end{aligned}$$

□

3.3. Conditional Probability Functions and Independence of Random Variables

Definition 3.3.1. Let X and Y have a joint probability function $f_{X,Y}(x,y)$. Then the conditional distribution of X given $(Y = y^*)$ denoted by $f_{X/Y}(x/y^*)$ and is given by

$$f_{X/Y}(x/y^*) = \frac{f_{X,Y}(x, y^*)}{f_Y(y^*)}$$

Thus if X and Y are discrete then

$$f_{X/Y}(x/y^*) = \frac{f_{X,Y}(x, y^*)}{f_Y(y^*)} = \frac{P(X = x, Y = y^*)}{P(Y = y^*)}.$$

In general $f_{X,Y}(x, y^*)$ is the joint function evaluated at $Y = y^*$. $f_Y(y^*)$ is the marginal probability function of Y evaluated at $Y = y^*$.

Similarly the conditional distribution of Y given $X = x^*$ given by

$$f_{Y/X}(y/x^*) = \frac{f_{X,Y}(x^*, y)}{f_X(x^*)}.$$

Example 3.3.2. Consider the following jointly distributed random variables X and Y .

Y	0	X 1	2	$f_Y(y)$
0	0.05	0.05	0.10	0.2
1	0.10	0.25	0.05	0.4
2	0.10	0.15	0.05	0.3
3	0.05	0.05	0.00	0.1
$f_X(x)$	0.30	0.50	0.20	

Table 3.1:

Find the conditional probability mass function of Y given that $X = 0$.

Solution. $f_{Y/X}(y/0)$ = conditional distribution of Y given $X = 0$

$$f_{Y/X}(y/0) = \frac{f_{X,Y}(y/0)}{f_X(0)} = \frac{P(X = 0, Y = y)}{P(X = 0)} = \frac{P(X = 0, Y = y)}{0.3}.$$

We have that

$$\begin{aligned} f_{Y/X}(y/0) &= \frac{P(X = 0, Y = 0)}{0.3}, & y = 0 \\ &= \frac{P(X = 0, Y = 1)}{0.3}, & y = 1 \\ &= \frac{P(X = 0, Y = 2)}{0.3}, & y = 2 \\ &= \frac{P(X = 0, Y = 3)}{0.3}, & y = 3. \end{aligned}$$

Therefore

Y	0	1	2	3
$f_{Y/X}(y/0)$	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{6}$

□

Example 3.3.3. Suppose that the joint distribution of X and Y is given by

$$f(x, y) = \begin{cases} \frac{e^{-x/y} e^{-y}}{y}, & 0 < x < \infty, \quad 0 < y < \infty \\ 0, & \text{otherwise} \end{cases}.$$

Find $P[X > 1/Y = 1]$.

Solution. We first obtain the conditional distribution of X given that $Y = y$.

$$f_{X/Y}(x/y) = \frac{f(x, y)}{f_Y(y)}$$

Then marginal $f_Y(y)$ is

$$f_Y(y) = \frac{1}{y} e^{-y} \int_0^\infty e^{-x/y} dx = \frac{1}{y} e^{-y} [-ye^{-x/y}]_0^\infty = e^{-y}.$$

Hence,

$$f_{X/Y}(x/y = 1) = \frac{e^{-x} e^{-1}}{e^{-1}} = e^{-x}.$$

Therefore

$$P[X > 1/Y = 1] = \int_1^\infty e^{-x} dx = [-e^{-x}]_1^\infty = e^{-1}.$$

□

Definition 3.3.4. Let X and Y have a joint probability function $f_{X,Y}(x, y)$, X and Y are said to be independent if and only if

$$f_{X,Y}(x, y) = f_X(x)f_Y(y)$$

where $f_X(x)$ and $f_Y(y)$ are the marginal probability functions of X and Y respectively.

Theorem 3.3.5. If X and Y have a joint probability function $f_{X,Y}(x, y)$ and X and Y are independent then

$$f_{X/Y}(x/y^*) = f_X(x) \quad \text{and} \quad f_{Y/X}(y/x^*) = f_Y(y).$$

Proof. In the independent case

$$f_{X/Y}(x/y^*) = \frac{f_{X,Y}(x, y)}{f_Y(y^*)} = \frac{f_{X,Y}(X = x, Y = y^*)}{f_Y(y^*)} = \frac{f_X(x) \cdot f_Y(y^*)}{f_Y(y^*)} = f_X(x).$$

□

Example 3.3.6. For the joint distribution of X and Y given by

$$f_{X,Y}(x, y) = \frac{\binom{3}{x} \binom{2}{y} \binom{3}{2-x-y}}{\binom{8}{2}}$$

the marginals of X and Y are

$$f_X(x) = \frac{\binom{3}{x} \binom{5}{2-x}}{\binom{8}{2}} \quad x = 0, 1, 2$$

and

$$f_Y(y) = \frac{\binom{2}{y} \binom{6}{2-y}}{\binom{8}{2}} \quad y = 0, 1, 2$$

X and Y are not independent because

$$f_{X,Y}(x, y) \neq f_X(x)f_Y(y).$$

Exercise 3.3.7. A minibus company receives calls from broken down buses and wrecker crew must haul the buses in for service. The joint distribution of the numbers of calls on Mondays and Tuesdays is given by:

Tuesdays	Mondays				
	0	1	2	3	4
0	0.02	0.04	0.06	0.04	0.04
1	0.02	0.04	0.06	0.04	0.04
2	0.01	0.02	0.03	0.02	0.02
3	0.04	0.08	0.12	0.08	0.08
4	0.01	0.02	0.03	0.02	0.02

Table 3.2:

Are the number of calls on Mondays and Tuesdays independent?

Theorem 3.3.8. If X and Y are independent random variables,

$$E(XY) = E(X) E(Y)$$

provided $E(X)$ and $E(Y)$ both exist.

Proof. Suppose that X and Y are continuous independent random variables. Then the joint

p.d.f can be written as $f_{X,Y}(x, y) = f_X(x) f_Y(y)$. Thus

$$\begin{aligned} E(XY) &= \int_x \int_y xy f_{X,Y}(x, y) dy dx \\ &= \int_x \int_y xy f_X(x) f_Y(y) dy dx \\ &= \left[\int_x x f_X(x) dx \right] \left[\int_y y f_Y(y) dy \right] \\ &= E(X) E(Y). \end{aligned}$$

□

Note 3.3.9. When random variables are not independent, a measure of the relationship between them, their *covariance*, enters into the picture.

Definition 3.3.10. If X and Y have a joint distribution function $f_{X,Y}(x, y)$, then the covariance of X and Y denoted by $Cov(X, Y)$ is given by

$$Cov(X, Y) = E\{(X - \mu_X)(Y - \mu_Y)\}$$

where $\mu_X = E(X)$ and $\mu_Y = E(Y)$.

Remark 3.3.11. For the random variables X and Y :

1. $Cov(X, Y) = E(XY) - E(X) E(Y) = E(XY) - \mu_X \mu_Y$
2. $Cov(X, Y) = Cov(Y, X)$
3. $Cov(X, X) = Var(X)$
4. $Cov(aX, Y) = a Cov(X, Y)$
5. Covariance is a measure of association.

Definition 3.3.12. If X and Y have joint distribution function $f_{X,Y}(x, y)$ then the correlation coefficient of X and Y denoted by ρ_{XY} and is given by

$$\rho_{XY} = \frac{Cov(X, Y)}{\sqrt{Var(X) Var(Y)}}.$$

$$\rho_{XY} = \frac{Cov(X, Y)}{\sqrt{\sigma_X^2 \sigma_Y^2}} = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$$

ρ_{XY} is a standardized measure of association $-1 \leq \rho_{XY} \leq 1$.

Remark 3.3.13. If X and Y are independent then, $\text{Cov}(X, Y) = 0$, hence, $\rho_{XY} = 0$.

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = E(X)E(Y) - E(X)E(Y) = 0$$

Question. If $\text{Cov}(X, Y) = 0$ does it mean that X and Y are independent?

Answer. Yes only if X and Y have joint normal distribution. Otherwise no! □

Theorem 3.3.14. The value of the correlation coefficient ρ_{XY} of X and Y is always between -1 and 1 .

$$-1 \leq \rho_{XY} \leq 1$$

Proof. Let

$$Q(t) = E\{[(X - \mu_X) + (Y - \mu_Y)t]^2\},$$

since $[(X - \mu_X) + (Y - \mu_Y)t]^2 \geq 0$

$$\implies E\{[(X - \mu_X) + (Y - \mu_Y)t]^2\} \geq 0$$

i.e $Q(t) \geq 0$

$$\begin{aligned} Q(t) &= E\{(X - \mu_X)^2 + 2(X - \mu_X)(Y - \mu_Y)t + t^2(Y - \mu_Y)^2\} \\ &= E(X - \mu_X)^2 + 2E[(X - \mu_X)t] + t^2E(Y - \mu_Y)^2 \\ &= \text{Var}(X) + 2\text{Cov}(X, Y)t + t^2\text{Var}(Y). \end{aligned}$$

Therefore

$$Q(t) = \text{Var}(Y)t^2 + 2\text{Cov}(X, Y)t + \text{Var}(X).$$

$Q(t)$ is a quadratic in t , since $Q(t) \geq 0$ implies its discriminant must be less than or equal to zero.i.e

$$[2\text{Cov}(X, Y)]^2 - 4\text{Var}(Y)\text{Var}(X) \leq 0$$

$$4[\text{Cov}(X, Y)]^2 \leq 4\text{Var}(X)\text{Var}(Y)$$

$$\implies \frac{[\text{Cov}(X, Y)]^2}{\text{Var}(X)\text{Var}(Y)} \leq 1$$

$$\implies \left[\frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} \right]^2 \leq 1$$

$$\rho_{XY}^2 \leq 1$$

Therefore, $-1 \leq \rho_{XY} \leq 1$. □

Example 3.3.15. A random variable X has a uniform distribution on $[-1, 1]$. Let $Y = X^2$. Find the correlation coefficient between X and Y .

Solution. Since $Y = X^2$

$$\begin{aligned}\text{Cov}(X, Y) &= E(XY) - E(X) E(Y) \\ &= E(X^3) - E(X) E(X^2).\end{aligned}$$

Since X has the probability density function

$$f(x) = \begin{cases} \frac{1}{2} & \text{if } -1 \leq x \leq 1, \\ 0 & \text{otherwise,} \end{cases}$$

we find that

$$E(X) = \int_{-1}^1 x, \frac{1}{2} dx = 0$$

and

$$E(X^3) = \int_{-1}^1 x^3 \frac{1}{2} dx = 0.$$

Therefore

$$\text{Cov}(X, Y) = 0.$$

Hence $\rho = 0$ although X and Y are clearly dependent. □

Results of this type complicate the interpretation of ρ but essentially it is a measure of the degree of linear dependence between X and Y .

Example 3.3.16. If X and Y have joint distribution given on the table.

Y	X			$f_Y(y)$
	0	1	2	
0	3/28	9/28	3/28	15/28
1	3/14	3/14	0	12/28
2	1/28	0	0	1/28
$f_X(x)$	10/28	15/28	3/28	

Find ρ_{XY} .

Solution. To find

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \cdot \text{Var}(Y)}},$$

we need to find

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y),$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

and

$$\text{Var}(Y) = E(Y^2) - [E(Y)]^2.$$

The marginals of X and Y are

X	0	1	2
$f_X(x)$	10/28	15/28	3/28

and

Y	0	1	2
$f_Y(y)$	15/28	12/28	1/28

. Then

$$E(X) = \sum_{x=0}^2 x f_X(x) = \frac{15}{28} + \frac{6}{28} = \frac{21}{28} = \frac{3}{4},$$

$$E(X^2) = \sum_{x=0}^2 x^2 f_X(x) = \frac{15}{28} + \frac{12}{28} = \frac{27}{28},$$

Hence

$$\text{Var}(X) = \frac{27}{28} - \frac{9}{16} = \frac{45}{112} = \frac{9}{4} \cdot \frac{5}{28}.$$

Also

$$E(Y) = \sum_{y=0}^2 y f_Y(y) = \frac{12}{28} + \frac{2}{28} = \frac{14}{28} = \frac{1}{2},$$

$$E(Y^2) = \sum_{y=0}^2 y^2 f_Y(y) = \frac{12}{28} + \frac{4}{28} = \frac{16}{28} = \frac{4}{7}$$

$$\text{Var}(Y) = \frac{4}{7} - \frac{1}{4} = \frac{16-7}{28} = \frac{9}{28}.$$

$$E(XY) = 1 \times \frac{3}{14} = \frac{3}{14}.$$

The covariance

$$\text{Cov}(X, Y) = \frac{3}{4} - \frac{3}{4} \cdot \frac{1}{2} = \frac{3}{2} \left(\frac{1}{7} - \frac{1}{4} \right) = \frac{-9}{2 \times 28}.$$

Therefore

$$\rho_{XY} = \frac{\frac{-9}{2 \times 28}}{\sqrt{\frac{9}{4} \cdot \frac{5}{28} \cdot \frac{9}{28}}} = \frac{\frac{-9}{2 \times 28}}{\frac{9}{28} \cdot \frac{1}{2} \cdot \sqrt{5}} = \frac{-1}{\sqrt{5}}.$$

□

Exercise 3.3.17. Suppose that

$$f_{XY}(x, y) = \frac{2}{3}(x + 2y), \quad 0 \leq x \leq 1, \quad 0 \leq y \leq 1.$$

Find ρ_{XY} .

Definition 3.3.18. If X and Y have a joint distribution $f_{XY}(x, y)$ then the conditional expectation is given as follows

$$E(X/Y = y^*) = \begin{cases} \sum_X x \cdot f_{X/Y}(x/y^*) & \text{if } X \text{ is discrete} \\ \int_X x \cdot f_{X/Y}(x/y^*) & \text{if } X \text{ is continuous} \end{cases}$$

where $f_{X/Y}(x/y^*)$ is the conditional distribution of X given $Y = y^*$.

$$E(Y/X = x^*) = \begin{cases} \sum_Y y \cdot f_{Y/X}(y/x^*) & \text{if } Y \text{ is discrete} \\ \int_Y y \cdot f_{Y/X}(y/x^*) & \text{if } Y \text{ is continuous} \end{cases}$$

Example 3.3.19. Consider the following jointly distributed random variables

X	Y			$f_X(x)$
	0	1	2	
0	0.05	0.05	0.10	0.20
1	0.10	0.25	0.05	0.40
2	0.10	0.15	0.05	0.30
3	0.05	0.05	0.00	0.10
$f_Y(y)$	0.30	0.50	0.20	

Find:

(a). $E(Y/X = 1)$.

Solution. The distribution $f_{Y/X}(y/1)$

Y	0	1	2
$f_{Y/X}(y/1)$	0.25	0.625	0.125

Then the conditional expectation

$$E(Y/X = 1) = 0(0.25) + 1(0.625) + 2(0.125) = 0.875$$

□

(b). $E(X/Y = 0)$

Solution. The distribution $f_{X/Y}(x/0)$

X	0	1	3	3
$f_{X/Y}(x/0)$	0.167	0.333	0.333	0.167

The conditional expectation

$$E(X/Y = 0) = 0(0.167) + 1(0.333) + 2(0.333) + 3(0.167) = 1.5$$

□

Example 3.3.20. Consider the joint p.d.f of X and Y given by

$$f_{X,Y}(x,y) = \begin{cases} \frac{12}{5} x(2-x-y) & 0 < x < 1, \quad 0 < y < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Find $E(X/Y = 1)$.

Solution. The marginal

$$\begin{aligned} f_Y(1) &= \int_0^1 \frac{12}{5} x(2-x-1) dx \\ &= \frac{12}{5} \int_0^1 (x-x^2) dx \\ &= \frac{12}{5} \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 \\ &= \frac{12}{5} \left(\frac{1}{6} \right). \end{aligned}$$

Then we get

$$f_{X/Y}(x/1) = \frac{f_{X,Y}(x,1)}{f_Y(1)} = \frac{\frac{12}{5}(x-x^2)}{\frac{12}{5} \cdot \frac{1}{6}} = 6(x-x^2).$$

Therefore

$$\begin{aligned} E[X/Y = 1] &= \int_0^1 x f_{X/Y}(x/1) dx \\ &= \int_0^1 6(x-x^2) dx \\ &= 6 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 \\ &= 6 \left[\frac{1}{2} - \frac{1}{3} \right] \\ &= 1. \end{aligned}$$

□

3.3.1 Special Joint Distribution Function (Bivariate Normal)

X and Y are said to have the bivariate normal distribution if their joint probability density function is given by

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp \left\{ -\frac{1}{2(1-\rho^2)} \left(\frac{(x-\mu_X)^2}{\sigma_X^2} - \frac{2\rho(x-\mu_X)(y-\mu_Y)}{\sigma_X \cdot \sigma_Y} + \frac{(y-\mu_Y)^2}{\sigma_Y^2} \right) \right\}$$

for $(x,y) \in \mathbb{R}^2$, $-\infty < x < \infty$, $-\infty < y < \infty$, and $\rho =$ correlation coefficient of X and Y .

$$\mu_X = E(X), \quad \sigma_X^2 = \text{var}(X)$$

$$\mu_Y = E(Y), \quad \sigma_Y^2 = \text{var}(Y)$$

If $\rho = 0$

$$\begin{aligned} f_{X,Y}(x,y) &= \frac{1}{2\pi\sigma_X\sigma_Y} \exp \left\{ -\frac{1}{2} \left(\frac{(x-\mu_X)^2}{\sigma_X^2} + \frac{(y-\mu_Y)^2}{\sigma_Y^2} \right) \right\} \\ &= \frac{1}{2\pi\sigma_X\sigma_Y} \exp \left\{ -\frac{1}{2} \frac{(x-\mu_X)^2}{\sigma_X^2} \right\} \cdot \exp \left\{ -\frac{1}{2} \frac{(y-\mu_Y)^2}{\sigma_Y^2} \right\} \\ &= \frac{1}{\sqrt{2\pi\sigma_X^2}} \exp \left\{ -\frac{1}{2} \frac{(x-\mu_X)^2}{\sigma_X^2} \right\} \cdot \frac{1}{\sqrt{2\pi\sigma_Y^2}} \exp \left\{ -\frac{1}{2} \frac{(y-\mu_Y)^2}{\sigma_Y^2} \right\} \\ &= f_X(x) \cdot f_Y(y) \end{aligned}$$

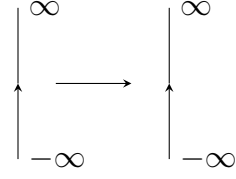
where $X \sim N(\mu_X, \sigma_X^2)$, $Y \sim (\mu_Y, \sigma_Y^2)$

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f_{X,Y}(x,y)dy, \quad \text{Let } z = \frac{x-\mu_X}{\sigma_X} \\ &= \int_{-\infty}^{\infty} \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp \left\{ -\frac{1}{2(1-\rho^2)} \left(z^2 - 2\rho z \cdot \frac{(y-\mu_Y)}{\sigma_Y} + \frac{(y-\mu_Y)^2}{\sigma_Y^2} \right) \right\} dy \end{aligned}$$

$$\text{Let } v = \frac{y - \mu_Y}{\sigma_Y}$$

 $\implies$

$$dv = \frac{dy}{\sigma_Y}$$



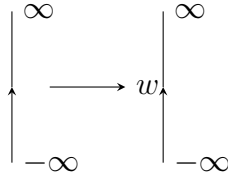
$$= \frac{1}{2\pi\sigma_X} \exp\left\{-\frac{z^2}{2(1-\rho^2)}\right\} \int_{-\infty}^{\infty} \frac{1}{\sigma_Y\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2}(v^2 - 2\rho zv)\right\} dv$$

$$= \frac{1}{2\pi\sigma_X} \int_{-\infty}^{\infty} \frac{1}{\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}(v^2 - 2\rho zv + (\rho z)^2 - (\rho z)^2)\right\} dv$$

$$= \frac{\exp\left\{-\frac{z^2}{2(1-\rho^2)}\right\} \cdot \exp\left\{-\frac{1(\rho^2 z^2)}{2(1-\rho^2)}\right\}}{2\pi\sigma_X} \int_{-\infty}^{\infty} \frac{1}{\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}(v - \rho z)^2\right\} dv$$

$$= \frac{\exp\left\{-\frac{z^2}{2}\right\}}{\sqrt{2\pi\sigma_X^2}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi(1-\rho^2)}} \exp\left\{-\frac{1}{2(1-\rho^2)}(v - \rho z)^2\right\} dv$$

$$\text{Let } w = \frac{v - \rho z}{\sqrt{1-\rho^2}} \implies dw = \frac{dv}{\sqrt{1-\rho^2}} \implies \sqrt{1-\rho^2}dw = dv$$



$$\begin{aligned}
&= \frac{\exp\left\{-\frac{z^2}{2}\right\}}{\sqrt{2\pi\sigma_X^2}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi} \cdot \sqrt{1-\rho^2}} e^{-\frac{1}{2}w^2} \cdot \sqrt{1-\rho^2} dw \\
&= \frac{e^{-\frac{z^2}{2}}}{\sqrt{2\pi\sigma_X^2}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}w^2} dw \quad \longrightarrow \quad \text{the inside is the pdf of } N(0,1) \\
&= \frac{e^{-\frac{z^2}{2}}}{\sqrt{2\pi\sigma_X^2}} = \frac{e^{-\frac{1}{2}\left(\frac{x-\mu_X}{\sigma_X}\right)^2}}{\sqrt{2\pi\sigma_X^2}} \sim N(\mu_X, \sigma_X^2)
\end{aligned}$$

It can also be shown that the condition distribution of X given $Y = y^*$ is

$$N\left(\mu_X + \rho \frac{\sigma_X}{\sigma_Y}(y^* - \mu_Y), \sigma_X^2(1 - \rho^2)\right)$$

, distribution of Y given $X = x^*$ is

$$N\left(\mu_Y + \rho \frac{\sigma_Y}{\sigma_X}(x^* - \mu_X), \sigma_Y^2(1 - \rho^2)\right).$$

To determine the conditional density of X given that $Y = y$, we will continually collect all factors that don not depend on x and represent them by the constants C_i .

$$\begin{aligned}
f_{X/Y}(x|y) &= \frac{f_{X,Y}(x,y)}{f_Y(y)} \\
&= C_1 f_{X,Y}(x,y) \\
&= C_2 \exp\left\{-\frac{1}{2(1-\rho^2)} \left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 - 2\rho \frac{x(y-\mu_Y)}{\rho_X \rho_Y}\right]\right\} \\
&= C_3 \exp\left\{-\frac{1}{2\sigma_X^2(1-\rho^2)} \left[x^2 - 2x \left(\mu_X + \rho \frac{\sigma_X}{\sigma_Y}(y - \mu_Y)\right)\right]\right\} \\
&= C_4 \exp\left\{-\frac{1}{2\sigma_X^2(1-\rho^2)} \left[x - \left(\mu_X + \rho \frac{\sigma_X}{\sigma_Y}(y - \mu_Y)\right)\right]^2\right\}
\end{aligned}$$

From the last equation we can conclude that, given $Y = y$, the random variable X is normally distributed with mean $\mu_X + \rho \frac{\sigma_X}{\sigma_Y}(y - \mu_Y)$ and the variance $\sigma_X^2(1 - \rho^2)$.

3.4. Joint Moment Generating Function

Definition 3.4.1. Let X and Y have a joint probability function $f_{X,Y}(x, y)$ then the joint moment generating function of X and Y denoted by

$$M_{X,Y}(t_1, t_2) = E(e^{t_1X+t_2Y})$$

given by

$$E(e^{t_1X+t_2Y}) = \begin{cases} \sum_X \sum_Y e^{t_1X+t_2Y} P(X=x, Y=y) & \text{discrete} \\ \int_X \int_Y e^{t_1X+t_2Y} f_{X,Y}(x, y) dx dy & \text{continuous} \end{cases}$$

Remark 3.4.2.

1. If $M(t_1, t_2)$ exists then the m.g.f of X is given by

$$M_X(t_1) = E(e^{t_1X}) = M(t_1, 0)$$

and the m.g.f of Y is given by

$$M_Y(t_2) = E(e^{t_2Y}) = M(0, t_2).$$

2. If X and Y are independent the $M_{X,Y}(t_1, t_2) = M_X(t_1)M_Y(t_2)$.

Proof. If X and Y are independent then $f_{X,Y}(x, y) = f_X(x)f_Y(y)$.

$$\begin{aligned} E(e^{t_1X+t_2Y}) &= \int_X \int_Y e^{t_1X+t_2Y} f_X(x)f_Y(y) dx dy \\ &= \int_X e^{t_1X} f_X(x) dx \int_Y e^{t_2Y} f_Y(y) dy \\ &= M_X(t_1) M_Y(t_2). \end{aligned}$$

□

Example 3.4.3. Find the joint moment generating function for X and Y with joint p.d.f.

$$f_{X,Y}(x, y) = \frac{x+y}{3}, \quad 0 < x < 1, \quad 0 < y < 2.$$

Solution.

$$\begin{aligned}
M_{X,Y}(t_1, t_2) &= E(e^{t_1 X + t_2 Y}) \\
&= \int_0^2 \int_0^1 e^{xt_1 + yt_2} \left(\frac{x+y}{3} \right) dx dy \\
&= \frac{1}{3} \int_0^2 e^{yt_2} \int_0^1 e^{xt_1} (x+y) dx dy \\
&= \frac{1}{3} \int_0^2 e^{yt_2} \int_0^1 (xe^{xt_1} + ye^{xt_1}) dx dy \\
&= \frac{1}{3} \int_0^2 e^{yt_2} \left[\frac{xe^{xt_1}}{t_1} \Big|_0^1 - \int_0^1 \frac{e^{xt_1}}{t_1} dx + \frac{y}{t_1} e^{xt_1} \Big|_0^1 \right] dy \\
&= \frac{1}{3} \int_0^2 e^{yt_2} \left[\frac{e^{t_1}}{t_1^2} - \frac{e^{xt_1}}{t_1^2} \Big|_0^1 + \frac{ye^{t_1}}{t_1} - \frac{y}{t_1} \right] dy \\
&= \frac{1}{3} \int_0^2 e^{yt_2} \left[\frac{e^{t_1}}{t_1} + \frac{ye^{t_1}}{t_1} - \frac{y}{t_1} - \frac{e^{t_1}}{t_1^2} + \frac{1}{t_1^2} \right] dy \\
&= \frac{1}{3t_1^2} \int_0^2 e^{yt_2} \left[t_1 e^{t_1} - e^{t_1} + 1 + yt_1 e^{t_1} - yt_1 \right] dy \\
&= \frac{1}{3t_1^2} \left[(1 + t_1 e^{t_1} - e^{t_1}) \int_0^2 e^{yt_2} dy + t_1 (e^{t_1} - 1) \int_0^2 ye^{yt_2} dy \right] \\
&= \frac{1}{3t_1^2} \left[(1 + t_1 e^{t_1} - e^{t_1}) \frac{e^{yt_2}}{t_2} \Big|_0^2 + t_1 (e^{t_1} - 1) \left[\frac{ye^{yt_2}}{t_2} \Big|_0^2 - \int_0^2 \frac{e^{yt_2}}{t_2} dy \right] \right] \\
&= \frac{1}{3t_1^2} \left[(1 + t_1 e^{t_1} - e^{t_1}) \left(\frac{e^{2t_2}}{t_2} - \frac{1}{t_2} \right) + t_1 (e^{t_1} - 1) \left(\frac{2e^{2t_2}}{t_2} - \frac{e^{yt_2}}{t_2^2} \Big|_0^2 \right) \right] \\
&= \frac{1}{3t_1^2} \left[(1 + t_1 e^{t_1} - e^{t_1}) \left(\frac{e^{2t_2} - 1}{t_2} \right) + t_1 (e^{t_1} - 1) \left(\frac{2e^{2t_2}}{t_2} - \frac{e^{2t_2}}{t_2^2} + \frac{1}{t_2^2} \right) \right].
\end{aligned}$$

Therefore

$$M_{X,Y}(t_1, t_2) = E(e^{Xt_1 + Yt_2}) = E(e^{Xt_1} e^{Yt_2}) = E\left(\sum_{k=0}^{\infty} \frac{(Xt_1)^k}{k!} \cdot \sum_{m=0}^{\infty} \frac{(Yt_2)^m}{m!}\right).$$

The marginals can be obtained by

$$\begin{aligned}
M_X(t_1) &= \lim_{t_2 \rightarrow 0} M_{X,Y}(t_1, t_2) \\
M_Y(t_2) &= \lim_{t_1 \rightarrow 0} M_{X,Y}(t_1, t_2).
\end{aligned}$$

□